

TECHNOLOGY



AWS Solution Architect

VPC, Networking, and Content Delivery



Learning Objectives

By the end of this lesson, you will be able to:

- Analyze the functionality of Network ACLs in controlling traffic at the subnet level within a VPC
- Configure VPC endpoints to enable private connections to AWS services, eliminating the need for public internet exposure
- Implement IAM policies to manage secure access to AWS resources effectively
- Evaluate and implement different routing methods to optimize network traffic management within a VPC



Learning Objectives

By the end of this lesson, you will be able to:

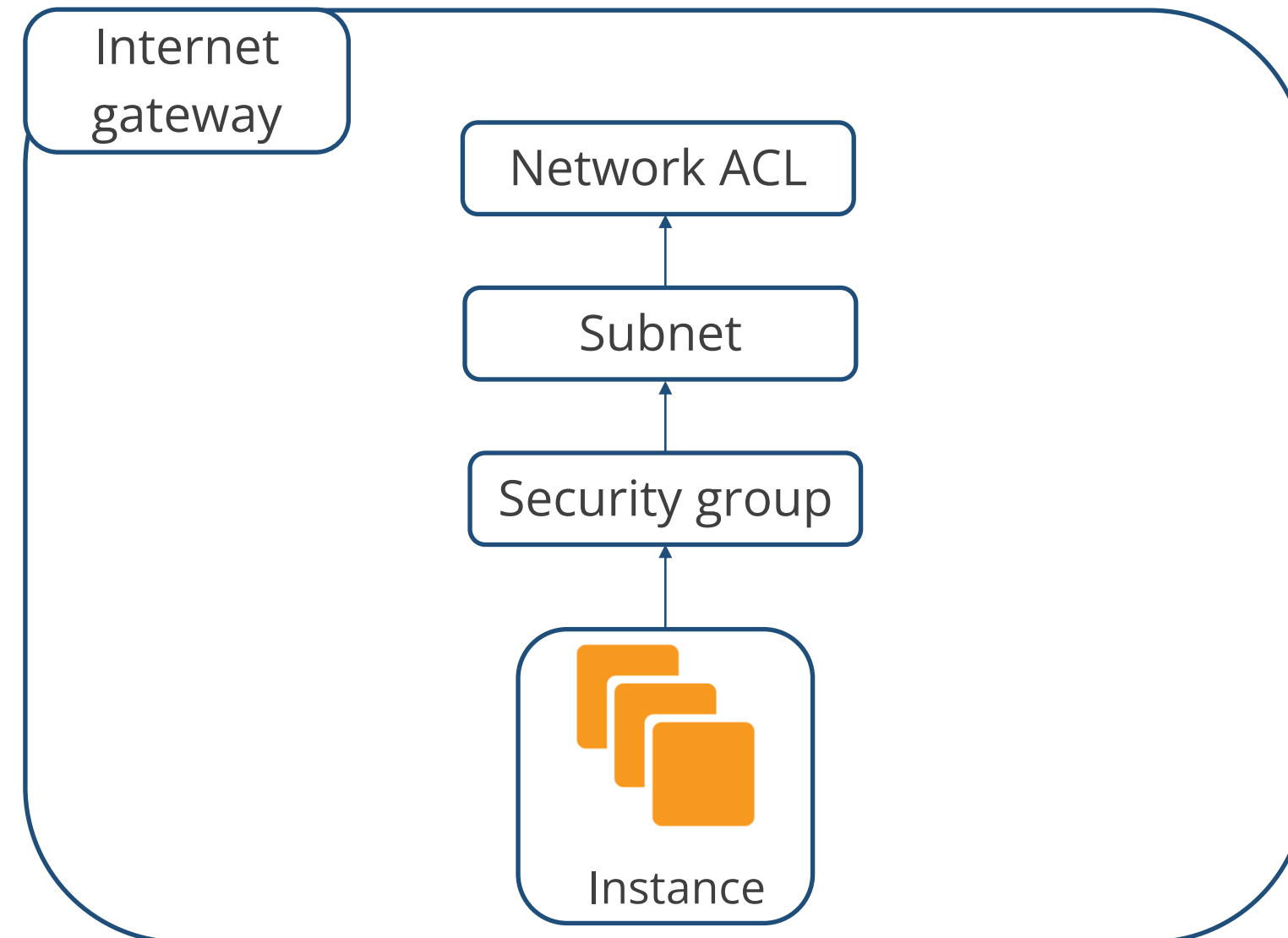
- Implement NAT and Internet Gateways to facilitate secure internet access for instances in private subnets
- Create and configure a CloudFront distribution to distribute content globally



Security Group and NACL

Security Group

It acts as a virtual firewall for EC2 instances, controlling both incoming and outgoing traffic.



Security Group: Rules

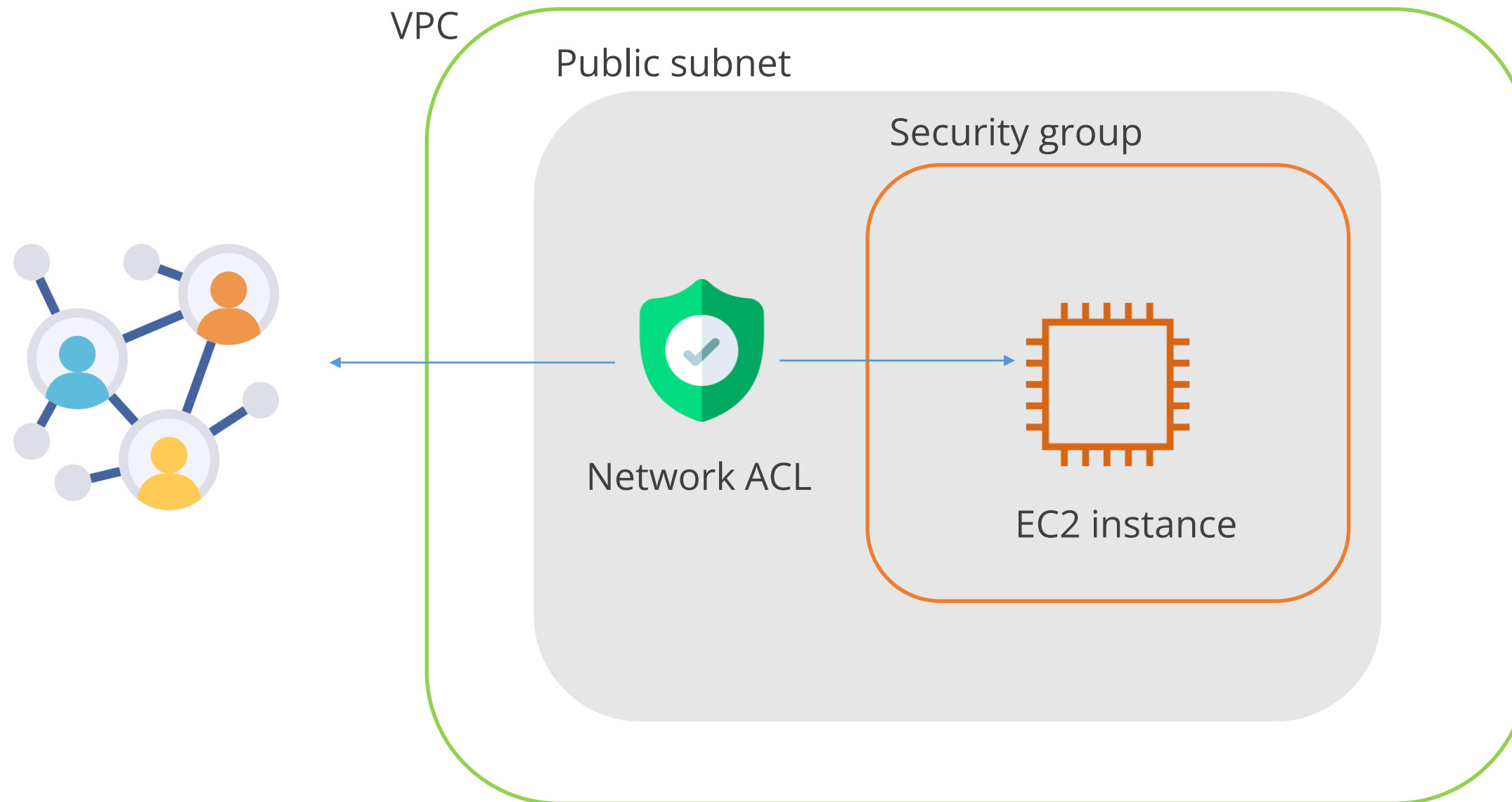
These are the following rules for security groups:



- By default, security groups include outbound rules that permit all outbound traffic.
- A user can filter traffic using security group rules based on protocols and ports.
- A user cannot construct security group rules that restrict access.
- A user can change a security group's rules, and all connected instances will be updated.

Network ACL (NACL)

It is an optional security feature. It acts as a firewall, controlling traffic that enters and leaves a subnet.



Network ACL (NACL): Rules

The rules for Network ACL include:



- **Rule number:** Rules rank from the lowest to the highest number. If traffic matches a lower-numbered rule, the system ignores any conflicting higher-numbered rule.
- **Type:** This category defines the traffic type, like SSH. Users can select a specific range or all traffic types.
- **Protocol:** Users can select any protocol with a standard number. For example, when using ICMP, they can identify its number and specify any or all its types and codes.

Network ACL (NACL): Rules

The rules for Network ACL include:



- **Port reach:** This specifies the port or range of ports that the network traffic uses. For example, HTTP traffic uses port **80** by default.
- **Source:** For inbound rules, this factor indicates the traffic's origin using the CIDR range.
- **Destination:** For outbound rules, this defines the traffic's destination, also specified using the CIDR range.
- **Allow or Deny:** This rule determines if the system should allow or deny the traffic.

Network ACL vs. Security Group

The differences between network ACL and security group are as follows:

Security group	Network ACL
• Operates at the instance level • Is stateful • Supports only allow rules	• Acts as a firewall at the subnet level • Is stateless • Supports both allow and deny rules

IP Addresses

The types of IP addresses in Amazon VPC are as follows:

Private IP address

Public IP address

Elastic IP address

- A private IP address is unreachable over the internet.
- Users can use it to communicate between instances within the same VPC.
- When users stop an instance, the public IP associated with it is released.

IP Addresses

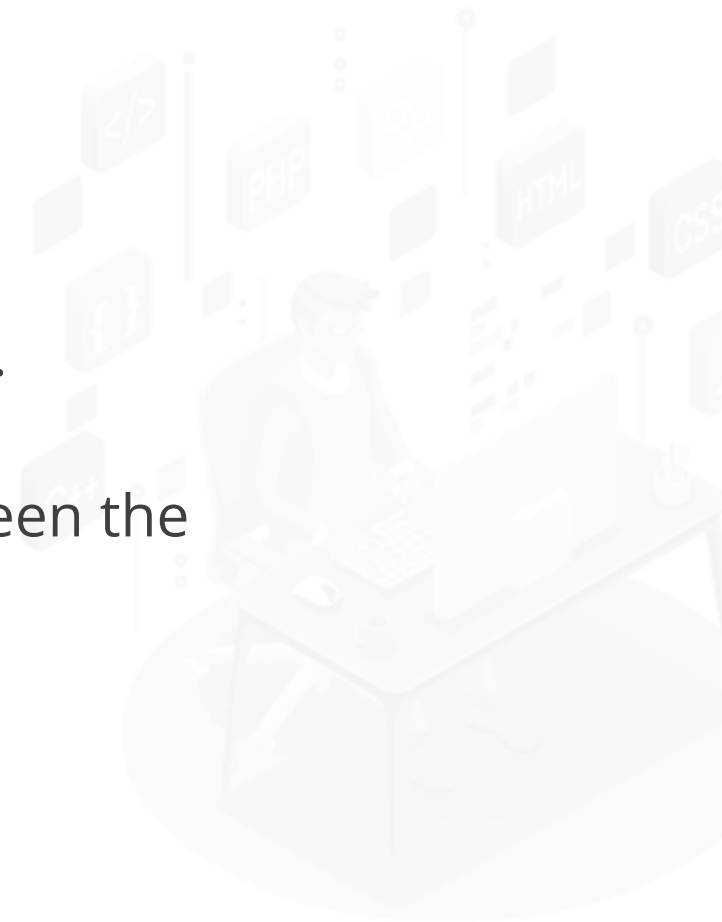
The types of IP addresses in Amazon VPC are as follows:

Private IP address

Public IP address

Elastic IP address

- A public IP address is reachable from the internet.
- Users can use this address to communicate between the instances and the internet.



IP Addresses

The types of IP addresses in Amazon VPC are as follows:

Private IP address

Public IP address

Elastic IP address

- An Elastic IP address is a public IP address that users can allocate to their accounts.
- Users can associate or disassociate it with instances as needed.
- It stays with the user's account until they decide to release it.

Multiple IP Addresses

Benefits of assigning multiple IP addresses to an instance in the VPC:

- Host multiple websites on one server using different SSL certificates. Each certificate can be associated with a unique IP address.
- Operate network appliances like firewalls or load balancers that need multiple IP addresses for each network interface.
- Redirect internal traffic to a standby instance if the primary one fails. This is done by reassigning the secondary IP address to the standby instance.

Creating Key Pair for EC2



Duration:10 min

Problem Statement:

You have been assigned a task to create a key pair for an EC2 instance to be used as a security credential for user authorization.

Outcome:

You will be able to create a key pair for an EC2 instance to be used as a security credential for user authorization.

Note: Refer to the demo document for detailed steps:
01_Creating_Key Pair_for_EC2

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a key pair for an EC2 instance



Creating a Default VPC



Duration:10 min

Problem Statement:

You have been assigned a task to set up a secure and isolated network environment that forms the basis for hosting AWS resources.

Outcome:

You will be able to create a default VPC to securely host AWS resources, gaining essential networking skills within the AWS ecosystem.

Note: Refer to the demo document for detailed steps:
[02_Creating_a_Default_VPC](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a default VPC



Exploring Network ACL Firewall



Duration:10 min

Problem Statement:

You have been assigned a task to set up the prerequisites for Network ACL Firewall and use it to secure network traffic in your AWS environment.

Outcome:

You will be able to set up and explore the network ACL firewall to manage traffic in and out of your subnets.

Note: Refer to the demo document for detailed steps:
[03_Exploring_Network_ACL_Firewall](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a network ACL



Quick Check



You need to control inbound and outbound traffic at the subnet level in your VPC while allowing stateless filtering. Which feature should you use?

- A. Security Group
- B. Network ACL (NACL)
- c. Key Pair
- D. Default VPC

VPC Flow Logs

VPC Flow Logs

It is a feature that captures information about the IP traffic going to and from network interfaces in a VPC.



When a user creates a flow log for a subnet or VPC, it monitors all network interfaces within that subnet or VPC.

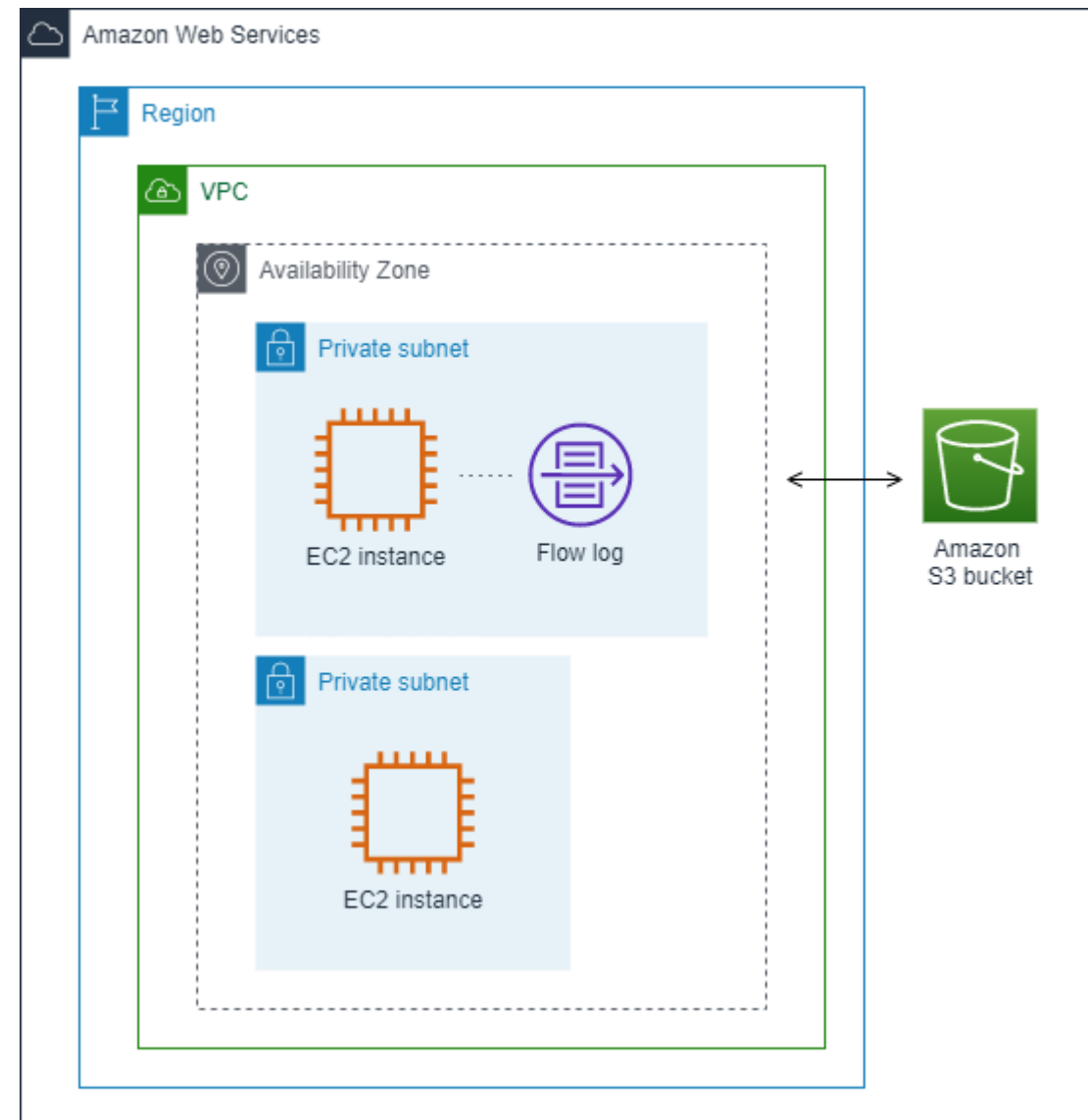
VPC Flow Logs

It can be published to two destinations: CloudWatch Logs and Amazon S3.



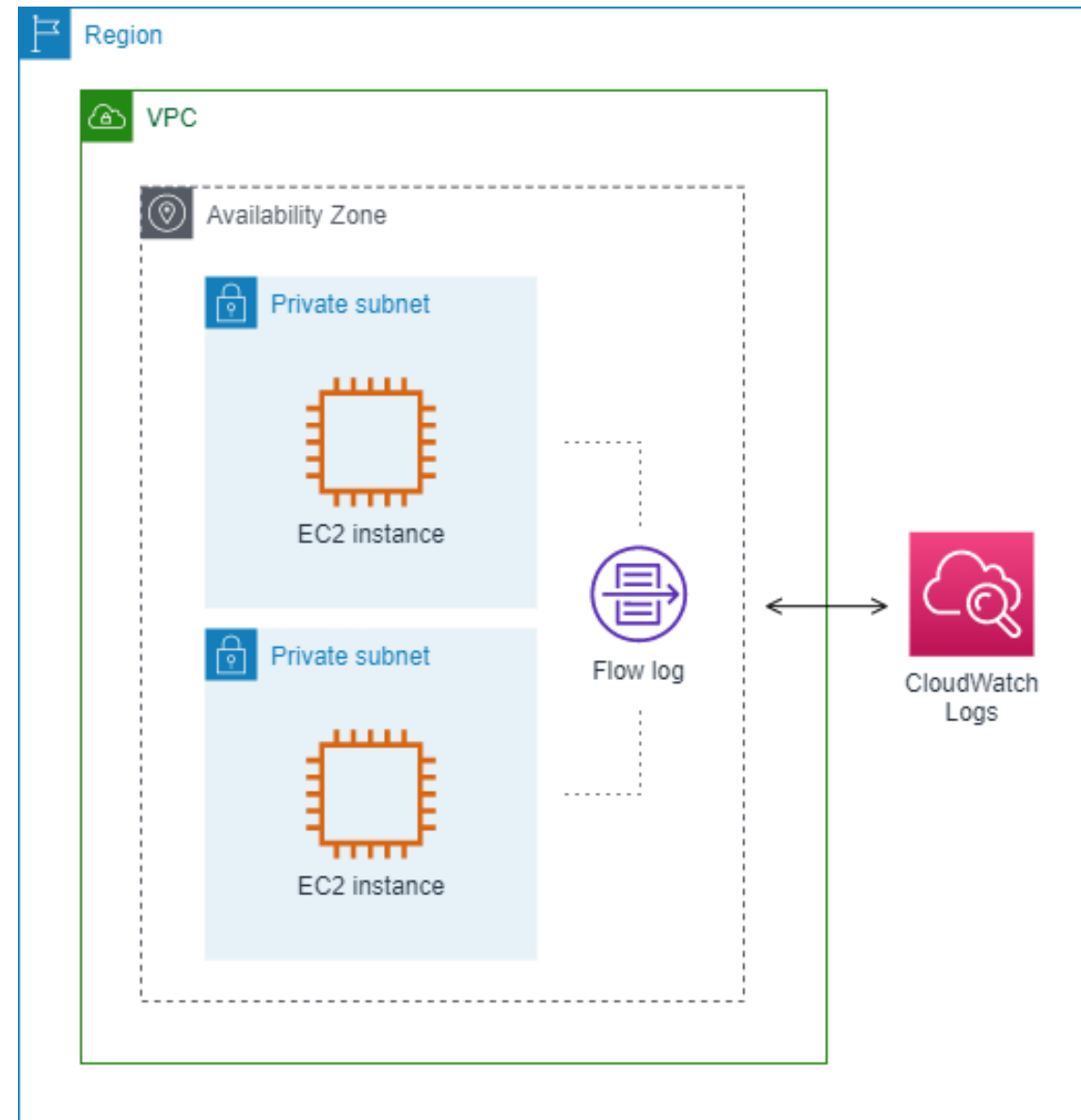
- **Amazon S3:** VPC Flow Logs collects and consolidates flow log records into log files.
- **CloudWatch Logs:** When flow log data is published to CloudWatch Logs, it populates a specific log group.

VPC Flow Logs in Amazon S3



- VPC Flow Logs gathers flow log records, consolidates them into log files, and then publishes these files every five minutes to an Amazon S3 bucket.
- Each of these log files captures the IP traffic data from the previous five minutes.

VPC Flow Logs in CloudWatch



- When a user publishes to CloudWatch Logs, the flow log data goes into a specific log group.
- Within this group, each network interface gets its unique log stream.
- Flow log records are then stored within these log streams.

VPC Flow Logs

Before a request in a VPC reaches an AWS-hosted EC2 instance endpoint, it must follow a sequence of steps.

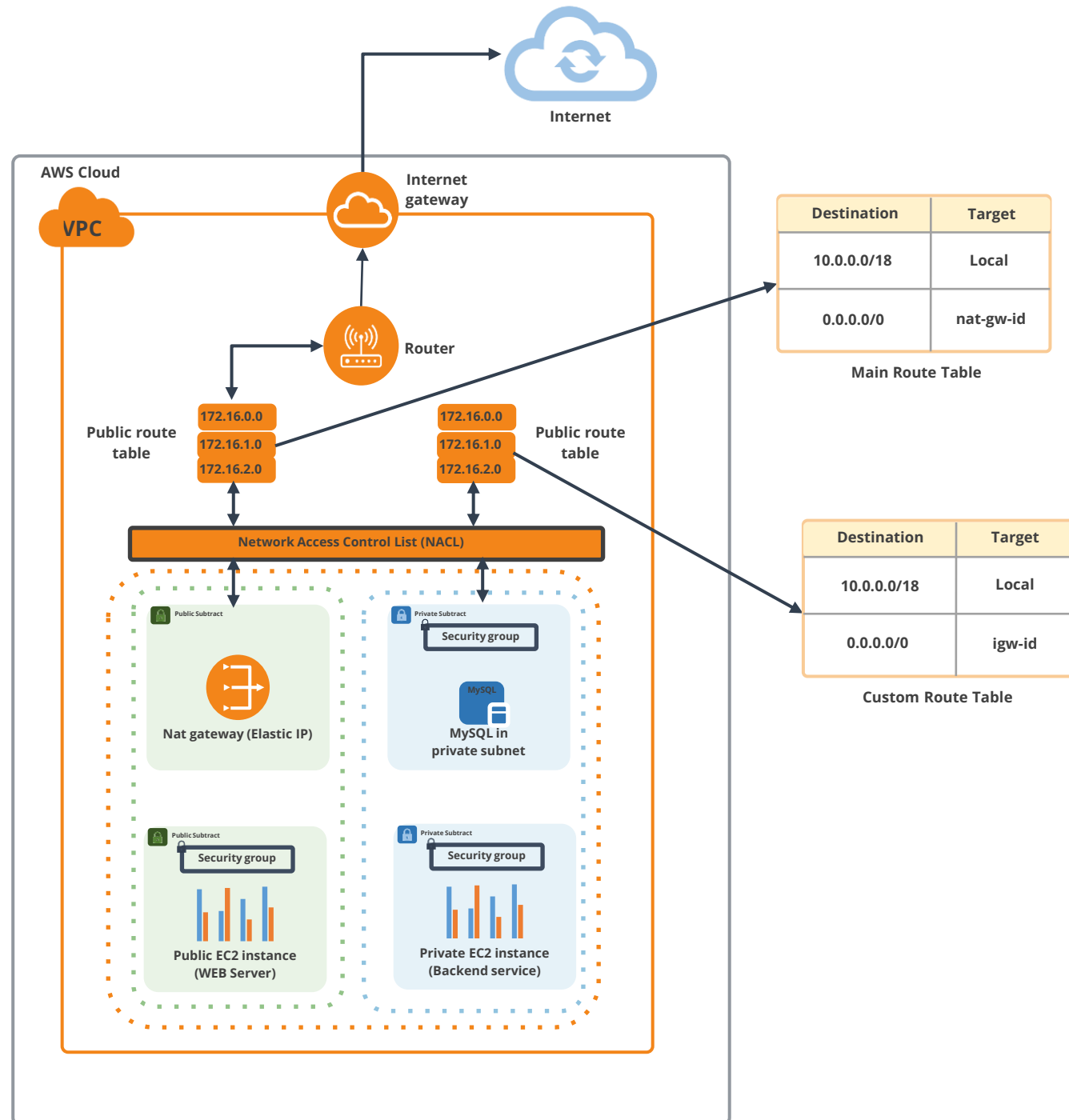


The VPC flow is typically explained using two components:

- The external request
- The private instance



VPC Flow Logs

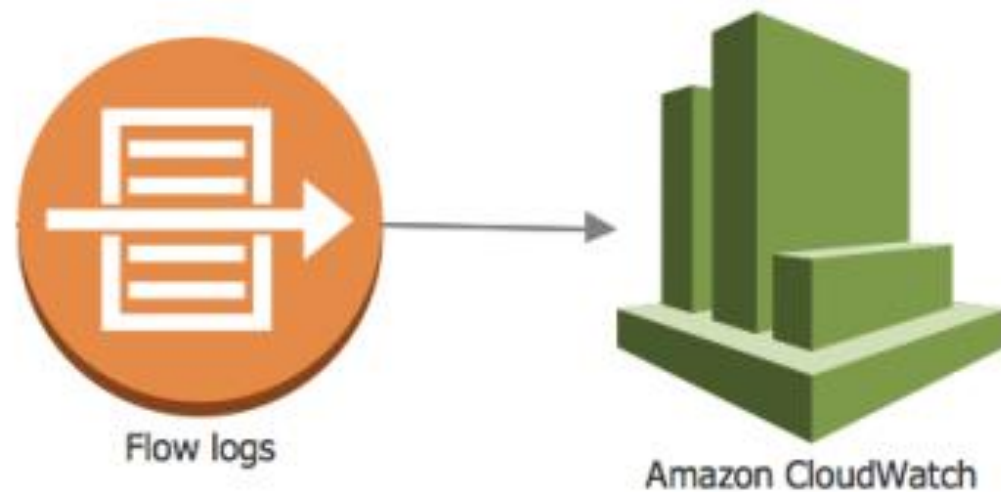


The image illustrates the workflow of components within a private instance. The following components are involved in processing an external request:

- Internet gateway
- Router or Route table
- Network ACL (associated with the subnet)
- Security group (linked to the instance)
- Public instance

IAM Roles for Publishing Flow Logs to CloudWatch Logs

To publish flow logs to a specific log group in CloudWatch Logs, users must ensure:



- The IAM role associated with the flow log has the required permissions.
- This IAM role resides within the AWS account.

IAM Roles for Publishing Flow Logs to CloudWatch Logs

For the user's IAM role, the attached policy must grant the following permissions:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "logs:CreateLogGroup",
        "logs:CreateLogStream",
        "logs:PutLogEvents",
        "logs:DescribeLogGroups",
        "logs:DescribeLogStreams"
      ],
      "Resource": "*"
    }
  ]
}
```



IAM Roles for Publishing Flow Logs to CloudWatch Logs

Ensure a trust relationship is set for the VPC Flow Logs service to assume the relevant IAM role.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "vpc-flow-logs.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```



IAM Roles for Publishing Flow Logs to CloudWatch Logs

Users are advised to use the **aws:SourceAccount** and **aws:SourceArn** condition keys to guard against the confused deputy problem.



Enhancing Trust Policy for Flow Logs

To improve security, users can append this condition block to their existing trust policy:

```
"Condition": {  
  "StringEquals": {  
    "aws:SourceAccount": "account_id"  
  },  
  "ArnLike": {  
    "aws:SourceArn": "arn:aws:ec2:region:account_id:vpc-flow-log/flow-log-id"  
  }  
}
```

Using Flow Log ARN in IAM Roles

The source account refers to the owner of the flow log, while the source ARN identifies the flow log's ARN.



If a user is unsure of the flow log ID, they can substitute that segment of the ARN with a wildcard. Later, they can update the policy after creating the flow log.

Setting Up an IAM Role for Flow Logs

Users can either update an existing role or follow a recommended procedure to create a new role specifically for flow logs.



Creating an IAM Role for Flow Logs

To create an IAM role for VPC flow logging, follow these steps:



- Go to the IAM console at: <https://console.aws.amazon.com/iam>
- In the navigation pane, select **Roles**
- Click **Create role**
- On the **Select role type** page, select **AWS service**

Creating an IAM Role for Flow Logs

To create an IAM role for VPC flow logging, follow these steps:



- In the **Choose a service that will use this role** list, select **VPC Flow Logs**
- Click **Next: Permissions**
- On the **Permissions** page, select the permission **AmazonVPCFullAccess**
- Click **Next: Tags**

Creating an IAM Role for Flow Logs

To create an IAM role for VPC flow logging, follow these steps:



- (Optional) To add tags to your role, select **Add tags** and enter the tag key and value
- Click **Next: Review**
- On the **Review** page, enter a name for the role and a description
- Click **Create role**

Permissions for IAM Users to Pass a Role

To allow users to pass a role in IAM for VPC flow log capture, they need specific permissions.

Below is the IAM policy for such permission:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:PassRole"
      ],
      "Resource": [
        "arn:aws:iam::account-id:role/flow-log-role-name"
      ]
    }
  ]
}
```



Creating a Flow Log in the VPC



Duration:10 min

Problem Statement:

You have been assigned a task to create a VPC flow log within the AWS framework and configure the logging settings, set filter criteria, and specify an Amazon S3 bucket.

Outcome:

You will be able to create a VPC flow log within the AWS framework, configure the logging settings, set filter criteria, and specify an Amazon S3 bucket for storing the logs.

Note: Refer to the demo document for detailed steps:
04 Creating a VPC Flow Log

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a flow log in the default VPC



Quick Check



You want to capture detailed information about the IP traffic going to and from your VPC and store this data in Amazon S3 for analysis. Which feature should you use?

- A. Security Group Logs
- B. Network ACLs
- C. VPC Flow Logs
- D. AWS CloudTrail

NAT and Internet Gateways

Internet Gateways

It is a VPC component that scales horizontally, is redundant, and offers high availability. This gateway enables communication between the VPC and the Internet.

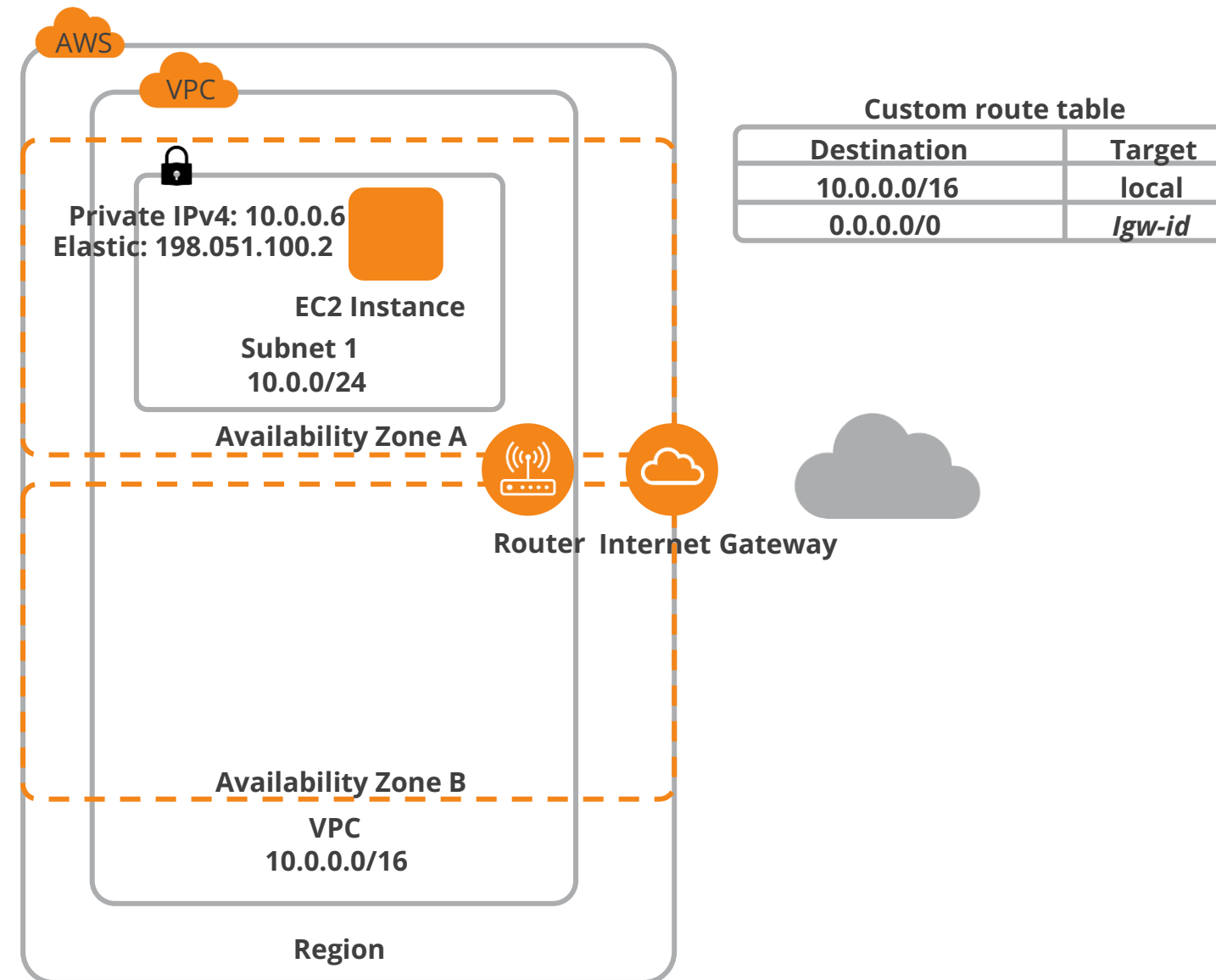
It serves two main purposes:

- It acts as a target in the VPC route tables for traffic that is routable on the internet.
- It handles network address translation (NAT) for instances with public IPv4 addresses.



Internet Gateways

The following diagram shows the communication between the Elastic IP address and the Internet:



Egress-only Internet Gateways

It is an internet gateway that users cannot associate with a security group.

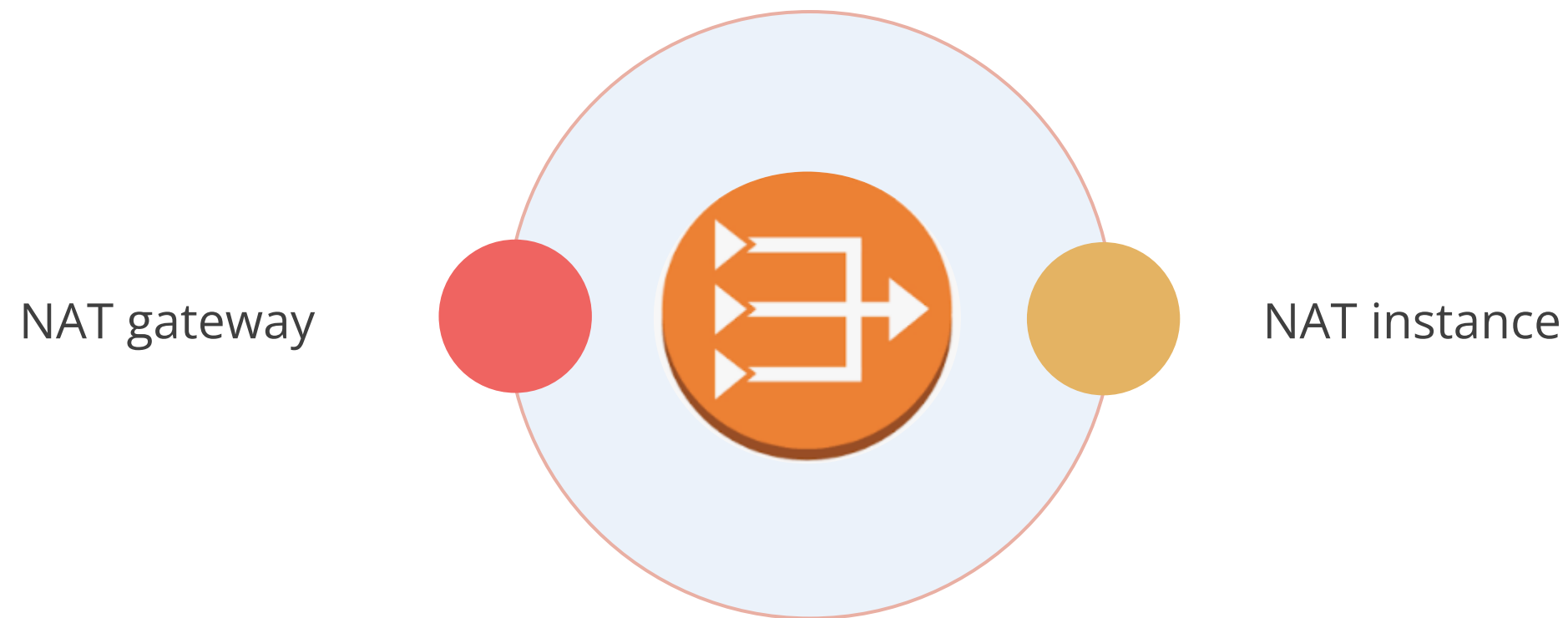


- Users can employ security groups for their instances within a private subnet to manage traffic in and out of those instances.
- To control the traffic entering and leaving the subnet, users can apply a network ACL.

Network Address Translation (NAT)

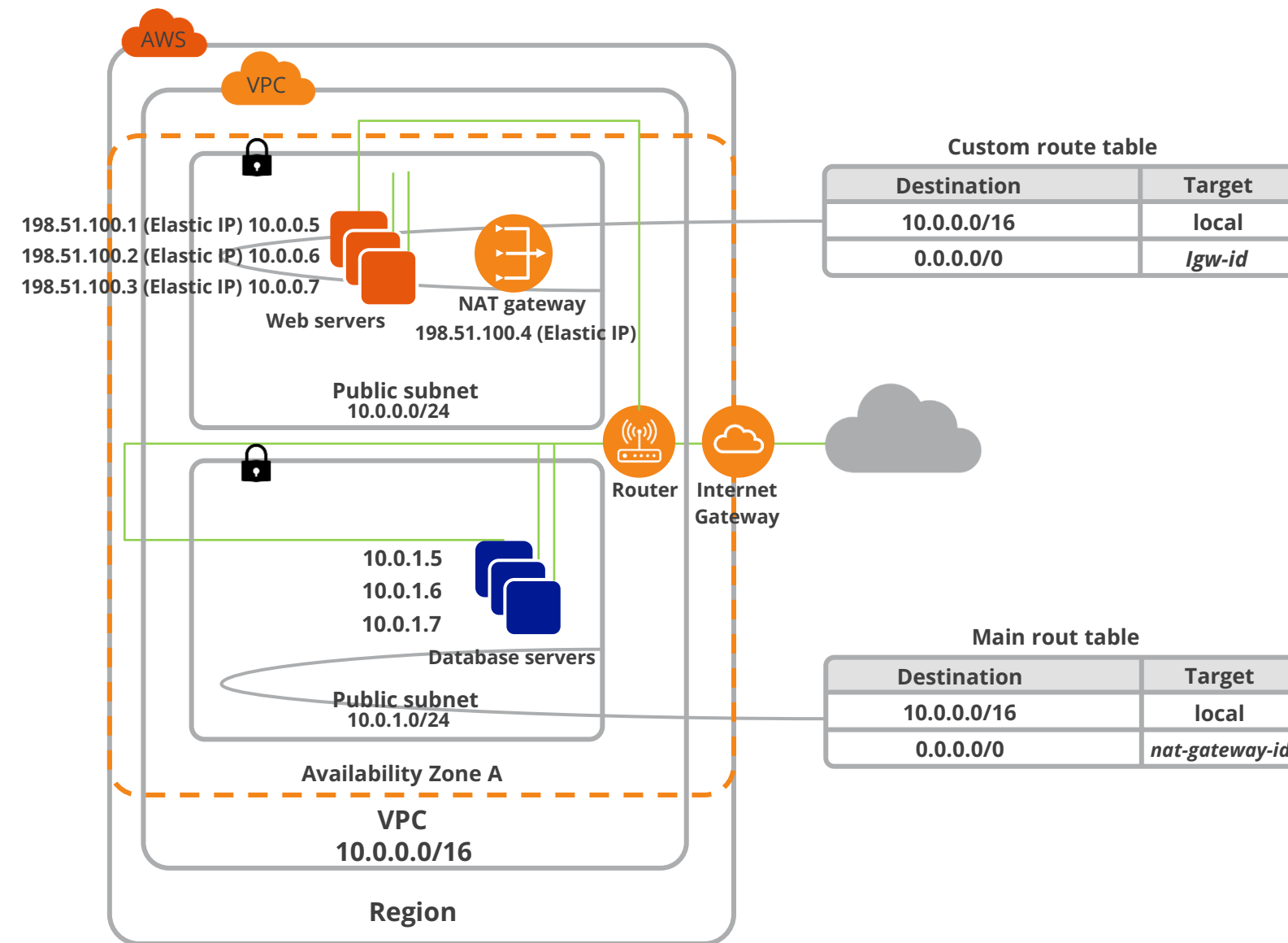
It allows instances in a private subnet to connect to the Internet or other AWS services. However, it prevents the Internet from initiating connections in these instances.

There are two main types of NAT devices:



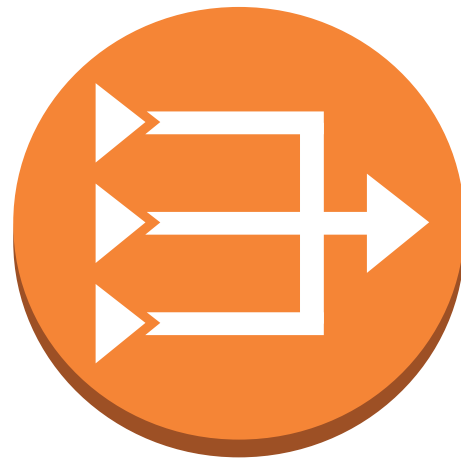
NAT Gateway

The following diagram illustrates the architecture of a VPC with a NAT gateway:



NAT Gateway: Characteristics

The NAT Gateway possesses the following key characteristics:



- Bandwidth and scalability
- Protocol support
- Network ACL control
- Concurrent connections

NAT Gateway: Limitations

While the NAT Gateway is a powerful tool, it does come with certain limitations:



- Single Elastic IP
- No security group association
- ClassicLink inaccessibility
- Routing constraints



VPC Endpoints: Overview

It enables users to connect the VPC to their AWS services without requiring an internet gateway, NAT device, VPN, or an AWS Direct Connect connection.

There are two types of VPC endpoints:



Amazon S3 supports both gateway endpoints and interface endpoints.

Interface VPC Endpoint

It provides users with the capability to establish connections to services powered by AWS PrivateLink, which encompasses a variety of AWS services.

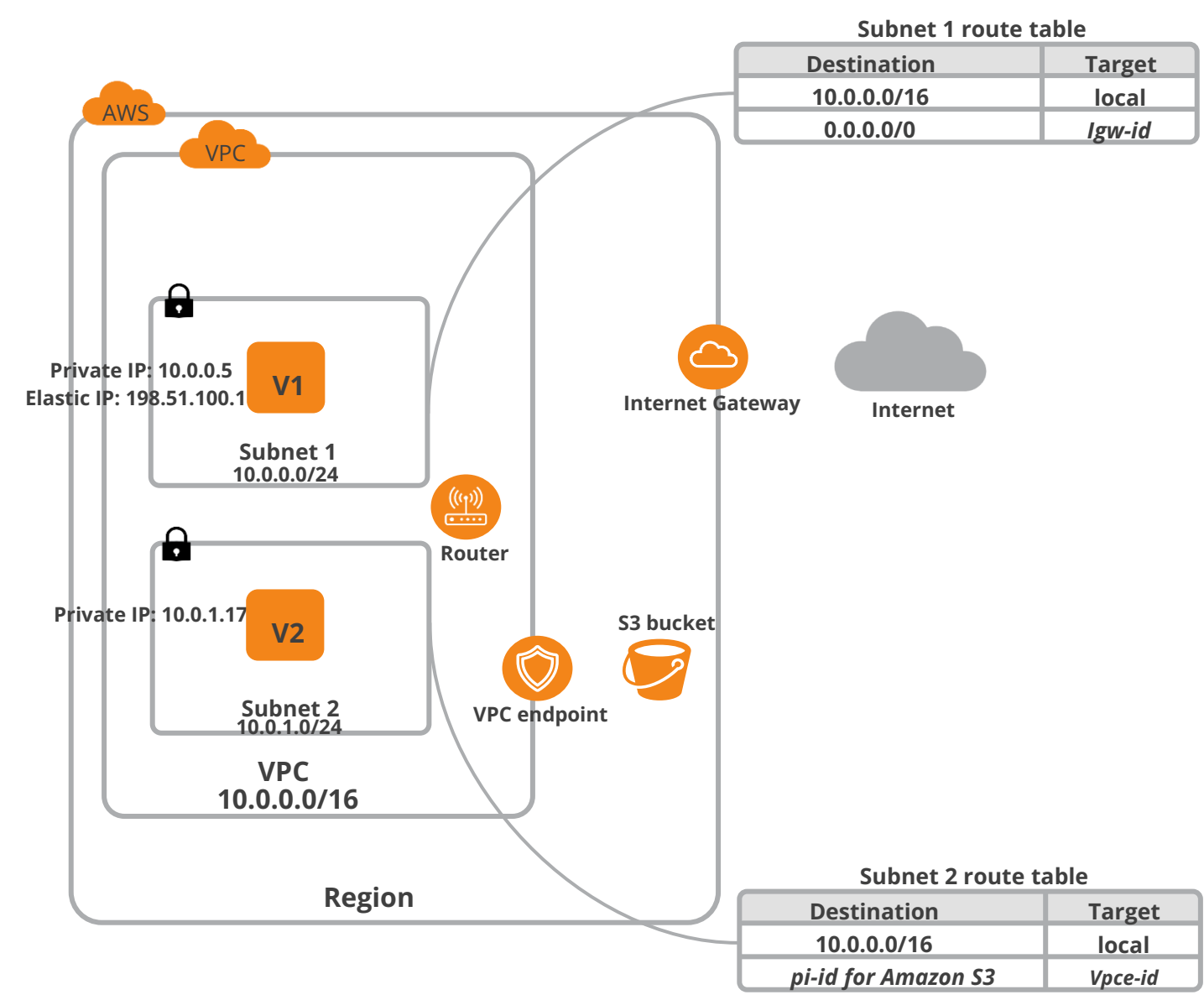


For each specified subnet within a user's VPC, the creation of an interface VPC endpoint results in the generation of an endpoint network interface in that subnet.



Accessing an AWS Service Using an Interface VPC Endpoint

An endpoint network interface is a requester-managed network interface.



- Users can view the VPC endpoint interface in their AWS account but cannot manage it themselves.
- Users are billed according to their hourly usage and data processing charges.

Interface Endpoint: Features

Interface VPC endpoints offer the following key features:

01

TCP traffic support: Interface VPC endpoints facilitate communication over the TCP protocol.

03

Publicly resolvable DNS names: The DNS names assigned to interface VPC endpoints can be resolved publicly.

02

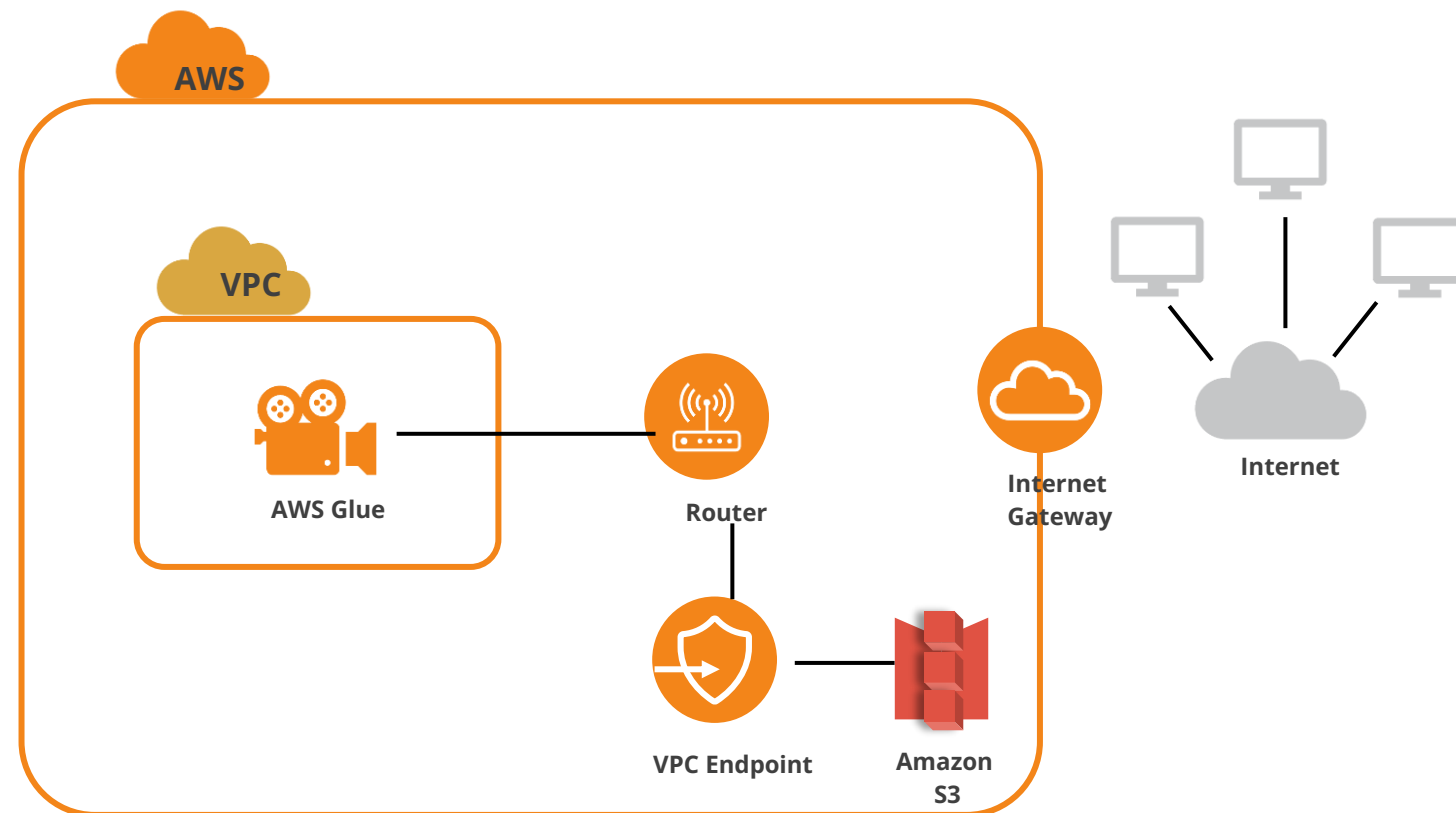
Selective response: Endpoints respond exclusively to traffic initiated by resources within the user's VPC.

04

Scalable bandwidth: Each interface endpoint boasts substantial bandwidth capabilities.

Gateway Endpoints

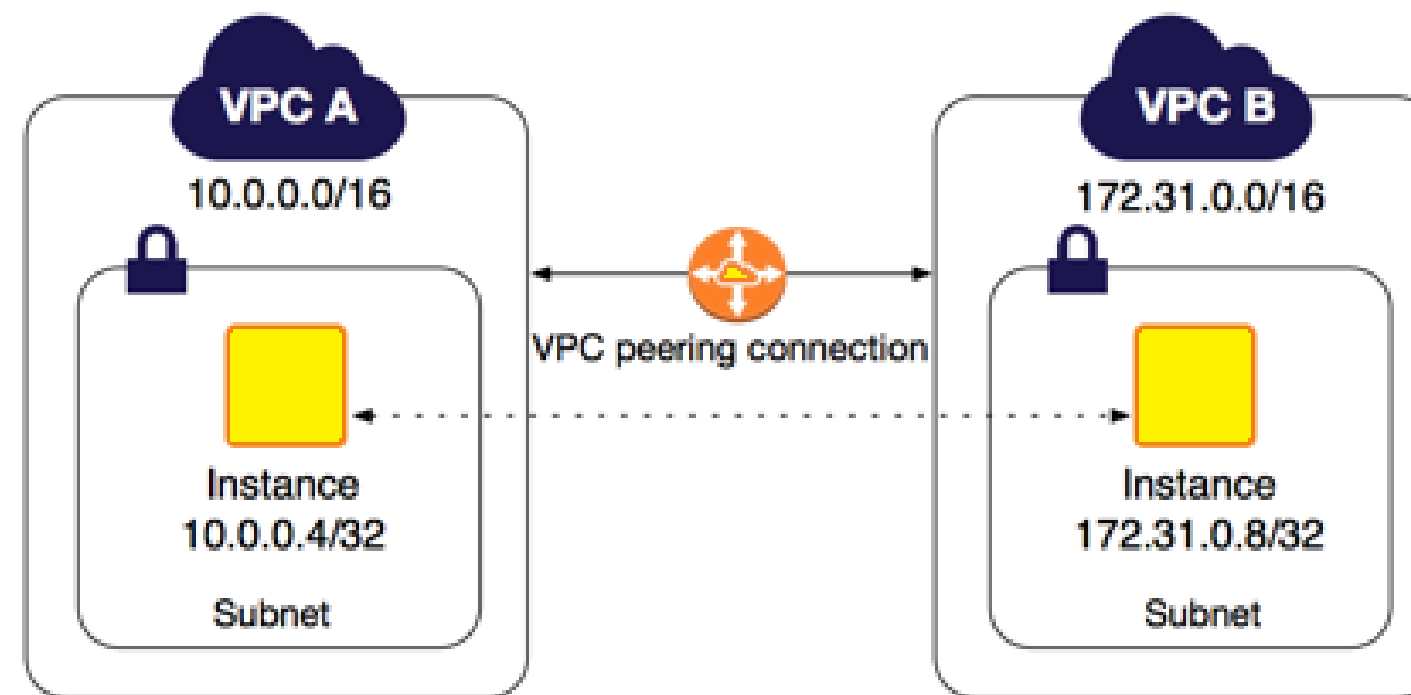
It offers dependable connectivity to Amazon S3 and DynamoDB, eliminating the need for an Internet gateway or NAT device within the user's VPC.



- However, it's important to note that gateway endpoints do not support AWS PrivateLink integration.
- The usage of gateway endpoints does not incur any additional charges.

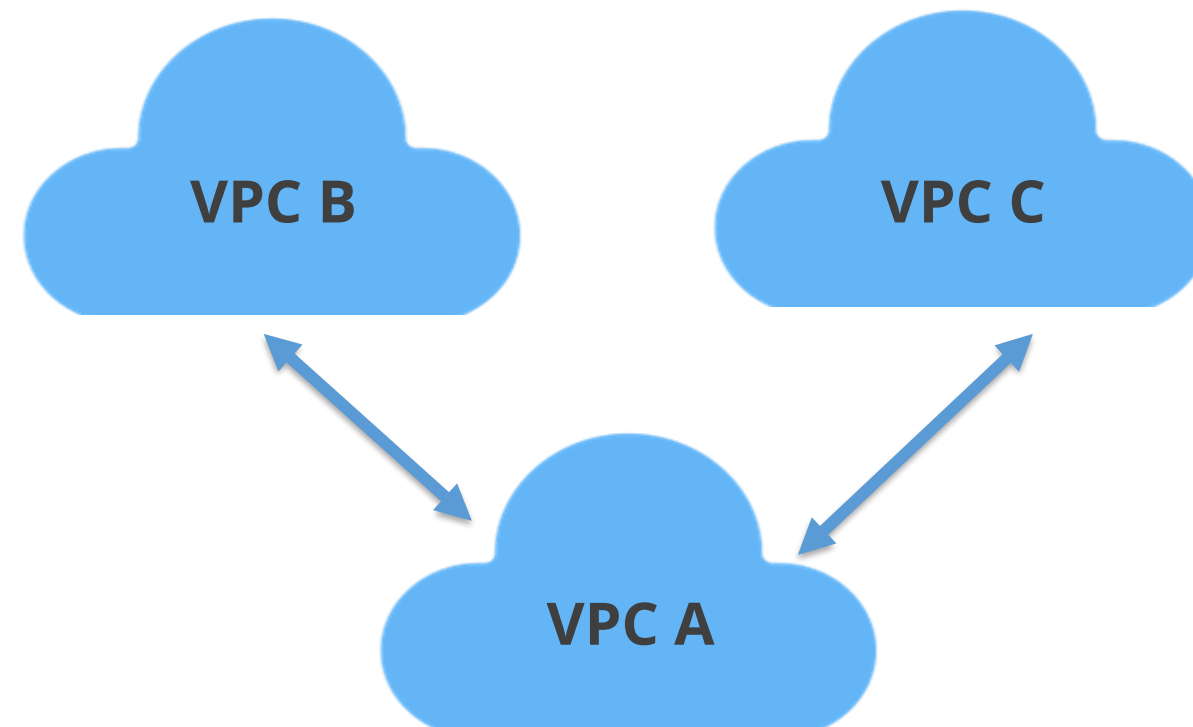
VPC Peering Connection

It establishes a direct and secure link between two VPCs. Users can create multiple VPC peering connections for each VPC they own.



Multiple VPC Peering Connections

It establishes a direct and secure link between two VPCs. Users have the ability to create multiple VPC peering connections for each VPC they own.



However, it's important to note that transitive peering relationships are not supported.



AWS Transit Gateway

It serves as a central hub, seamlessly connecting users with both Amazon Virtual Private Clouds (VPCs) and on-premises networks.

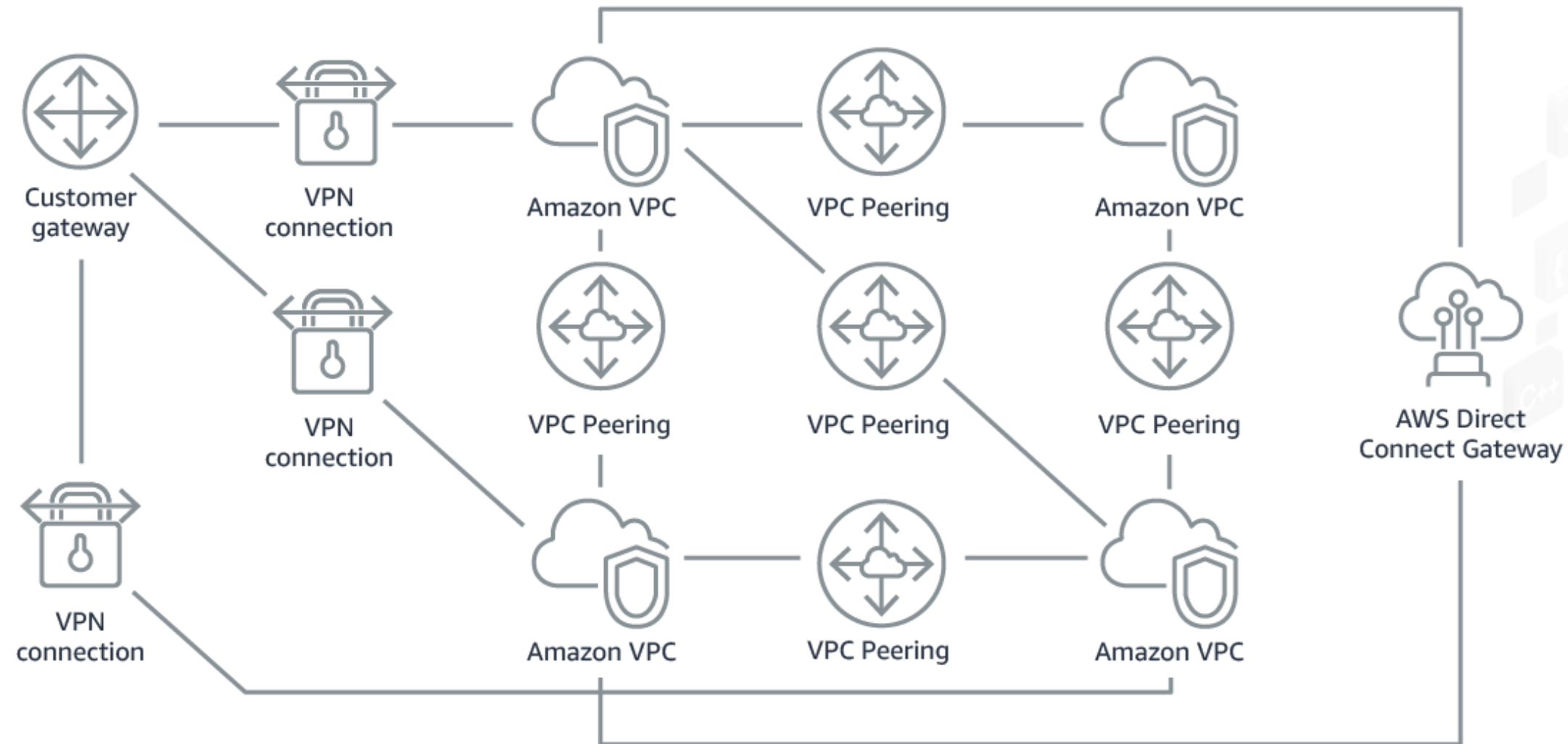


Simplifying network configurations and streamlining connectivity eradicate the need for complex peering relationships.



AWS Transit Gateway

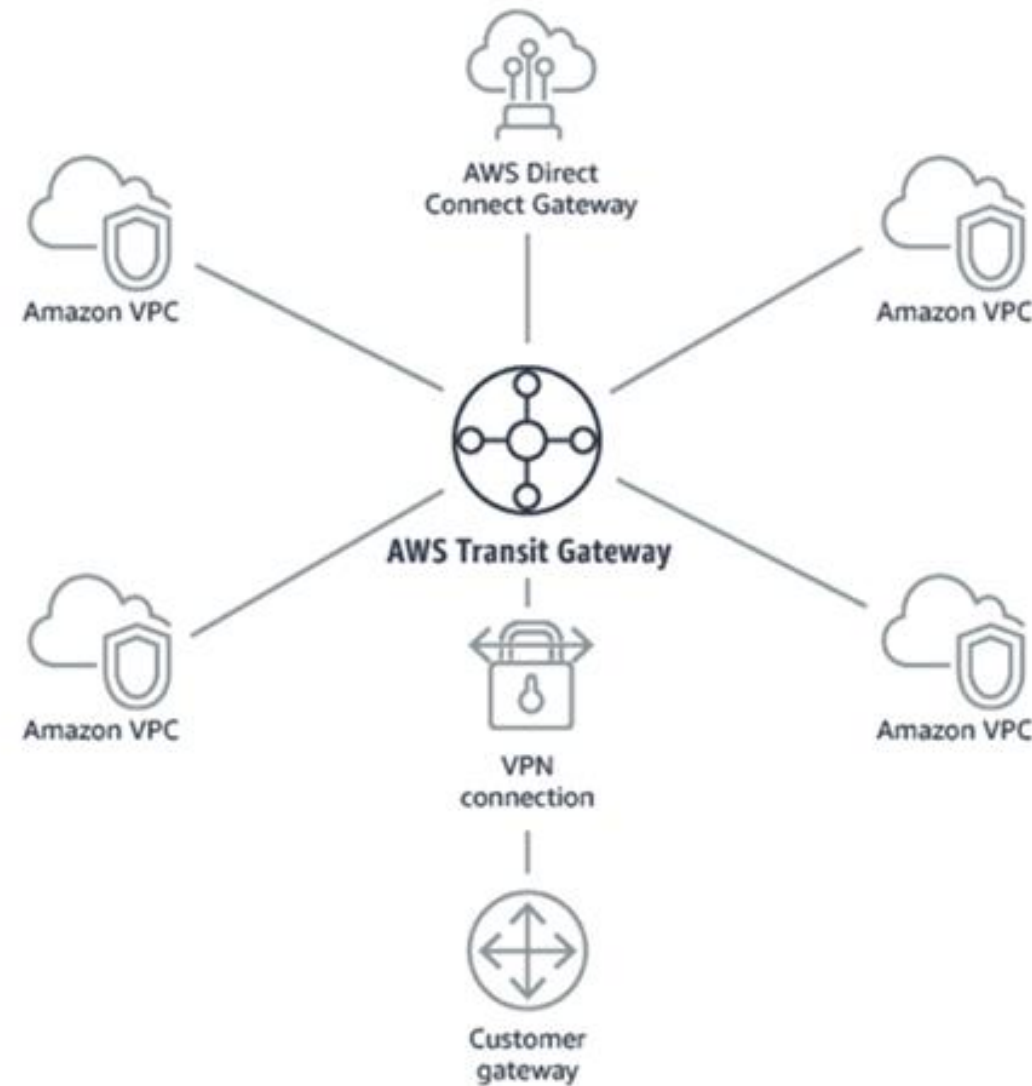
Below is a visual representation of an AWS transit gateway in an idle state:



Source: <https://aws.amazon.com/blogs/aws/new-use-an-aws-transit-gateway-to-simplify-your-network-architecture/>

AWS Transit Gateway

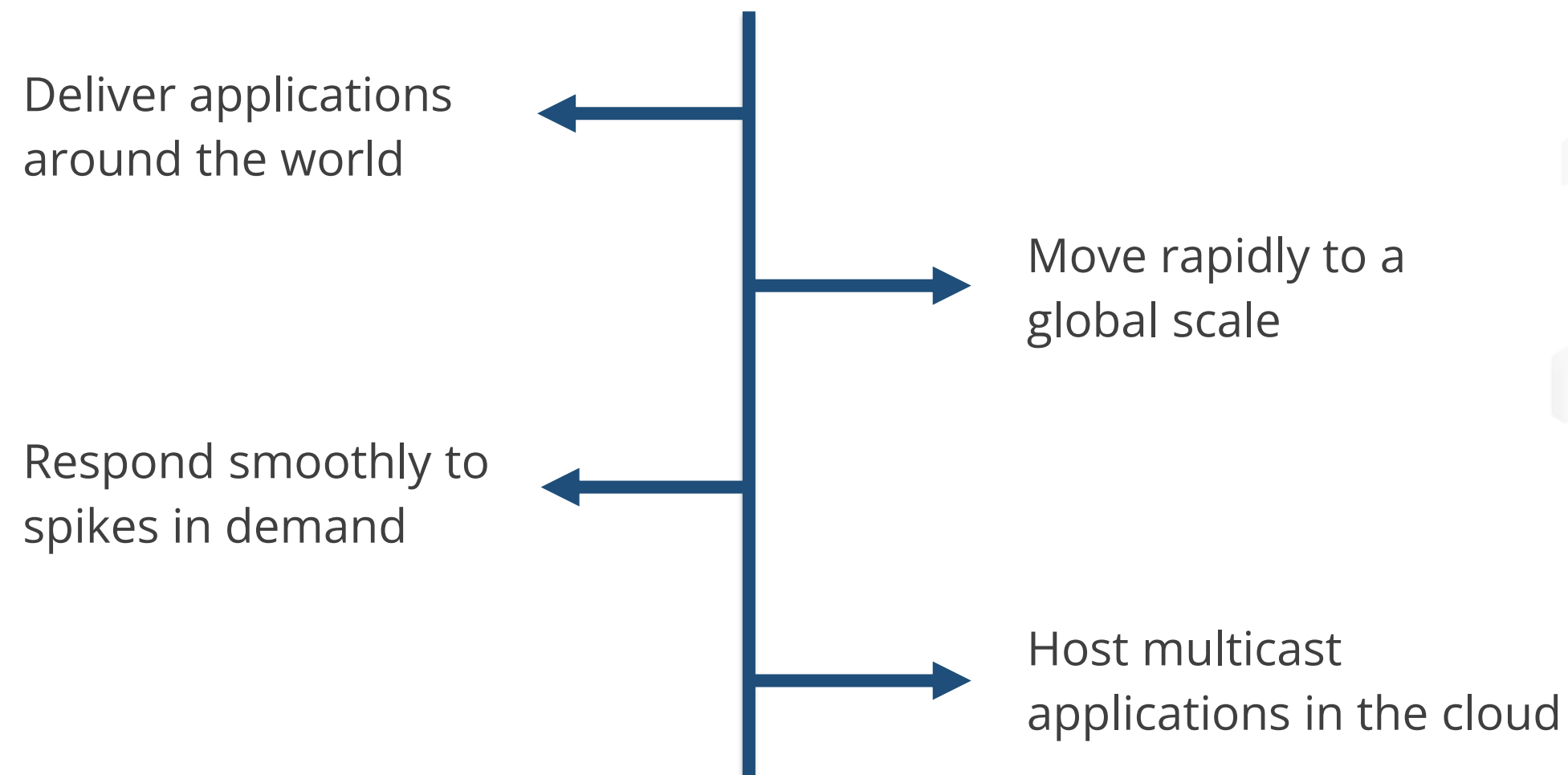
Below is a visual representation of an operational AWS transit gateway:



Source: <https://aws.amazon.com/blogs/aws/new-use-an-aws-transit-gateway-to-simplify-your-network-architecture/>

AWS Transit Gateway: Use Cases

The following are the use cases for the AWS Transit Gateway:



Configuring Subnets, Route Table, and NAT



Duration:10 min

Problem Statement:

You have been assigned a task to create a Virtual Private Cloud (VPC) on AWS, set up subnets, configure route tables, and create a Network Address Translation (NAT) gateway for secure and efficient network management.

Outcome:

You will be able to create a Virtual Private Cloud (VPC) on AWS, set up subnets, configure route tables, and create a Network Address Translation (NAT) gateway.

Note: Refer to the demo document for detailed steps:
05_Configuring_Subnets_Route_Table,_and_NAT

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a VPC
2. Create Internet gateways
3. Create Subnets
4. Create a route table and NAT



Creating a VPC Endpoint



Duration:10 min

Problem Statement:

You have been assigned a task to create a VPC Endpoint to establish a secure connection between a Virtual Private Cloud (VPC) and an Amazon S3 bucket.

Outcome:

You will be able to create a VPC Endpoint to establish a secure connection between a Virtual Private Cloud (VPC) and an Amazon S3 bucket.

Note: Refer to the demo document for detailed steps:
06_Creating_a_VPC_Endpoint

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create a VPC Endpoint



Configuring VPC Peering Connections



Duration:10 min

Problem Statement:

You have been assigned a task to connect two VPCs with different regions using a peering connection for secure connection.

Outcome:

You will be able to connect two VPCs in different regions using a peering connection.

Note: Refer to the demo document for detailed steps:
06_Creating_a_VPC_Endpoint

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create two VPCs with different regions
2. Configure Peer connection



Quick Check

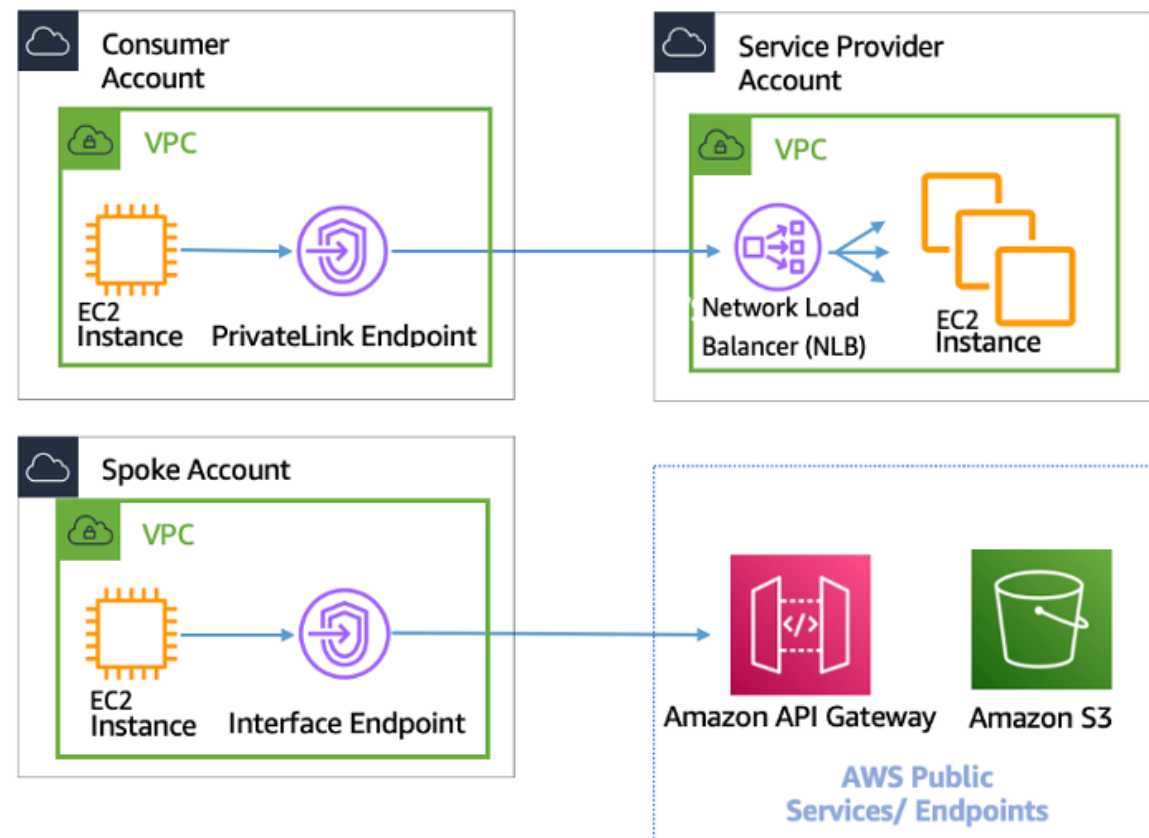


You want to allow instances in a private subnet to access the internet for software updates, while preventing incoming connections from the internet. Which feature should you use?

- A. Internet Gateway
- B. NAT Gateway
- c. Egress-only Internet Gateway
- D. Interface VPC Endpoint

Private Link and Bastion Host

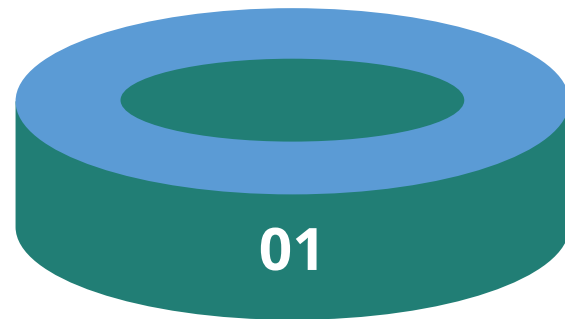
AWS PrivateLink



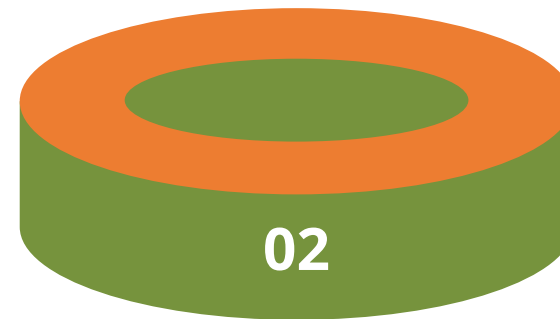
- AWS PrivateLink establishes secure and private connections between VPCs, AWS services, and on-premises networks.
- AWS PrivateLink simplifies the process of connecting services across diverse accounts and VPCs, offering a streamlined approach to enhancing the network architecture.

AWS PrivateLink: Benefits

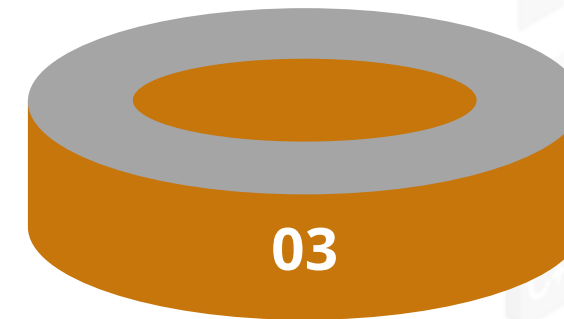
The benefits of private link are as follows:



Enhanced data security



Streamlined network management

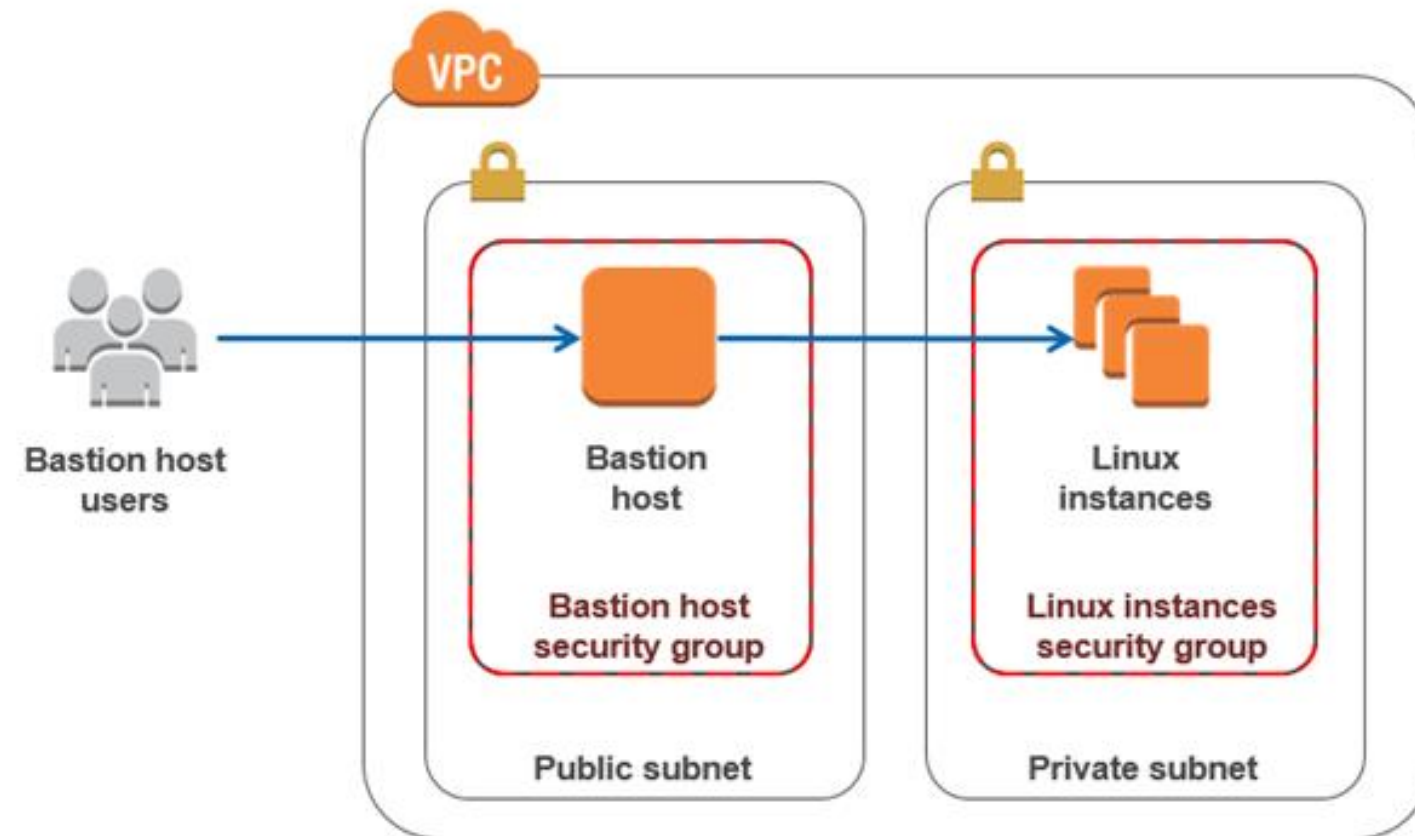


Smooth cloud migration



Bastion Host

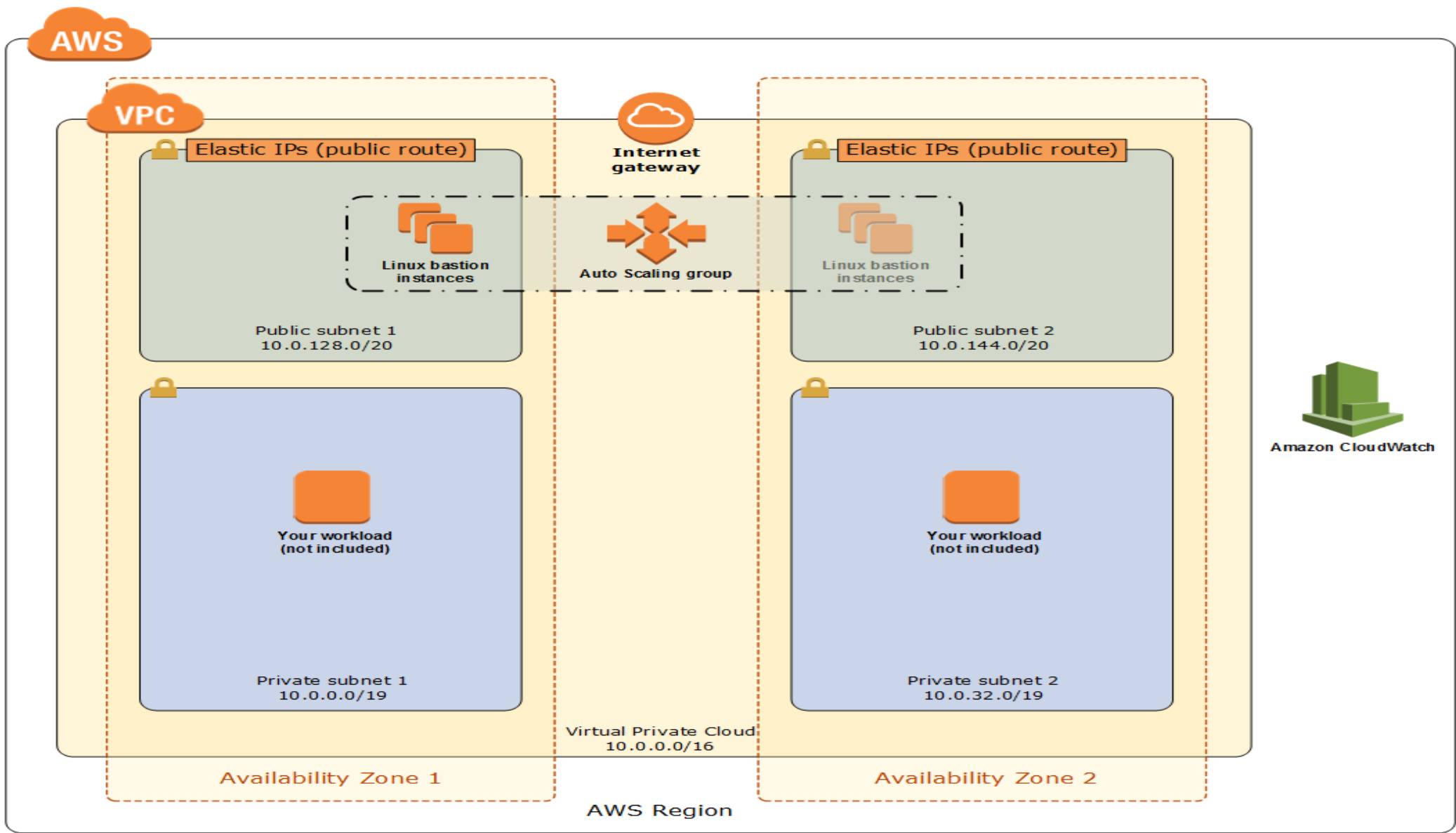
It is a specialized server designed to facilitate secure access between a private and an external network, such as the Internet.



Source: <https://aws.amazon.com/blogs/security/how-to-record-ssh-sessions-established-through-a-bastion-host/>

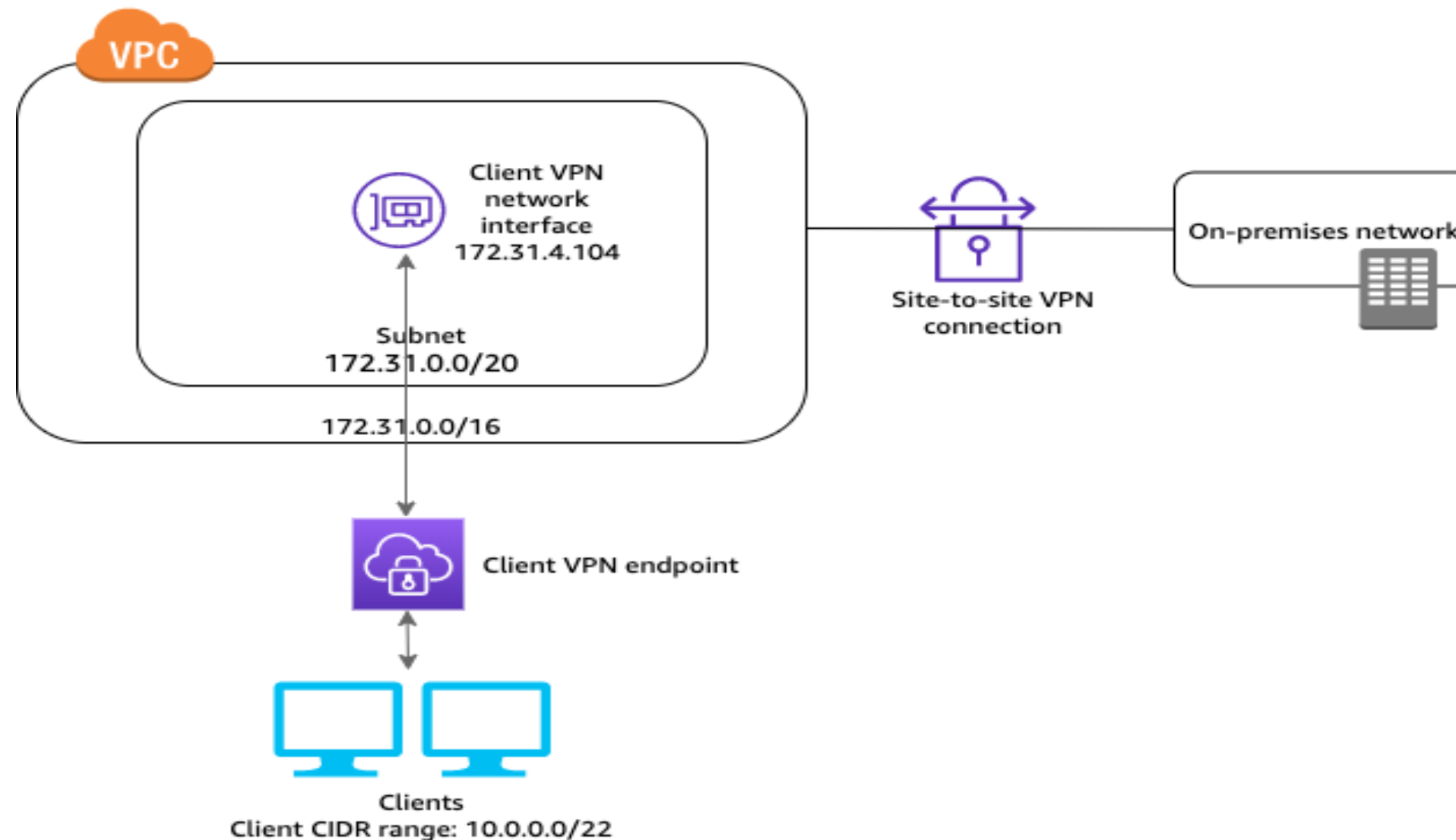
Linux Bastion Host Architecture

The diagram below illustrates the architecture of a Linux bastion host on AWS:



External Clients

The configuration for this scenario includes access to an on-premises network only.



The configuration mentioned above is advisable when users require clients to have access to resources within an on-premises network.

Steps for Accessing an On-Premises Network

The steps are as follows:



Create a VPC with at least one subnet. Identify the subnet in the VPC that a user wants to associate with the client VPN endpoint and note its IPv4 CIDR ranges.



Identify a suitable CIDR range for the client IP addresses that do not overlap with the VPC CIDR.



Review the rules and limitations for client VPN endpoints.

Steps to Implement the Configuration

These are the steps to implement the configuration of the client VPN:

01 Generate server and client certificates and keys

02 Create a Client VPN endpoint

03 Associate a target network

04 Add an authorization rule for the VPC

05 Provide access to the internet

06 Verify security group requirements

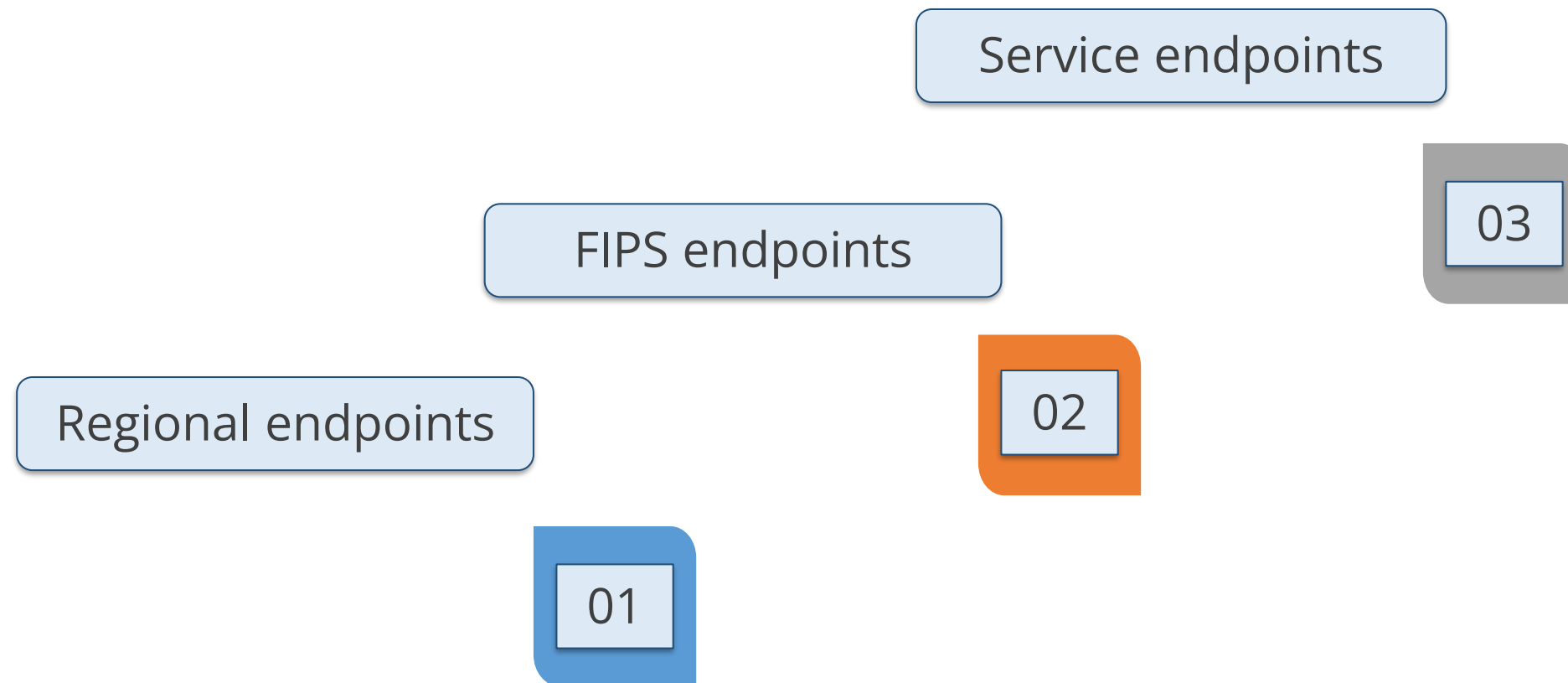
07 Download the Client VPN endpoint configuration file

08 Connect to the Client VPN endpoint

Public Endpoints

To connect programmatically to an AWS service, users use an endpoint. An endpoint is the URL of the entry point for an AWS web service.

The different types of endpoints are as follows:



Systems Manager

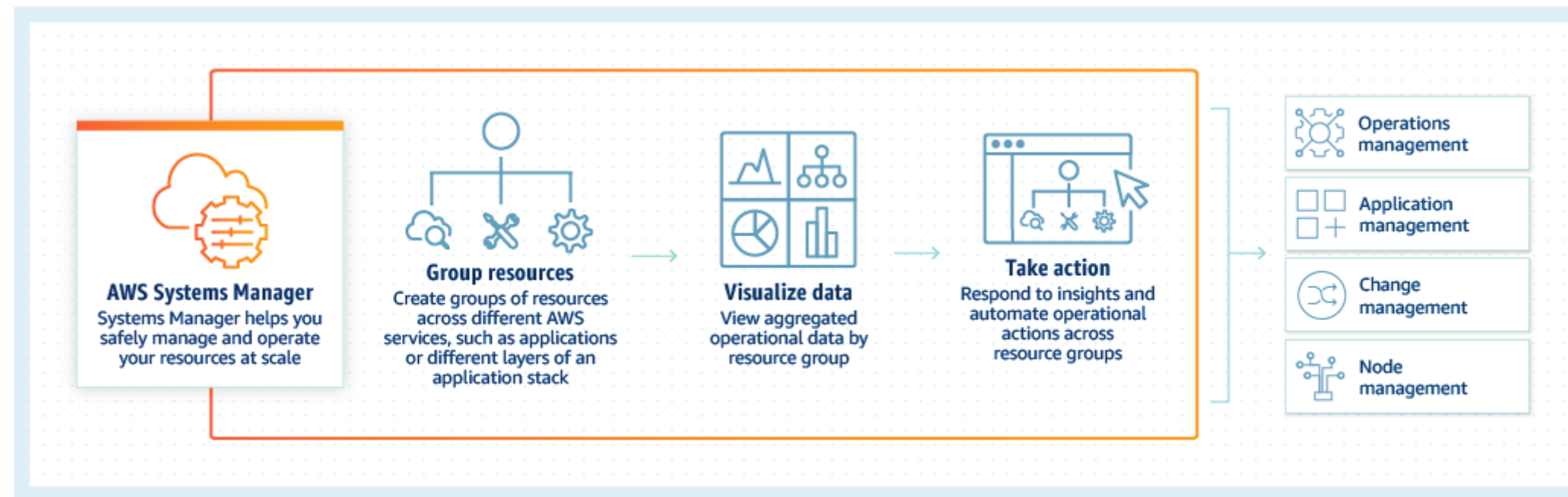
It is a powerful tool that grants users comprehensive visibility and control over their entire AWS infrastructure.



- It integrates with CloudWatch, which allows users to view the dashboard and operational data or report bugs.
- It also includes the run command to automate operational tasks such as security patching.
- It also organizes the inventory by grouping resources by application or environment.

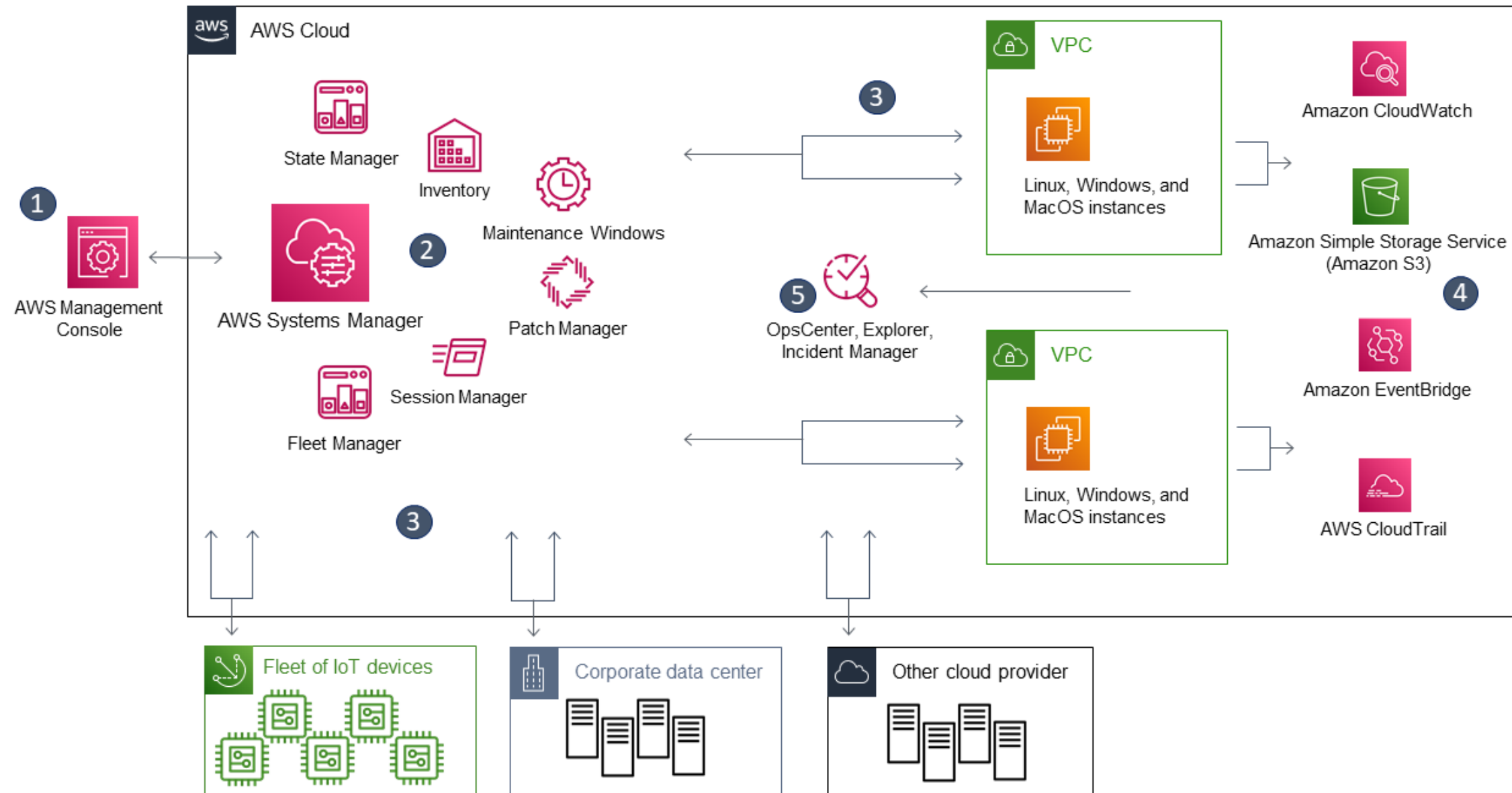
Systems Manager

AWS Systems Manager is a robust and secure end-to-end management solution tailored for hybrid cloud environments.



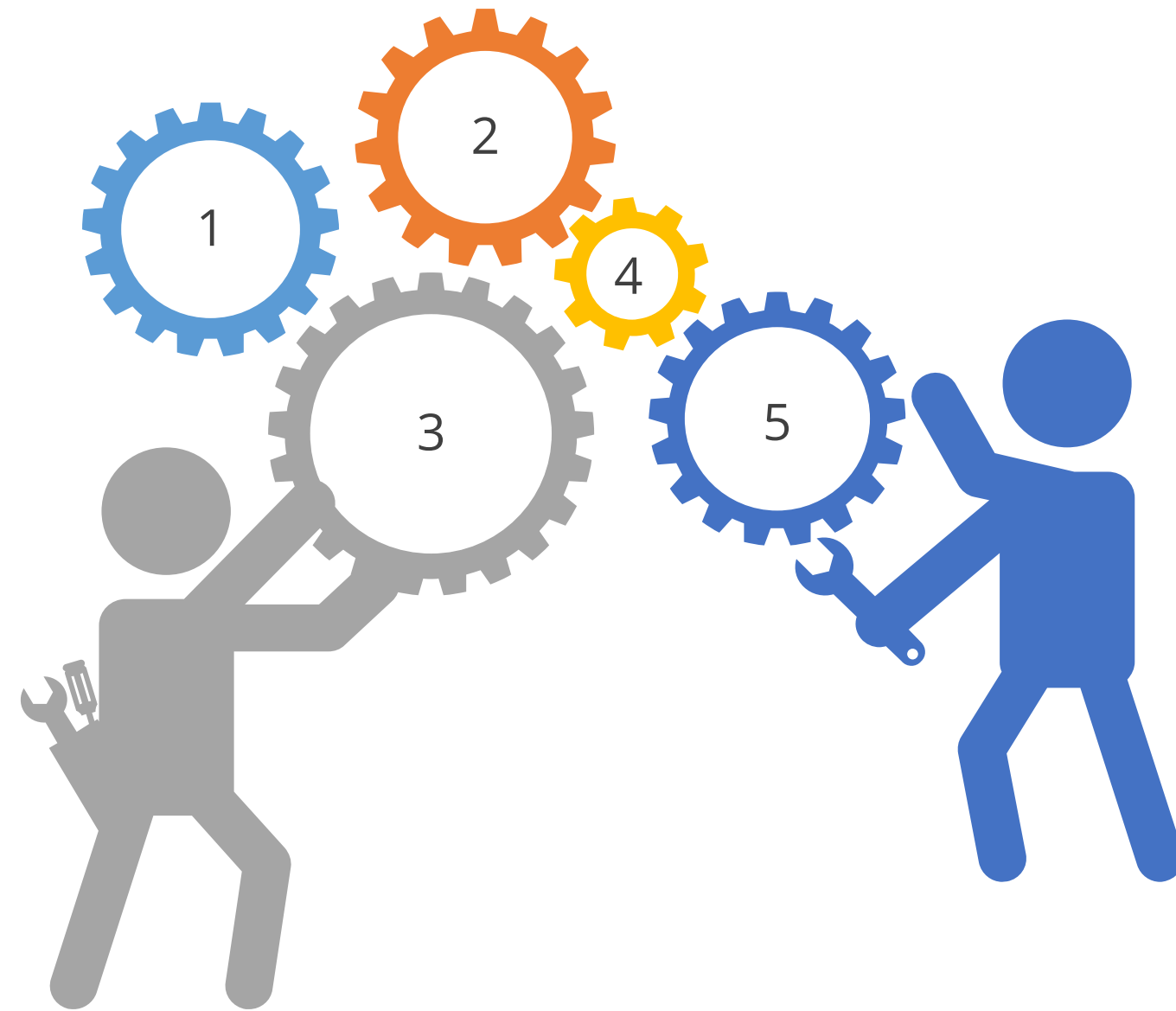
System Manager: Process Flow

Here is an example of the process flow within AWS Systems Manager:



System Manager: Works in Details

The key aspects are as follows:



Accessing system manager

Choosing a system manager

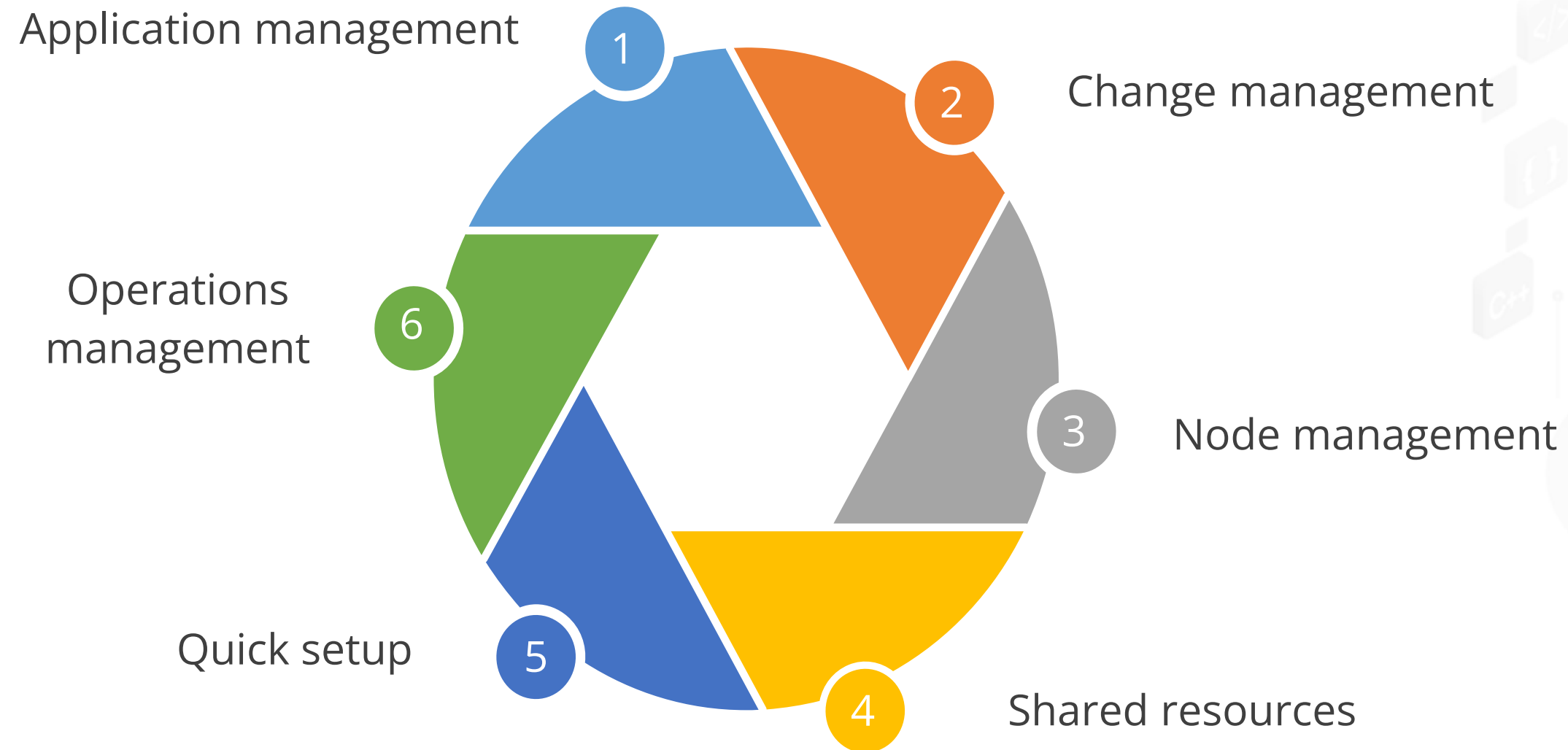
Verification and processing

Reporting

Leveraging system manager's
operations management
capabilities

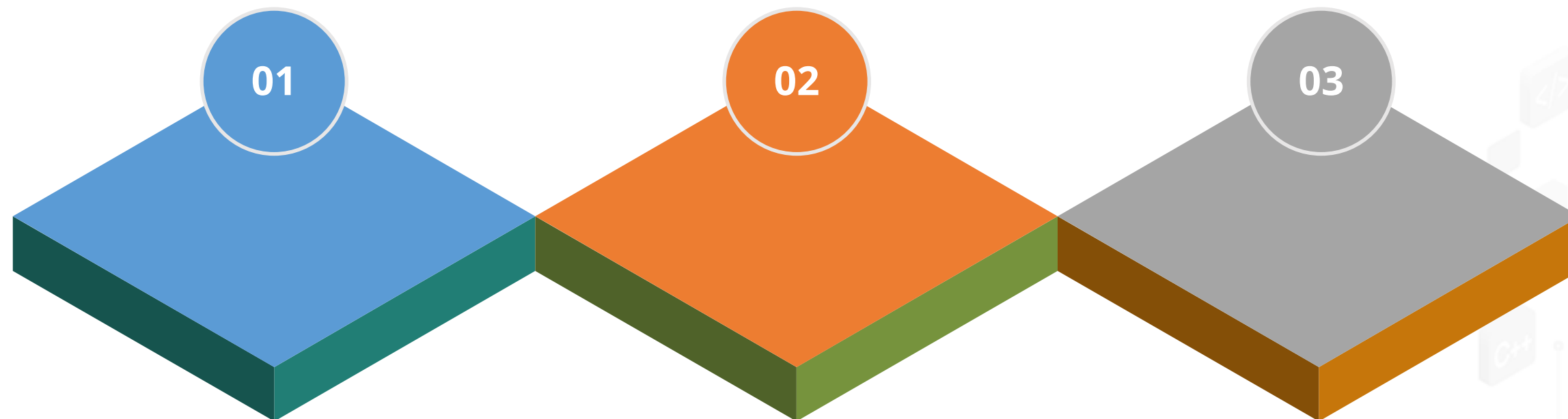
System Manager Capabilities

The capabilities of systems manager are divided into the following categories:



Application Management

The application management capabilities are as follows:



Application management:

This feature allows you to logically group AWS resources that you want to operate as a unit within your application.

App configuration:

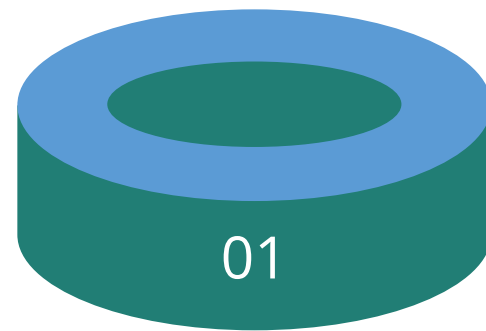
It assists in creating, managing, and deploying application configurations and feature flags.

Parameter store:

This service provides secure, hierarchical storage for configuration data and enables remote management.

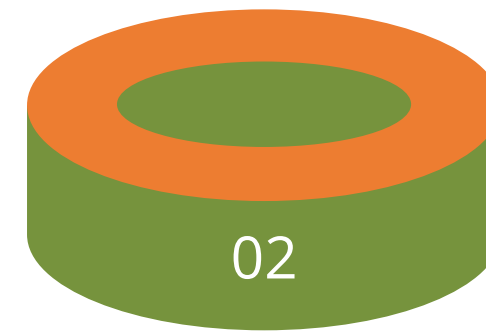
Change Management

The change management capabilities are as follows:



Change manager:

It is an enterprise change management framework for requesting, approving, implementing, and reporting the operational changes to user's application configuration and infrastructure.

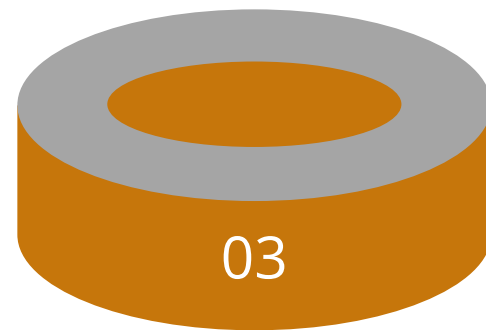


Automation:

You can utilize automation to streamline and automate common maintenance and deployment tasks.

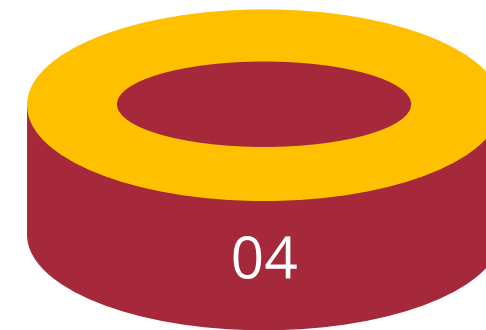
Change Management

The change management capabilities are as follows:



Change calendar:

This feature assists users in configuring date and time ranges during specified actions.

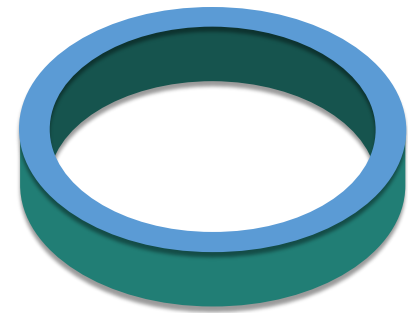


Maintenance windows:

These windows enable users to establish recurring schedules for managed instances to perform administrative tasks.

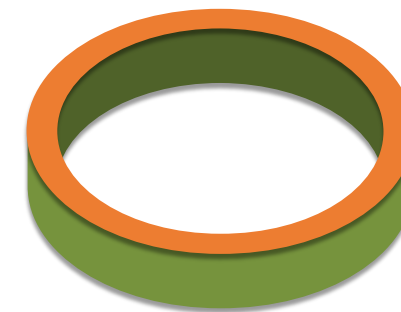
Node Management

The node management capabilities are as follows:



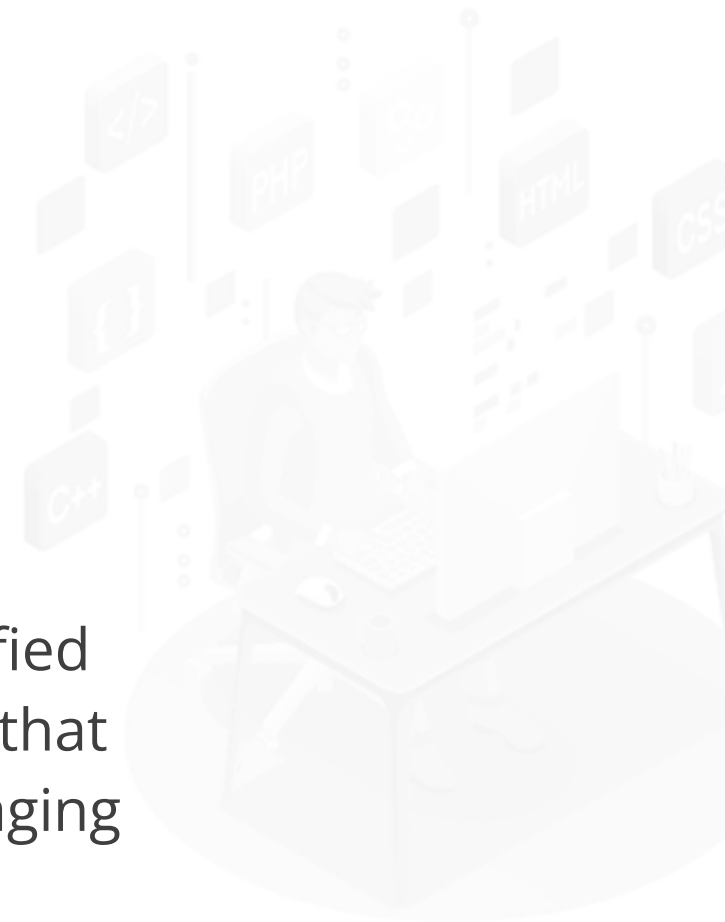
Use compliance:

This capability allows users to scan their fleet of managed nodes for patch compliance and configuration inconsistencies.



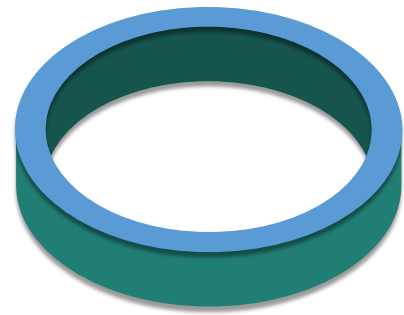
Fleet manager:

Fleet Manager provides a unified user interface (UI) experience that assists users in remotely managing the nodes.



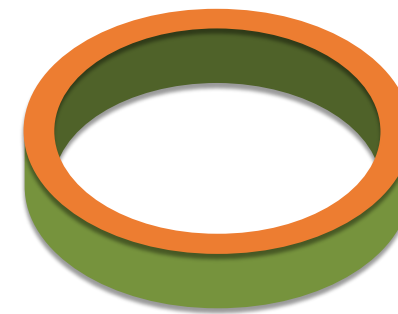
Node Management

The node management capabilities are as follows:



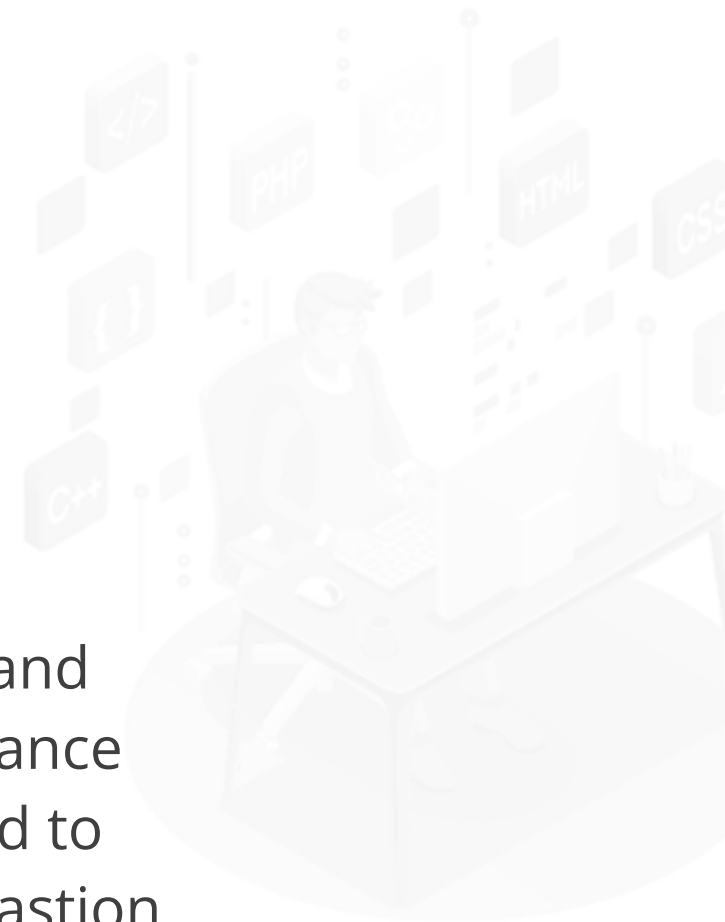
Inventory:

This capability automates the process of collecting software inventory from the managed nodes. Users can use inventory to gather metadata about applications, files, components, patches, and more.



Session manager:

This feature provides secure and auditable edge device and instance management without the need to open inbound ports, maintain bastion hosts, or manage SSH keys.

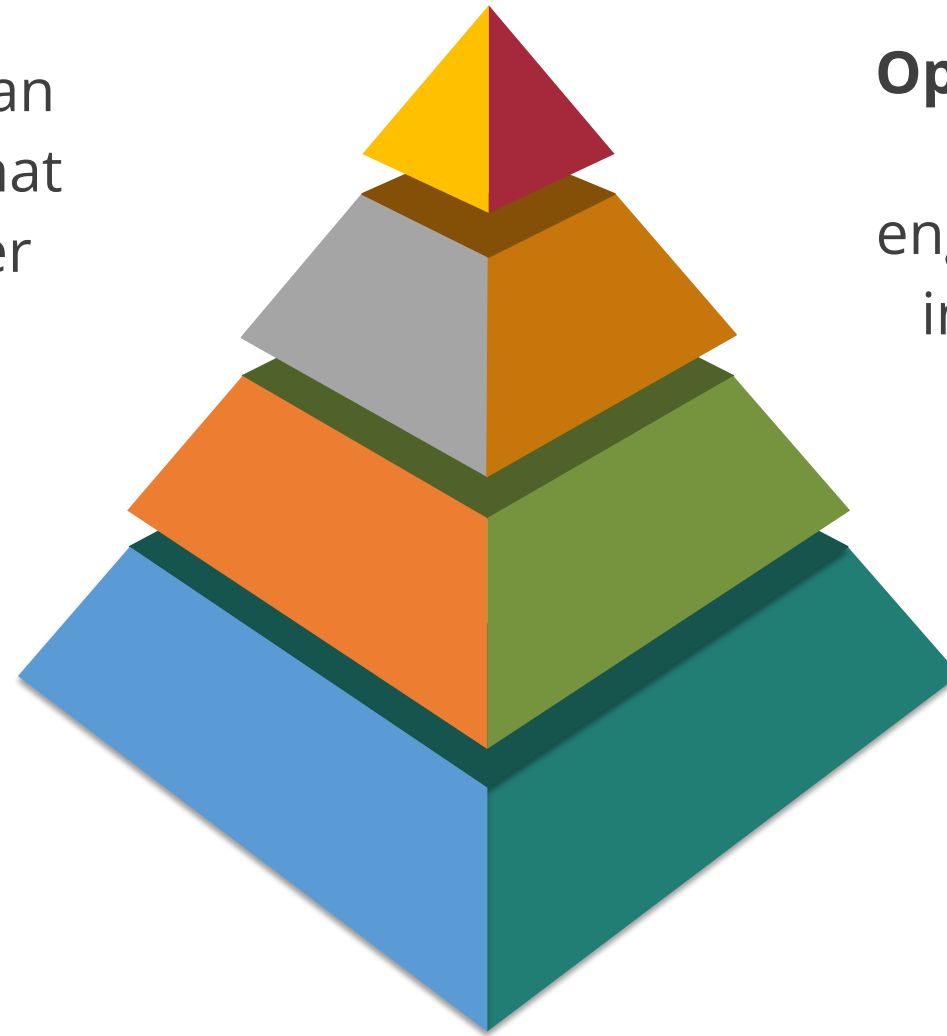


Operations Management

The operations management are as follows:

Incident manager: It serves as an incident management console that helps users mitigate and recover from incidents.

Explorer: This feature displays an aggregated view of operations data (Ops Data) for AWS accounts and across AWS Regions.

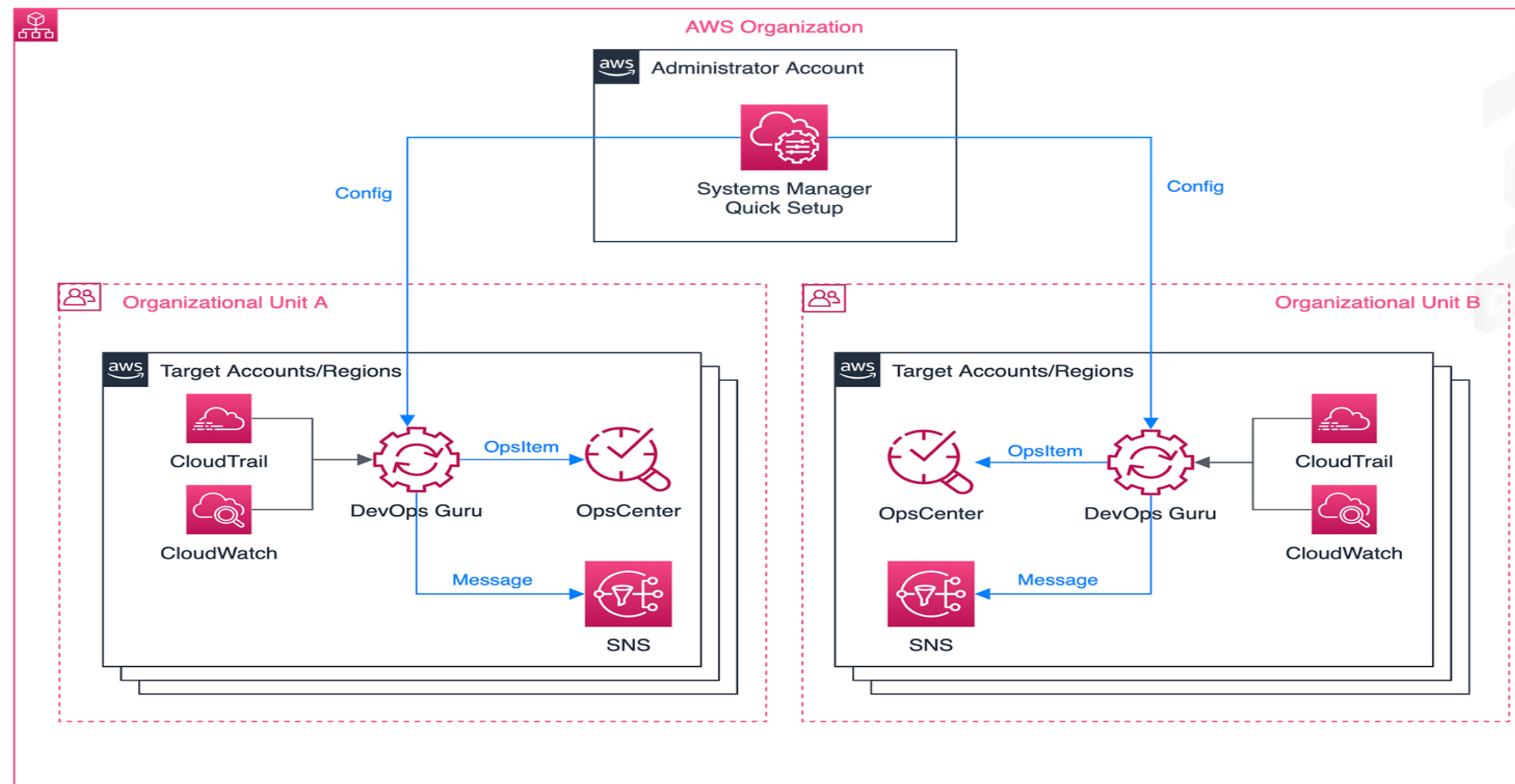


OpsCenter: It provides a central location where operations engineers can view, resolve, and investigate operational items related to AWS resources.

Amazon CloudWatch: Dashboards are customizable pages in the CloudWatch console that users can utilize to monitor resources in a single view.

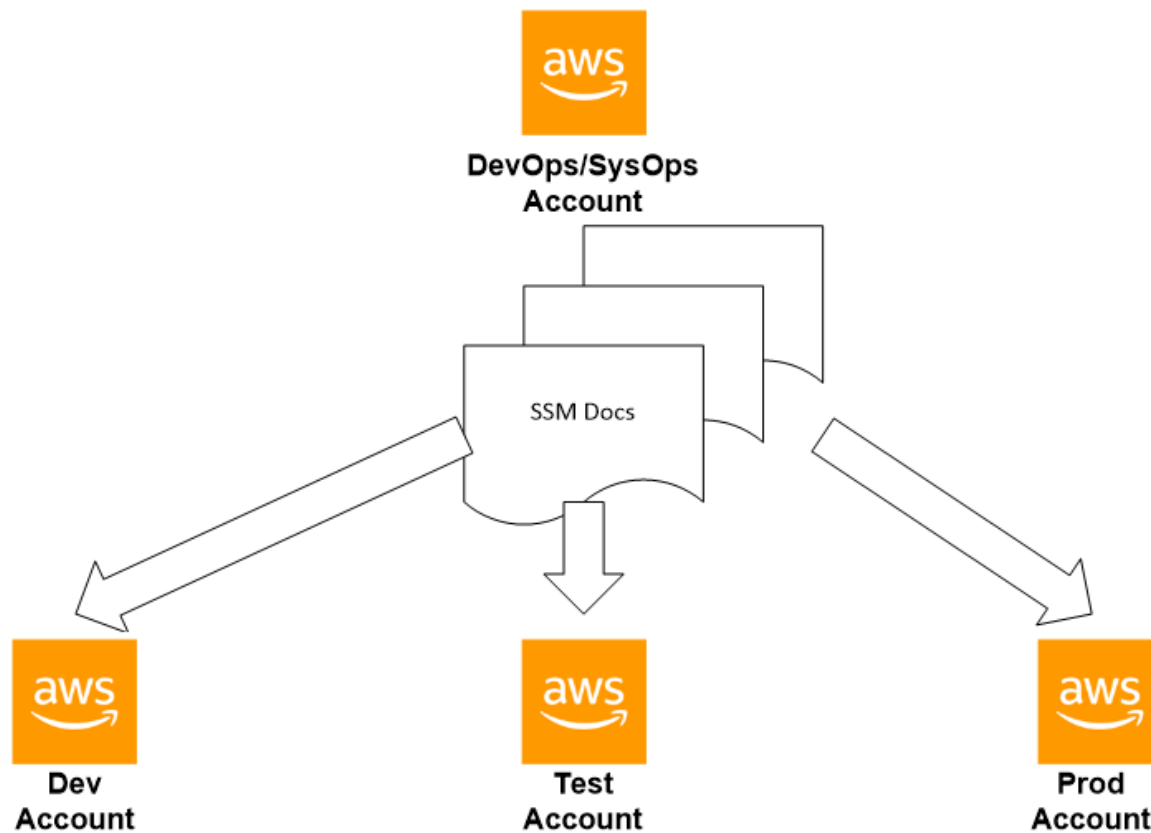
Quick Setup

It simplifies setting up services, including Systems Manager, by automating common or recommended tasks. These tasks include creating the required AWS Identity and Access Management (IAM) instance profile role.



Understanding SSM Documents

A systems manager document (SSM document) defines the actions that systems manager performs.



SSM document types include:

- **Command documents:** Used by the State Manager for Run Commands
- **Automation runbooks:** Used by Systems Manager Automation

Quick Check



You need to securely connect your on-premises network to a service running in AWS without exposing the service to the public internet. Which feature should you use?

- A. Internet Gateway
- B. NAT Gateway
- c. AWS PrivateLink
- D. Bastion Host

VPN Connections

VPN Connections

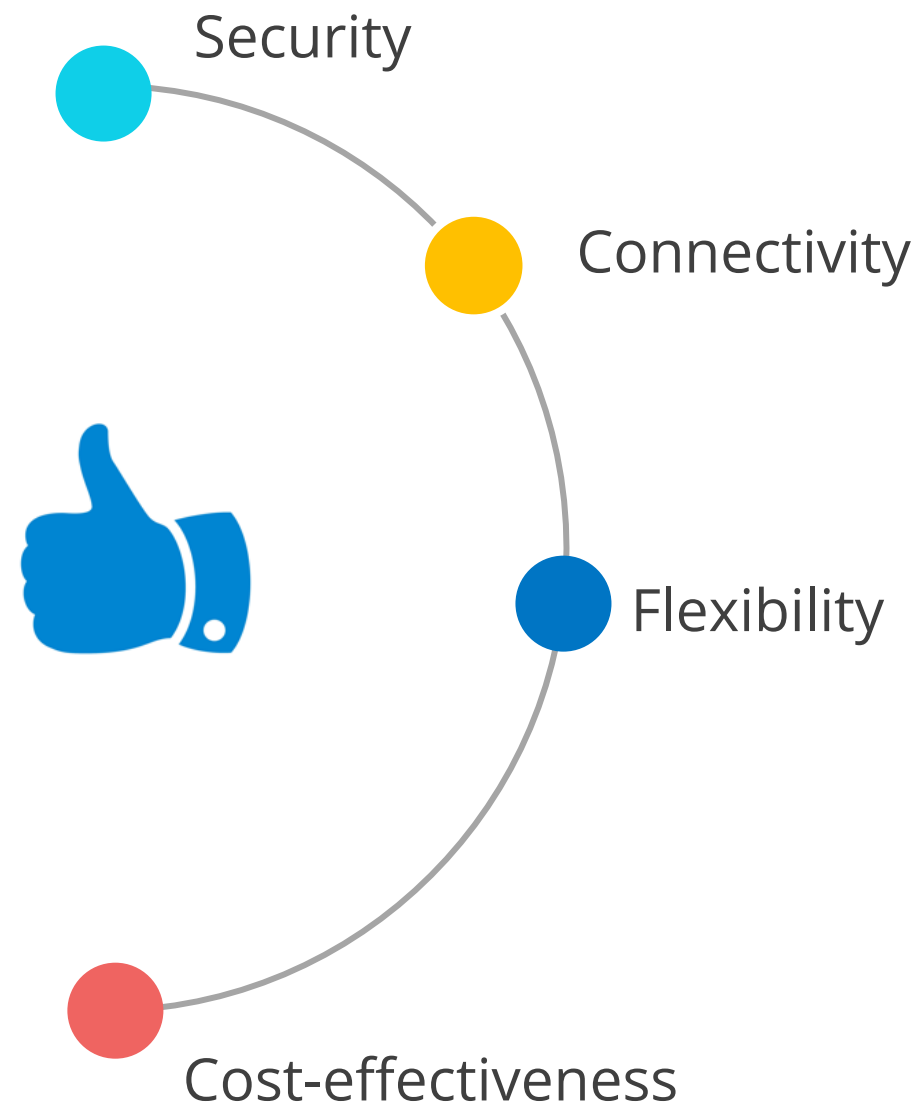
It establish secure connections between a user's on-premises networks, remote offices, client devices, and the AWS global network.

The following VPN connectivity options facilitate the connection of Amazon Virtual Private Cloud (VPC) to remote networks:

- 01 AWS Site-to-Site VPN
- 02 AWS Client VPN
- 03 AWS VPN CloudHub
- 04 Third-Party Software VPN Appliance

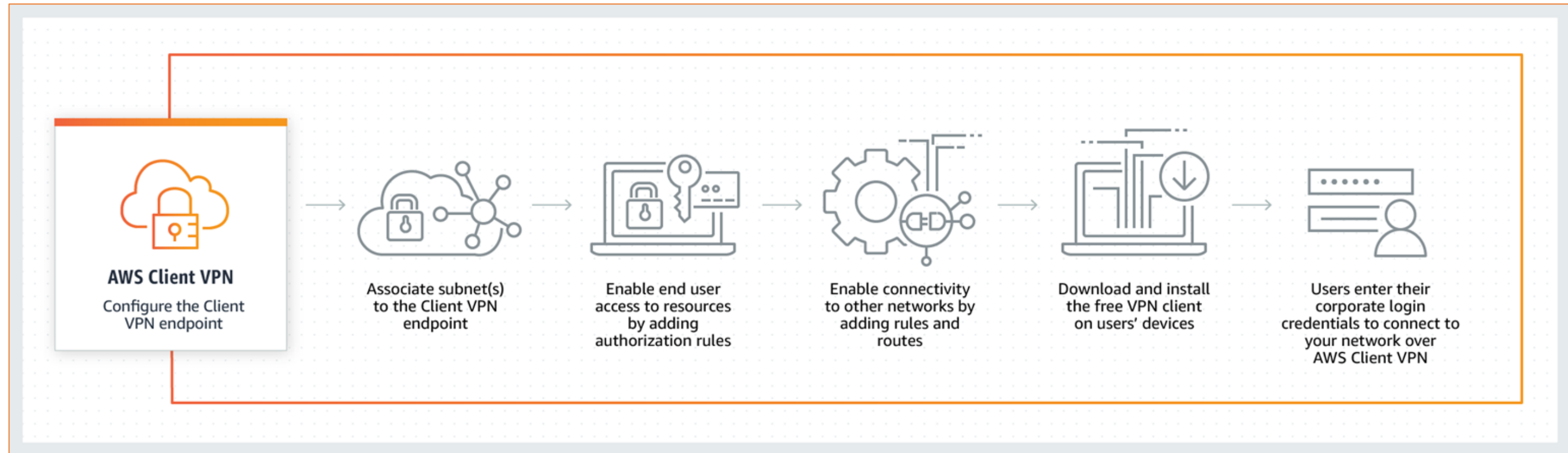
VPN Connections: Benefits

The benefits of the VPN connections are as follows:



Working of VPN Connections

The following diagram shows the working of VPN connections:

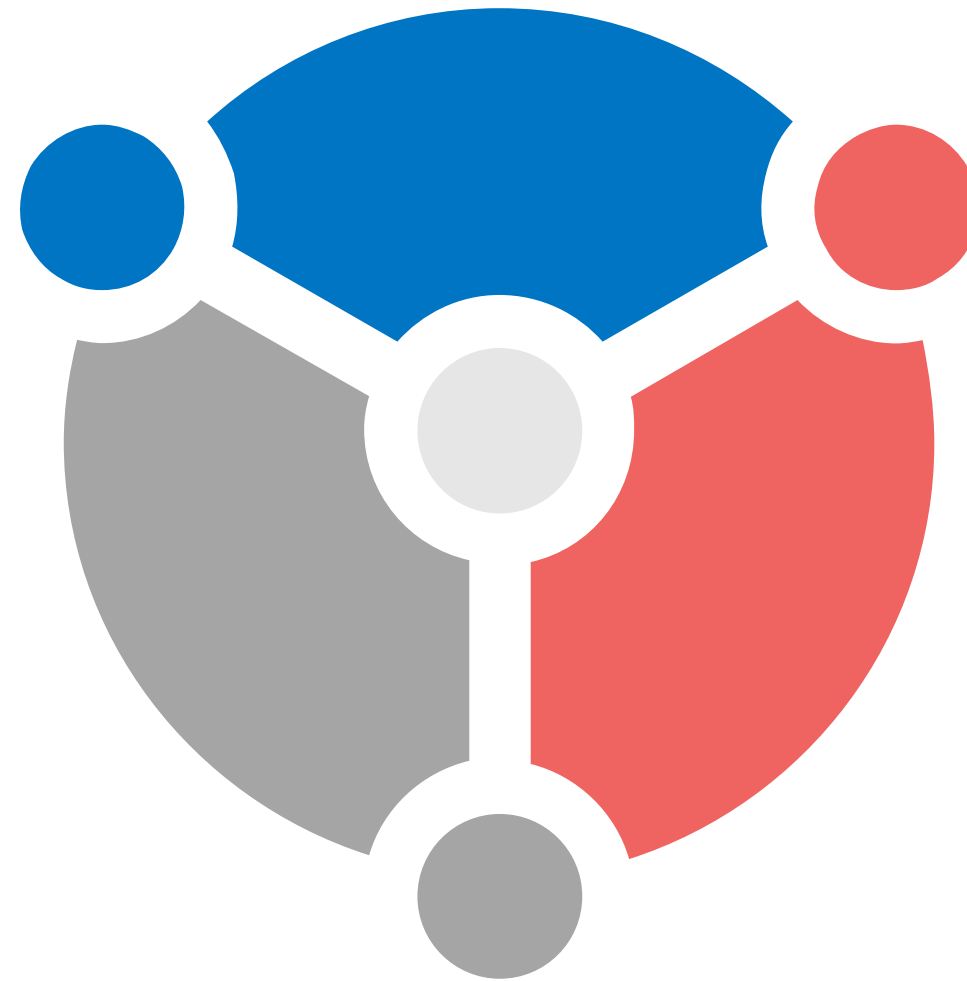


Source: <https://aws.amazon.com/vpn/>

Use Cases of VPN Connections

VPN connections offer versatile solutions that address several practical scenarios, such as:

Effortless
application access



Rapid remote
access scaling

Effortless deployment and
removal for temporary
workers



Direct Connect

It is a cloud service solution that simplifies the establishment of a dedicated network connection from the user's premises to AWS.

More about AWS direct connect:

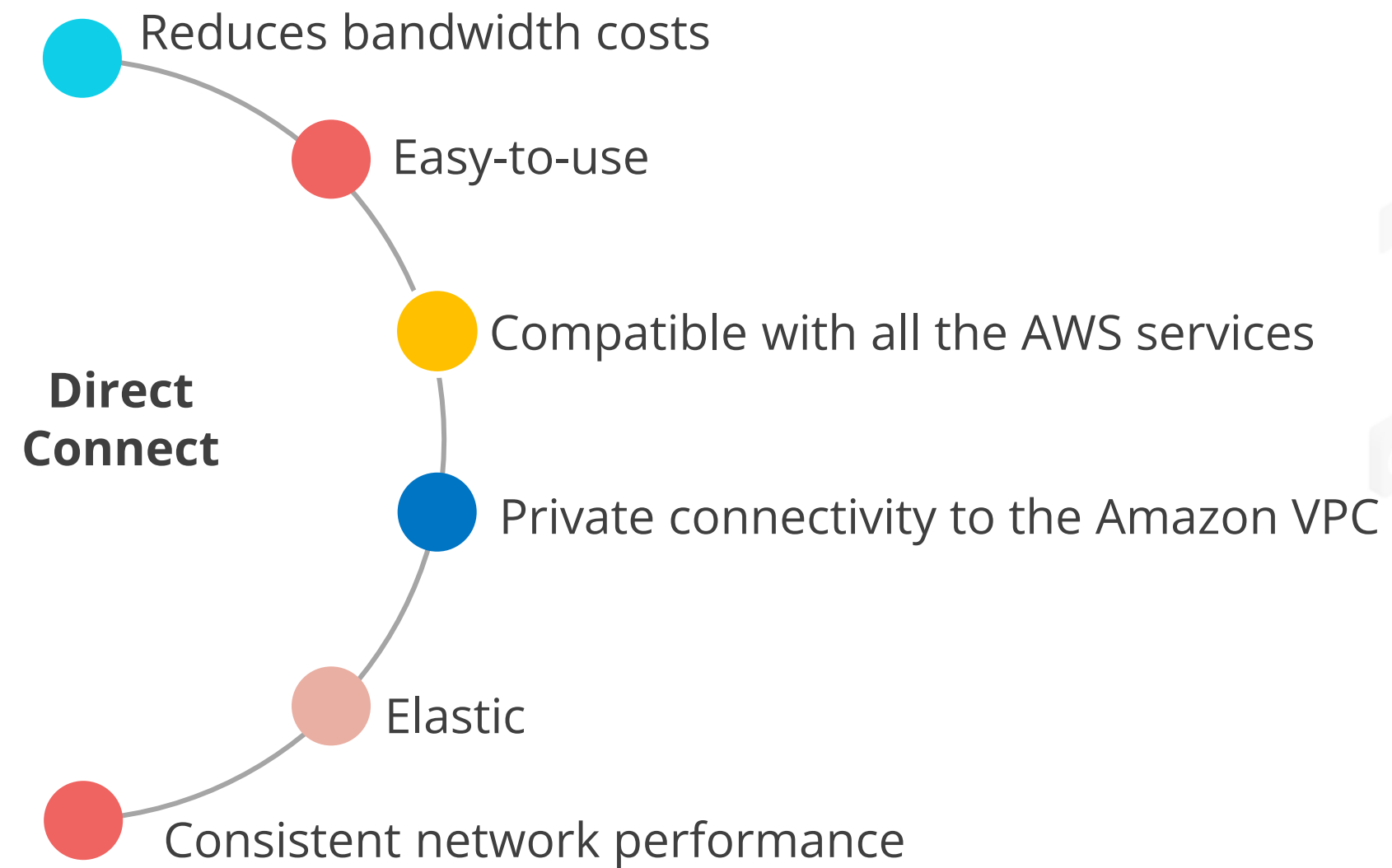
01

It facilitates the creation of private connectivity between AWS and data centers, offices, or colocation environments.

02

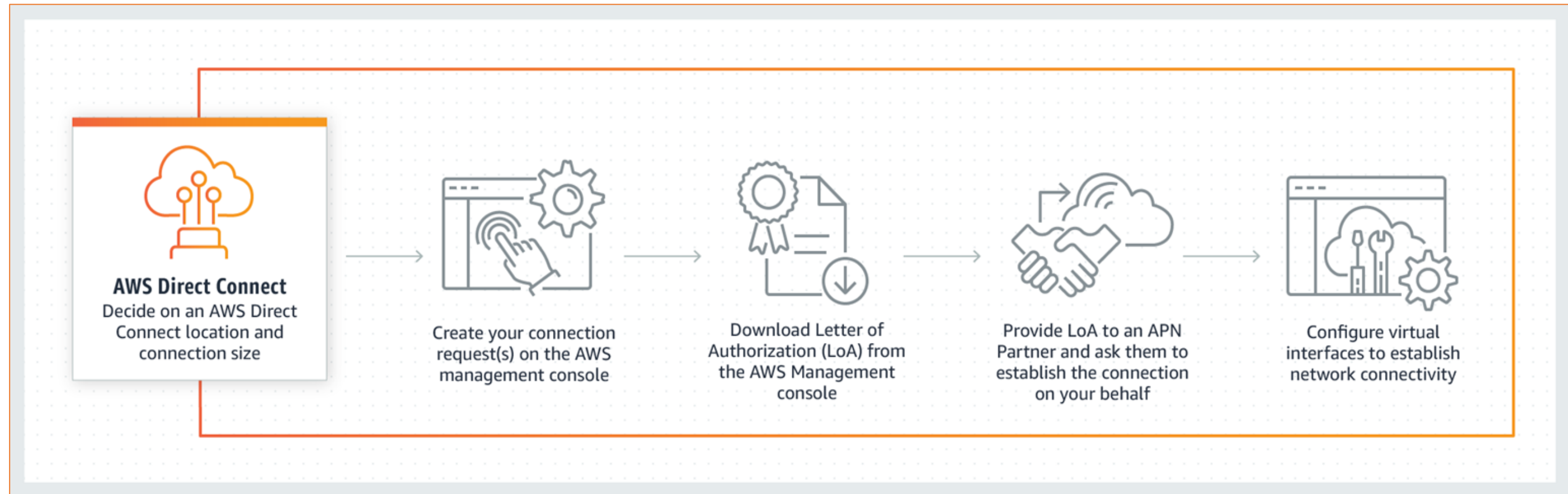
It establishes a dedicated connection between the user's network and one of the AWS Direct Connect locations.

Direct Connect: Benefits



Working of Direct Connect

The following diagram shows the working of direct connect:



Source: <https://aws.amazon.com/directconnect/?nc=sn&loc=1>

Working of Direct Connect

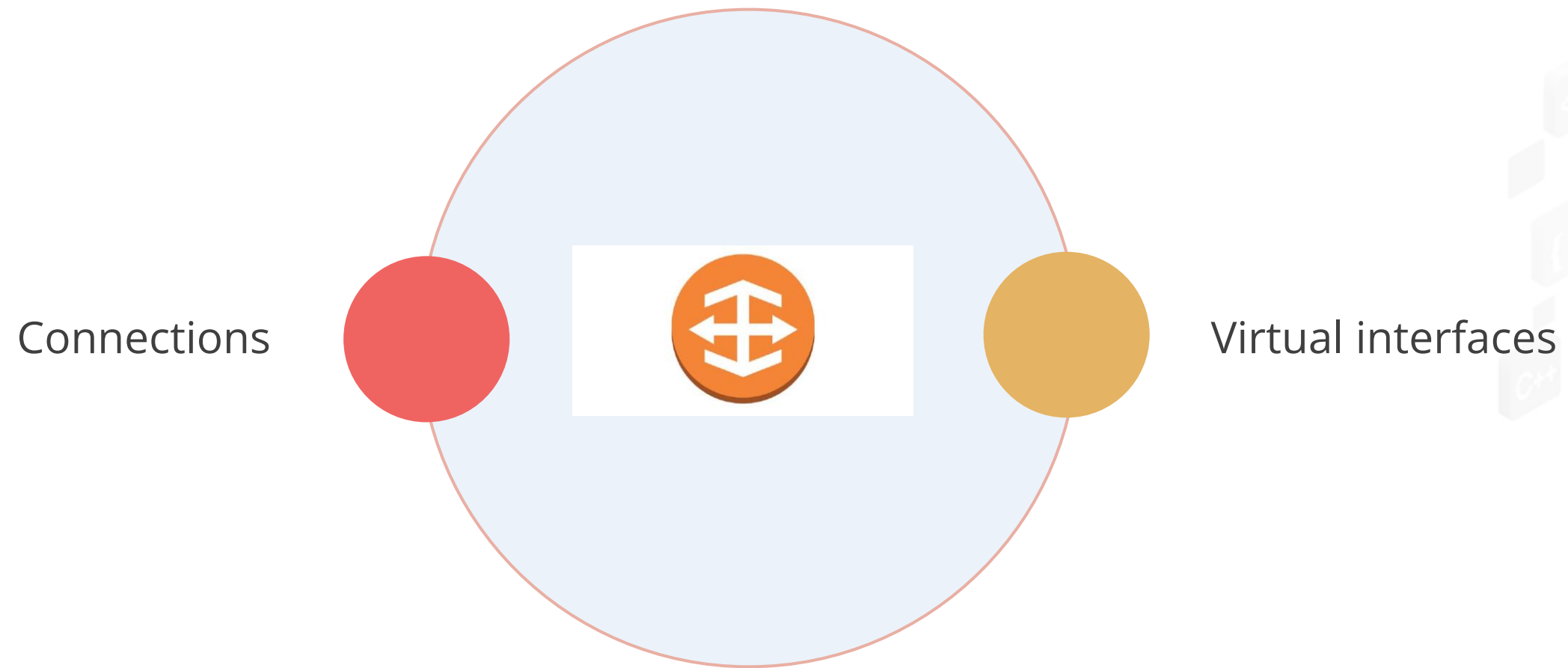
The diagram below illustrates the interaction of AWS direct connect with the network:



Source: <https://aws.amazon.com/directconnect/?nc=sn&loc=1>

Components of Direct Connect

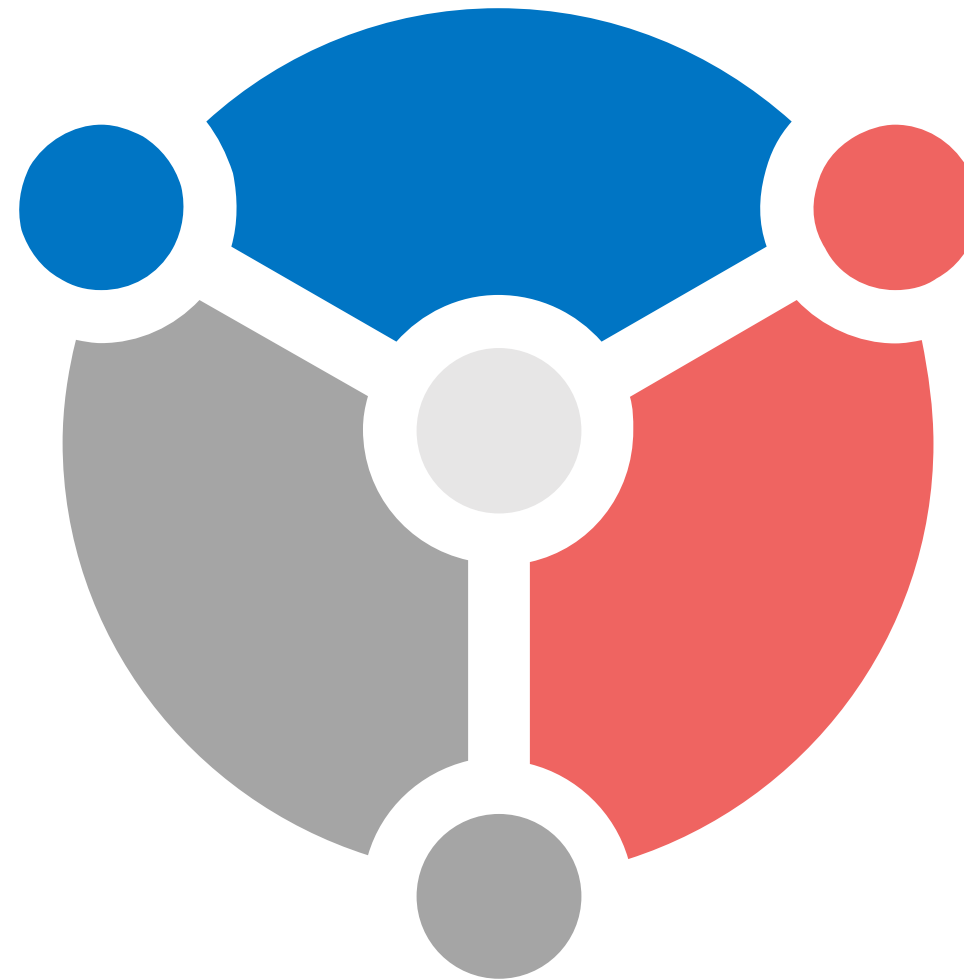
AWS direct connect comprises two essential components:



Use Cases of Direct Connect

It offers versatile solutions that address several practical scenarios, enhancing data transfer, performance, and operational efficiency.

Efficient handling of
large data sets



Real-time data feeds and
exchanges

Hybrid environments
integration



Direct Connect: Costs

The following are the costs associated with direct connect:

01 Dedicated connections

02 Hosted connections

03 Data transfer



Direct Connect: Costs

The following are the costs associated with direct connect:

04 TCO calculator

05 AWS pricing calculator

06 Economics resource center



Quick Check



You need a consistent, low-latency connection between your on-premises data center and AWS, with dedicated bandwidth for high-volume data transfers. Which service should you choose?

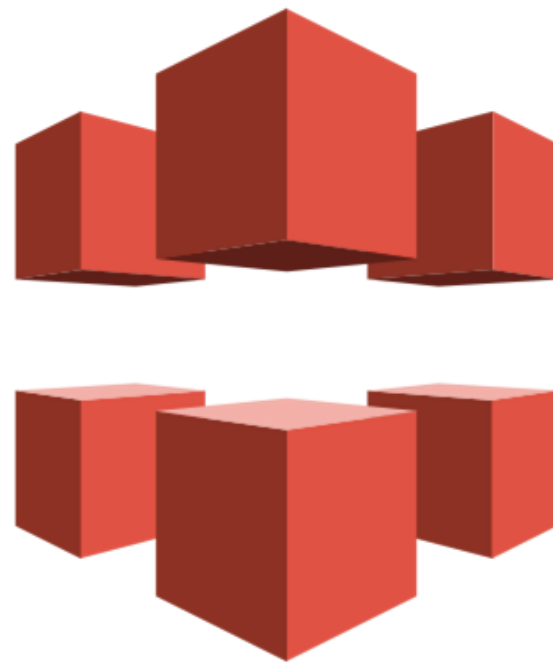
- A. VPN Connection
- B. Internet Gateway
- C. AWS Direct Connect
- D. NAT Gateway

TECHNOLOGY

CloudFront

CloudFront

It is a global content delivery network (CDN) service designed to securely and efficiently distribute content to end users, ensuring low latency and high transfer speeds.

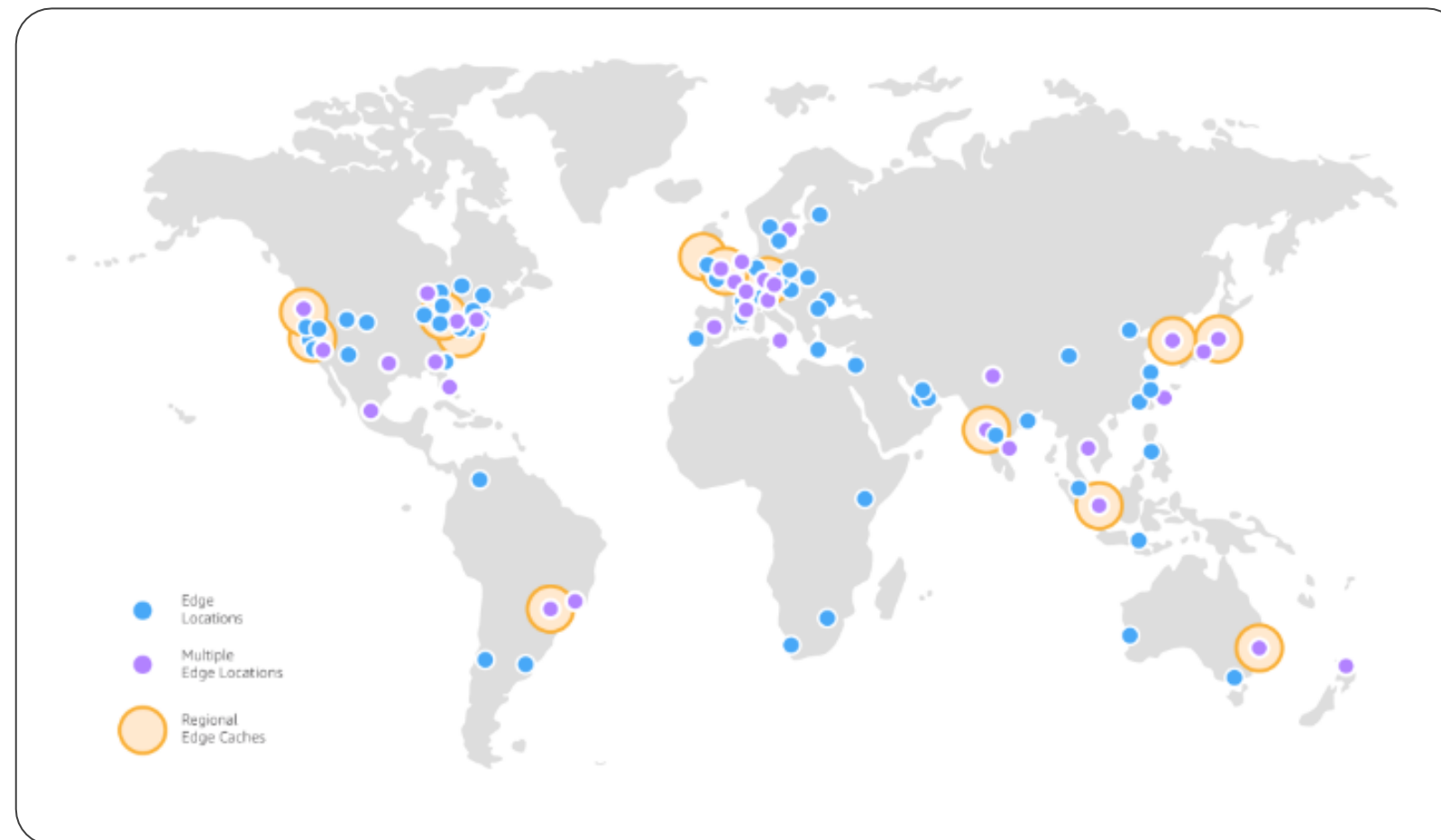


Amazon CloudFront



Amazon CloudFront Points of Presence (POPs)

Amazon CloudFront operates through a network of strategically located Points of Presence (POPs) to optimize content delivery, enhance user experiences, and ensure high availability.



Source: <https://aws.amazon.com/blogs/networking-and-content-delivery/400-amazon-cloudfront-points-of-presence/>

Amazon CloudFront Points of Presence (POPs)

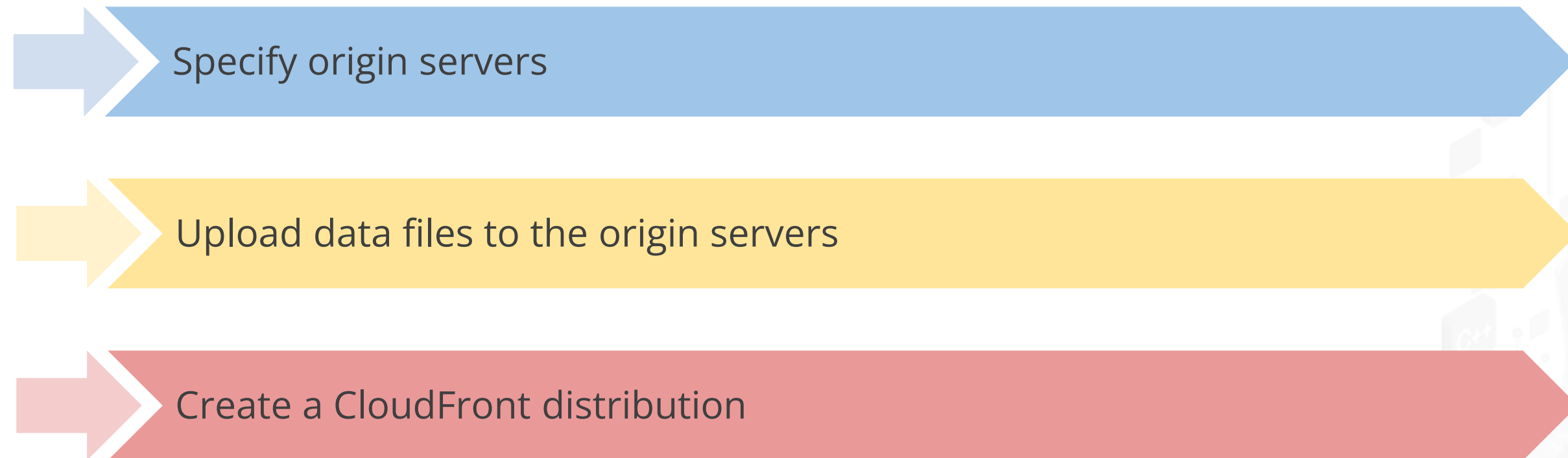
Points of Presence is a single data center.



They speed up website loading by caching web pages and images.

Content Delivery Using CloudFront

The process of establishing a CloudFront distribution to efficiently deliver content involves the following steps:



CloudFront assigns a domain name to the new distribution and sends the distribution's configuration to all its edge locations.

Content Delivery Using CloudFront

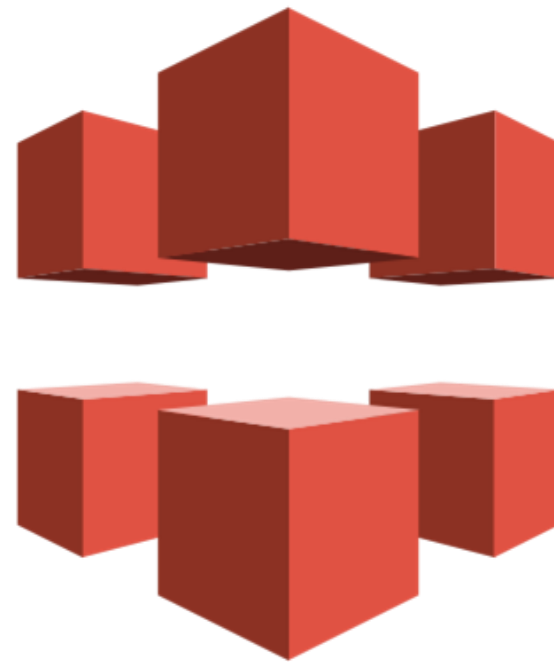
Here is a visual representation of setting up a CloudFront distribution to deliver content:



Source: <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

CloudFront With S3 origin

CloudFront provides seamless integration with Amazon S3, bolstering the security of S3 bucket access by utilizing CloudFront Origin Access Identity (OAI).



CloudFront: Use Cases

The use cases for Amazon CloudFront are as follows:



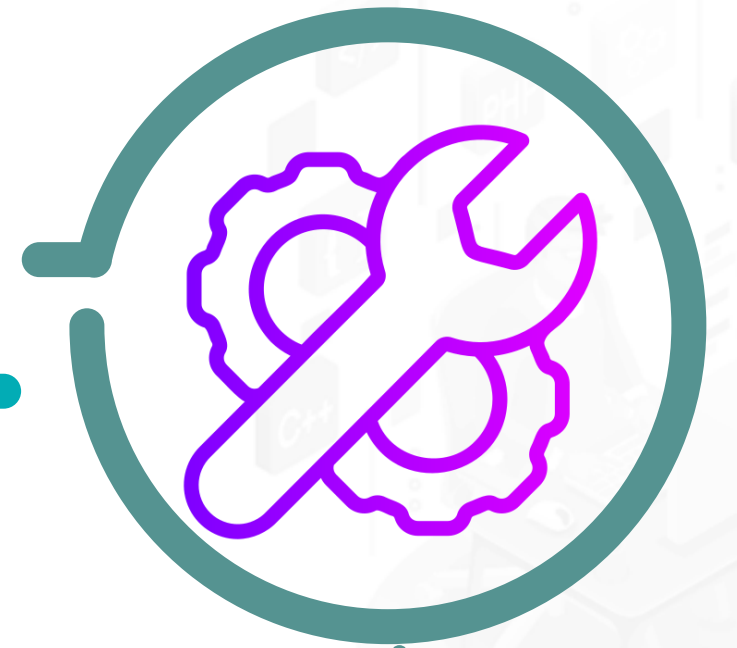
Accelerate static website
content delivery



Serve video on demand
or live streaming video

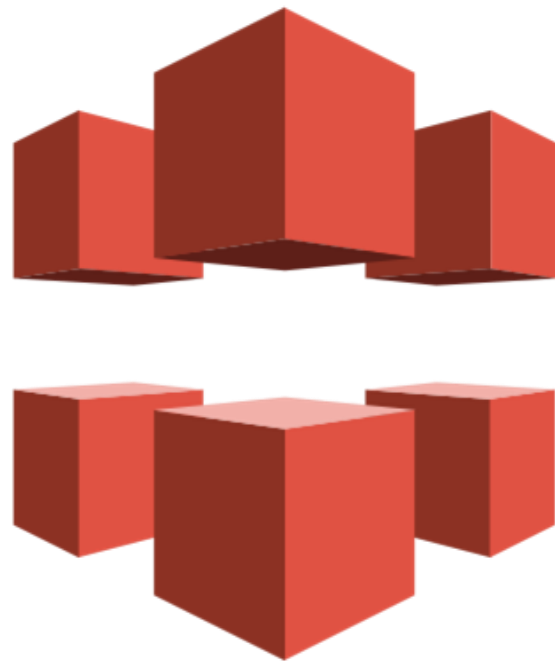


Encrypt specific fields
throughout system
processing



Customize the edge

CloudFront: Benefits



- CloudFront alleviates the load on the origin servers by caching content at its edge locations.
- It enables targeted content delivery based on user locations through its GEO targeting services.
- It utilizes HTTP and HTTPS protocols to ensure secure and rapid content delivery to end users.

Configuring CloudFront Distribution with S3 Origin



Duration:10 min

Problem Statement:

You have been assigned a task to host static web content in an Amazon S3 bucket, protected and accelerated by Amazon CloudFront for efficient content delivery.

Outcome:

You will be able to host static web content in an Amazon S3 bucket, protected and accelerated by Amazon CloudFront.

Note: Refer to the demo document for detailed steps:
08_Configuring_CloudFront_Distribution_with_S3 Origin

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create an S3 bucket
2. Upload a file
3. Configure Amazon CloudFront



Implementing Cache Configuration and Invalidation



Duration:10 min

Problem Statement:

You have been assigned a task to configure CloudFront distributions and perform cache invalidation to ensure timely updates of content served through Amazon CloudFront.

Outcome:

You will be able to configure CloudFront distributions and perform cache invalidation to ensure timely updates of content served through Amazon CloudFront.

Note: Refer to the demo document for detailed steps:
09_Creating _Cache Configuration and Invalidation

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Create Distributions
2. Create Invalidation



Configuring Origin Access Identity (OAI)



Duration:10 min

Problem Statement:

You have been assigned a task to block S3 access for the public and restrict access only through CloudFront using OAI.

Outcome:

You will be able to block S3 access for the public and restrict access only through CloudFront using OAI (Origin Access Identity).

Note: Refer to the demo document for detailed steps:
10_Configuring_Origin_Access_Identity (OAI)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Configure CloudFront access with OAI



Quick Check



You have a global audience and want to ensure low-latency access to your website content, which is stored in an S3 bucket. Which service should you use to distribute this content efficiently?

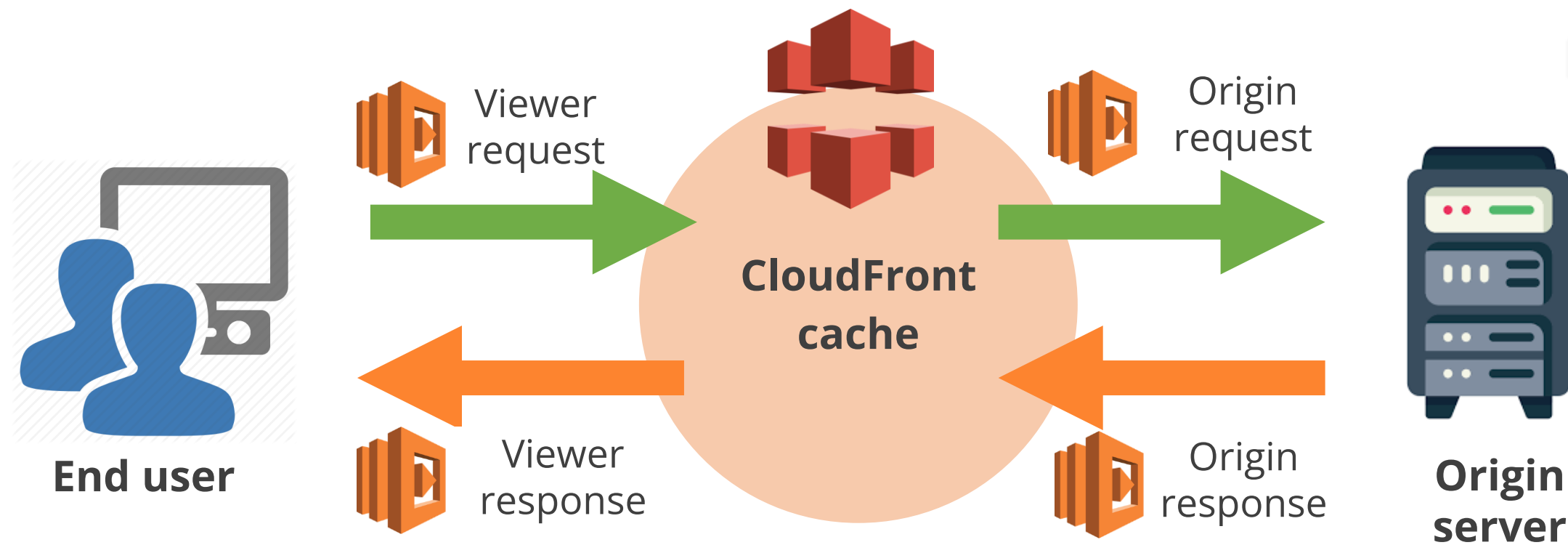
- A. Amazon S3 Transfer Acceleration
- B. Amazon Route 53
- C. Amazon CloudFront
- D. AWS Global Accelerator

TECHNOLOGY

Lambda@Edge

What Is Lambda@Edge?

It is an Amazon CloudFront feature, empowers users to access code closer to their application's end users, resulting in improved speed and reduced latency.



Lambda@Edge: Benefits

1

Makes the apps more responsive

2

Eliminates the need for server administration

3

Personalizes the content distribution



Working of Lambda@Edge

It operates seamlessly within the request cycle, making informed decisions that directly impact the delivery of CloudFront content.



Source: <https://aws.amazon.com/lambda/edge/>

Lambda@Edge: Use Cases

Following are a few important use cases of Lambda@Edge:



Simplify and reduce origin infrastructure:

Lambda@Edge can be employed as an extension or replacement for the origin.

Website security and privacy:

The users can trigger a Lambda function to add HTTP security headers to all origin responses without modifying the application code on the origin.

Lambda@Edge: Use Cases

Following are a few important use cases of Lambda@Edge:



Dynamic web application:

Lambda@Edge can be combined with other AWS services to build powerful web applications.

Search Engine Optimization (SEO):

Lambda@Edge can improve search engine optimization for a user's website.

Lambda@Edge: Use Cases

Following are a few important use cases of Lambda@Edge:



Intelligently route across origins and data centers :

The load on the origins can be balanced while improving user performance.

Bot mitigation at the edge:

Lambda@Edge can help users by blocking unwanted bots at the edge and letting authorized traffic go through.

Lambda@Edge: Use Cases

Following are a few important use cases of Lambda@Edge:



Improved user experience :

Website and web application content can be personalized without sacrificing performance.

Real-time image transformation:

A user can customize their experience by transforming images on the fly based on user characteristics.

Quick Check



You want to run custom code closer to your end users, in response to events generated by Amazon CloudFront, to customize the content delivered by your application. Which AWS service should you use?

- A. AWS Lambda
- B. Amazon CloudFront
- c. Lambda@Edge
- D. AWS Fargate

TECHNOLOGY

Global Accelerator

AWS Global Accelerator

It is a powerful solution designed to enhance application performance and availability for local and global users.



By utilizing a network of AWS Edge Locations, Global Accelerator ensures that the application delivers a seamless and optimized experience to users globally.



AWS Global Accelerator

It offers versatile solutions that provide substantial benefits to applications based on their type and requirements.



The types of Amazon Global Accelerator are:

- Standard accelerator
- Custom routing accelerator

AWS Global Accelerator

It provides static Anycast IPv4 addresses for enhanced global application availability and performance.



The supported endpoints include the following:

- Internet-facing Application Load Balancer (ALB)
- Internal Application Load Balancer (ALB)
- Internet-facing Network Load Balancer (NLB)
- Elastic IP
- EC2 Instance (with or without a Public IP)

How Global Accelerator Works

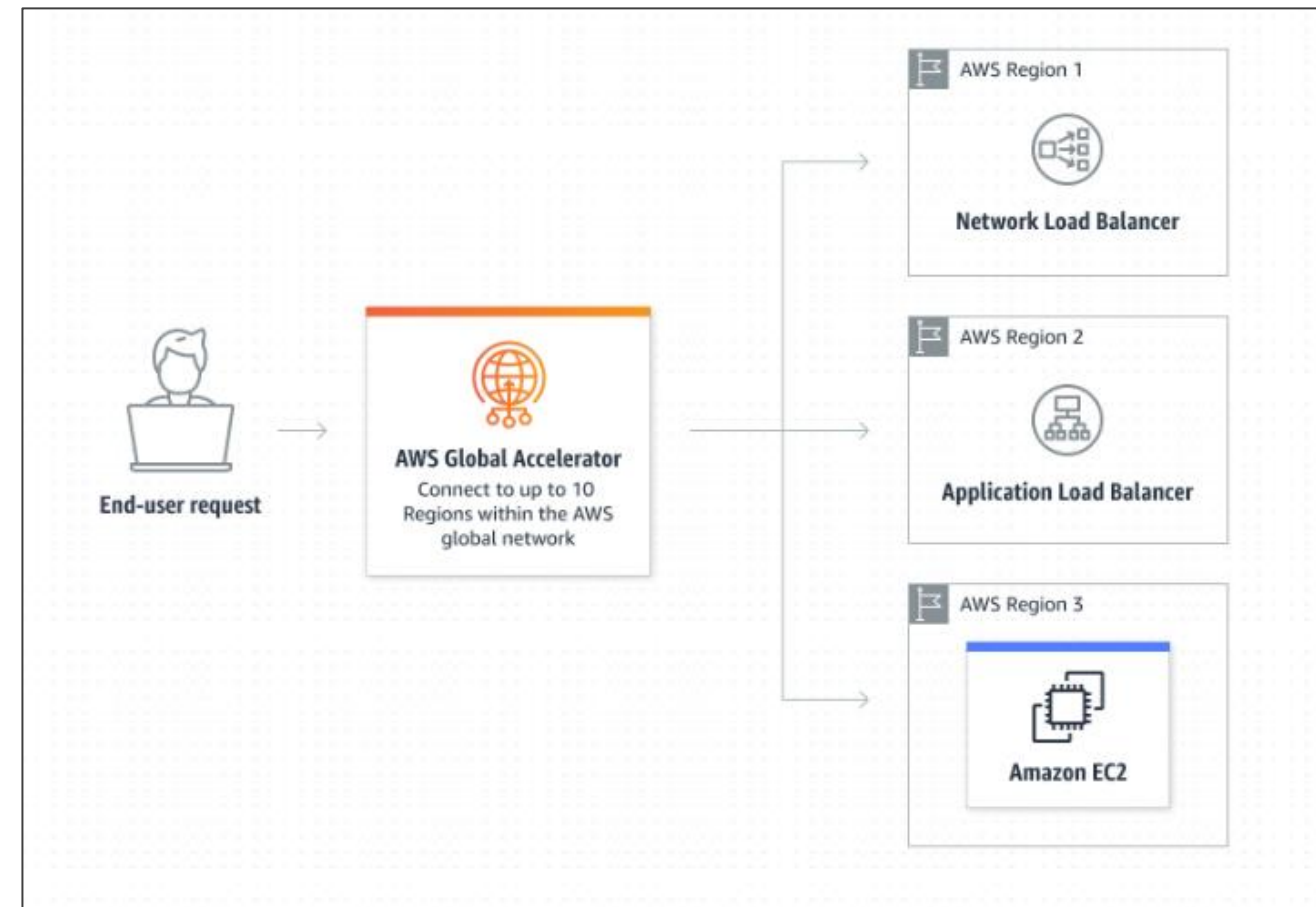
It operates by providing clients with static IP addresses that serve as a consistent and fixed entry point to the application.



- For standard accelerators, the traffic is directed to the fastest AWS endpoint based on various factors, such as the user's geographical location, the state of the endpoint, and the weights assigned to each endpoint.
- In the case of custom routing accelerators, clients are directed to specific Amazon EC2 instances and ports within a VPC subnet.

AWS Global Accelerator

It streamlines the path that TCP or UDP packets take when entering Amazon's network, effectively reducing the number of hops required.

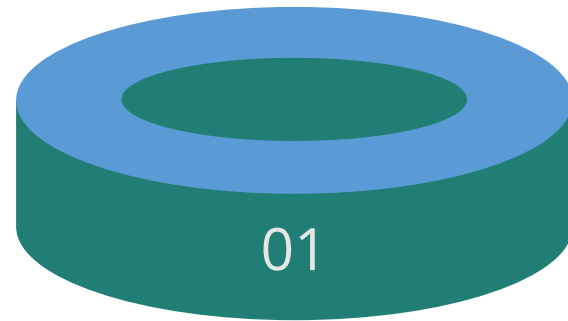


This optimization of the route for each client results in lower latencies and improved application performance.



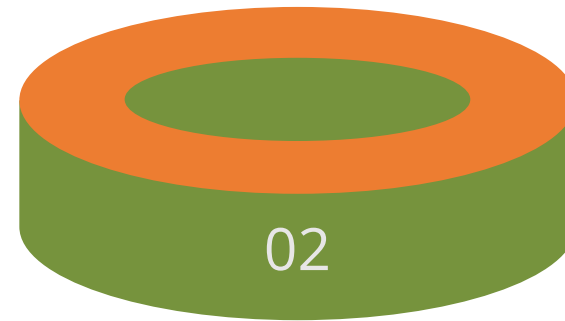
AWS Global Accelerator

The key benefits of utilizing the AWS Global Accelerator are as follows:



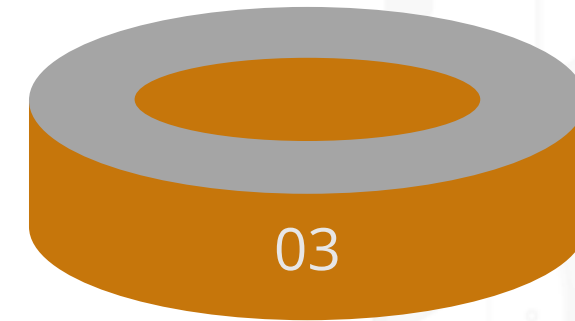
Global anycast network:

Leverages the extensive, high-performance AWS global network.



Static IP addresses:

Provides consistent entry points.

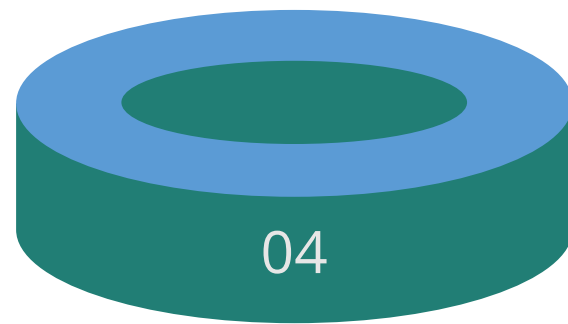


Accelerated traffic routing:

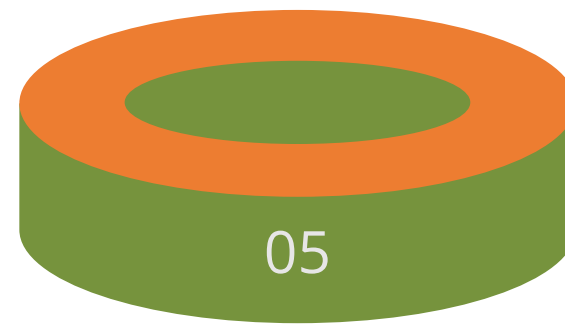
Routes traffic dynamically based on latency, health checks, and user-defined policies.

AWS Global Accelerator

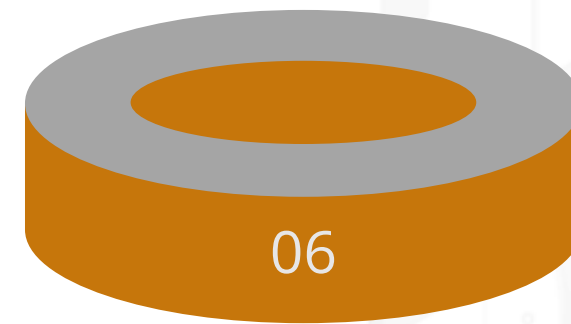
The key benefits of utilizing the AWS Global Accelerator are as follows:



Health checks and endpoint monitoring:
Monitors endpoint health continuously.



Global failover and disaster recovery:
Redirects traffic to healthy endpoints in another region during failures.



Flexible port mapping:
Supports TCP and UDP protocols and maps multiple ports to a single endpoint.

AWS Global Accelerator: Use Cases

AWS Global Accelerator is valuable for a range of use cases:

Content delivery
accelerates media and video delivery
via the nearest edge location.

Multi-region applications
elevate performance with high
availability and low latency.



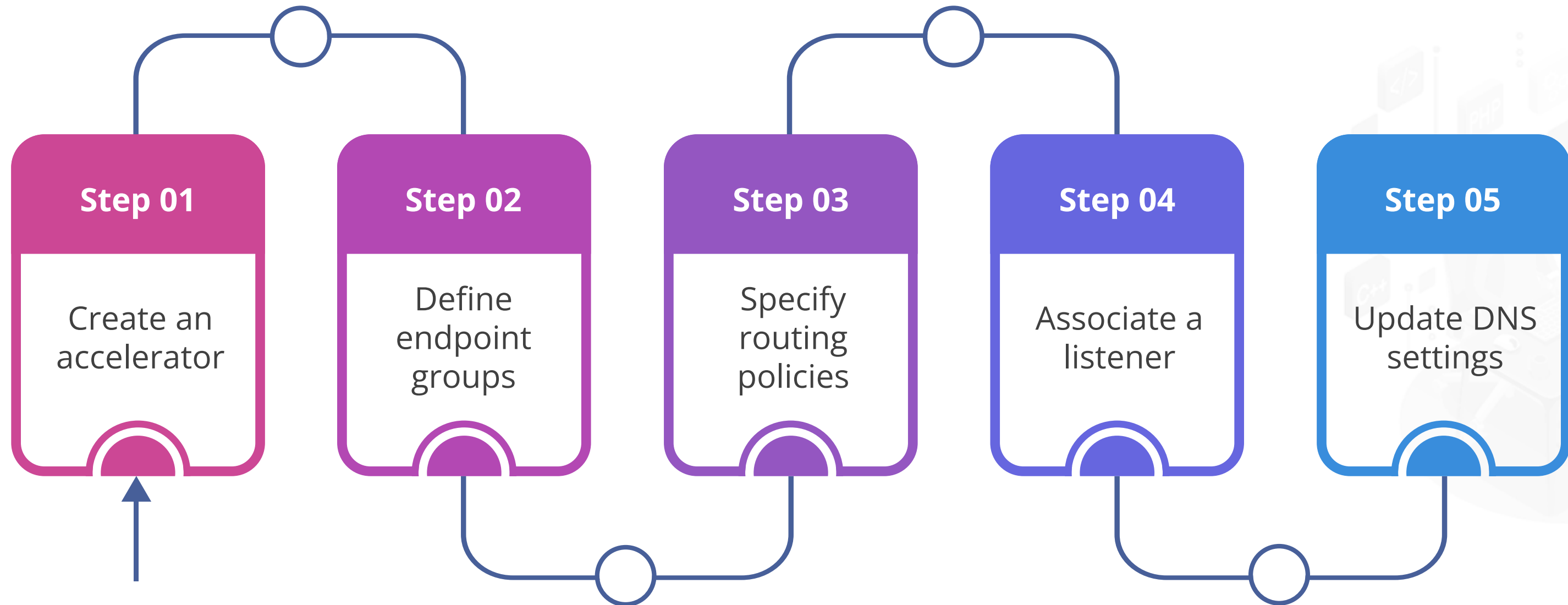
IoT applications
optimize real-time communication
and rapid responses.

Gaming
improves multiplayer experiences
through reduced lag.

Web applications
ensure fast and consistent web
experiences by intelligently routing
traffic.

AWS Global Accelerator

To begin using AWS Global Accelerator, follow these simple steps:



Creating a Global Accelerator



Problem Statement:

You have been assigned a task to create a Global Accelerator on AWS for enhancing the availability and performance of your applications by intelligently routing traffic across multiple regions and endpoints.

Outcome:

You will be able to create a Global Accelerator on AWS, enhancing application availability and performance by routing traffic across multiple regions and endpoints.

Note: Refer to the demo document for detailed steps:
[10_Creating_a_Global_Accelerator](#)

Assisted Practice: Guidelines

Steps to be followed are:

1. Create an EC2 web instance
2. Create a Global Accelerator



Quick Check



You have a global application that needs to improve availability and performance for users worldwide by directing traffic to the nearest and healthiest endpoints. Which service should you use?

- A. Amazon CloudFront
- B. AWS Global Accelerator
- C. Amazon Route 53
- D. Elastic Load Balancer

Amazon Route 53

Amazon Route 53

It is a scalable and highly available DNS web service provided by Amazon Web Services (AWS).



Its primary function is to translate user-friendly domain names into corresponding IP addresses, enabling users to access websites and applications effortlessly.



Amazon Route 53: Features

01

Resolver

02

Traffic flow

03

Latency based
routing

04

Geo DNS

05

Private DNS for VPC

06

DNS failover

Amazon Route 53: Features

07

Health checks and monitoring

08

Domain registration

09

CloudFront zone apex support

10

Amazon ELB integration

11

Management console

12

Weighted round robin

Amazon Route 53: Benefits

Some of the benefits of Amazon Route 53 are as follows:

01 Highly available and reliable

02 Flexible

03 Well designed

04 Easy-to-use

05 Fast performance

Amazon Route 53: Benefits

Some of the benefits of Amazon Route 53 are as follows:

06 Cost-effective

07 Secure

08 Scalable

09 Simplifies the hybrid cloud

Quick Check



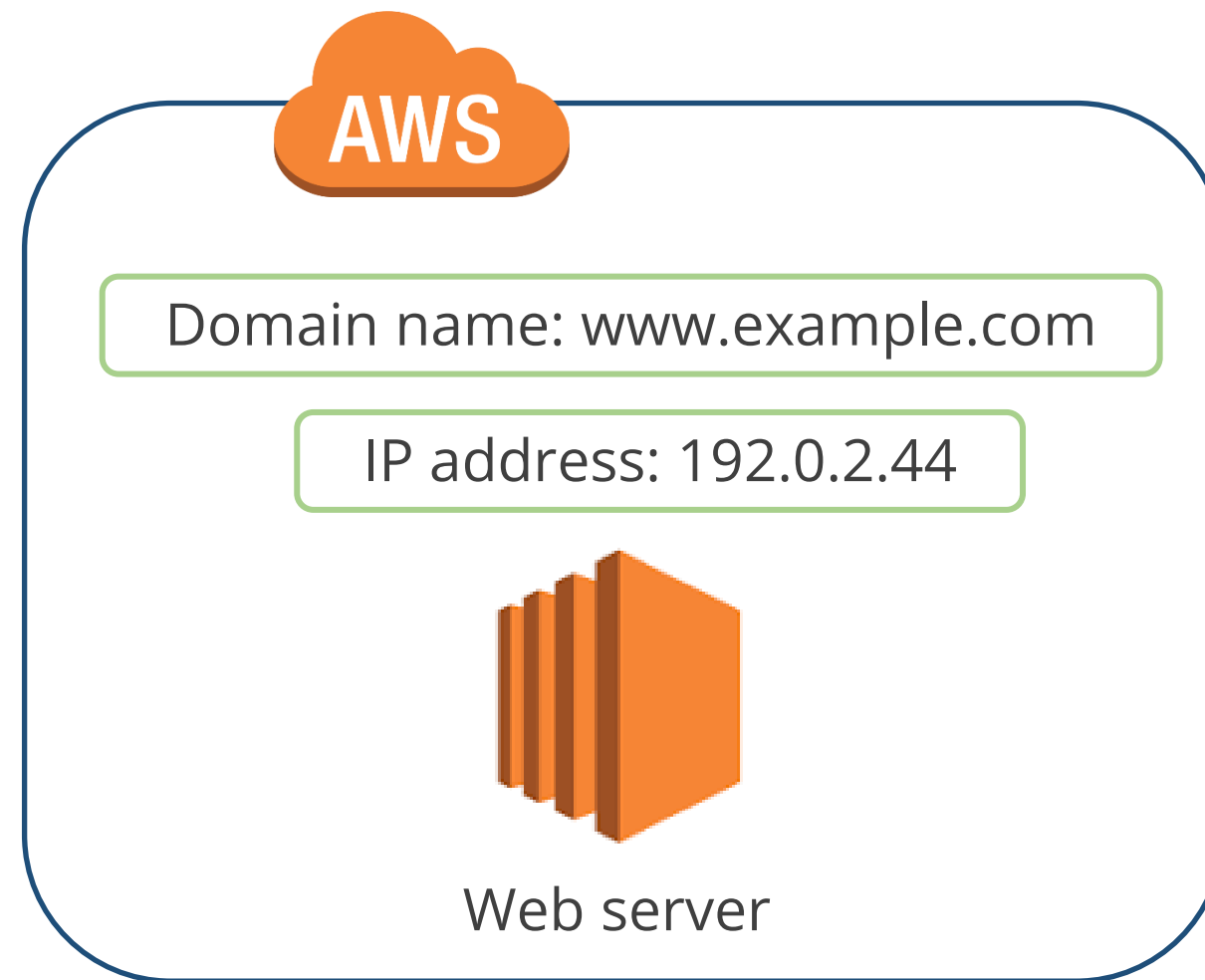
You need to route user traffic to different server locations based on the geographic location of the users. Which Amazon Route 53 routing policy should you use?

- A. Simple Routing
- B. Latency Routing
- C. Geolocation Routing
- D. Weighted Routing

Route 53 Routing Policies

Domain Name System (DNS)

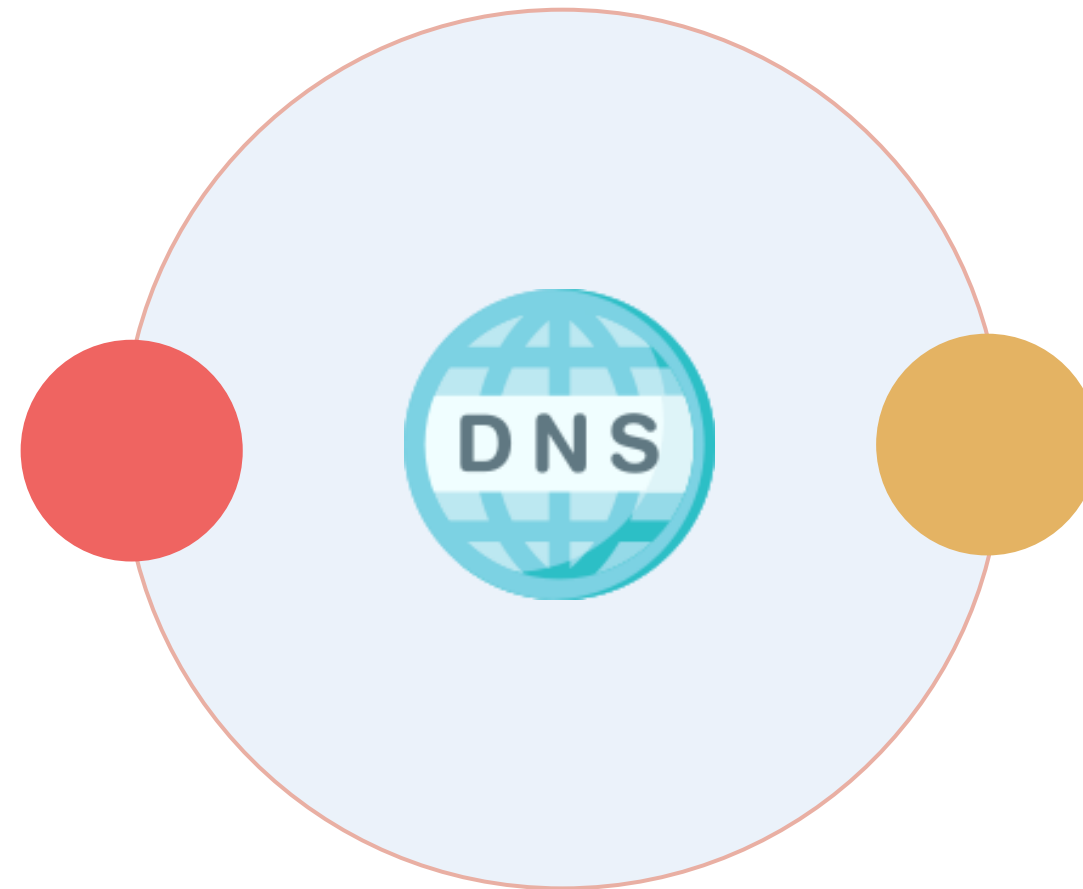
It translates human-readable domain names (**www.example.com**) to machine-readable IP addresses (**192.0.2.44**).



Domain Name System Services

These are the two types of DNS services:

Authoritative DNS



Recursive DNS



Domain Name System Workflow

The following diagram is an overview of how recursive and authoritative DNS services work together to route an end-user to a website or application:

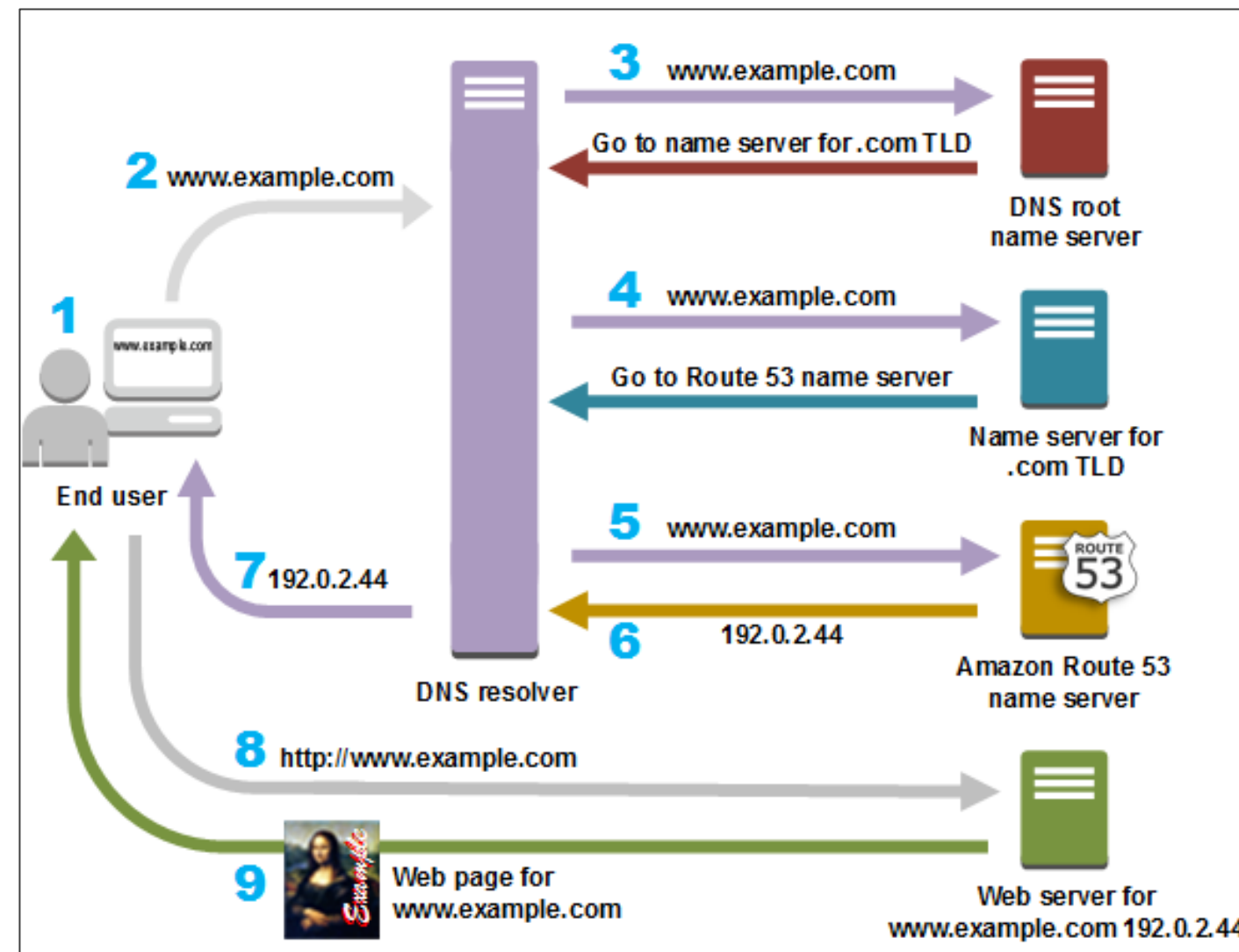
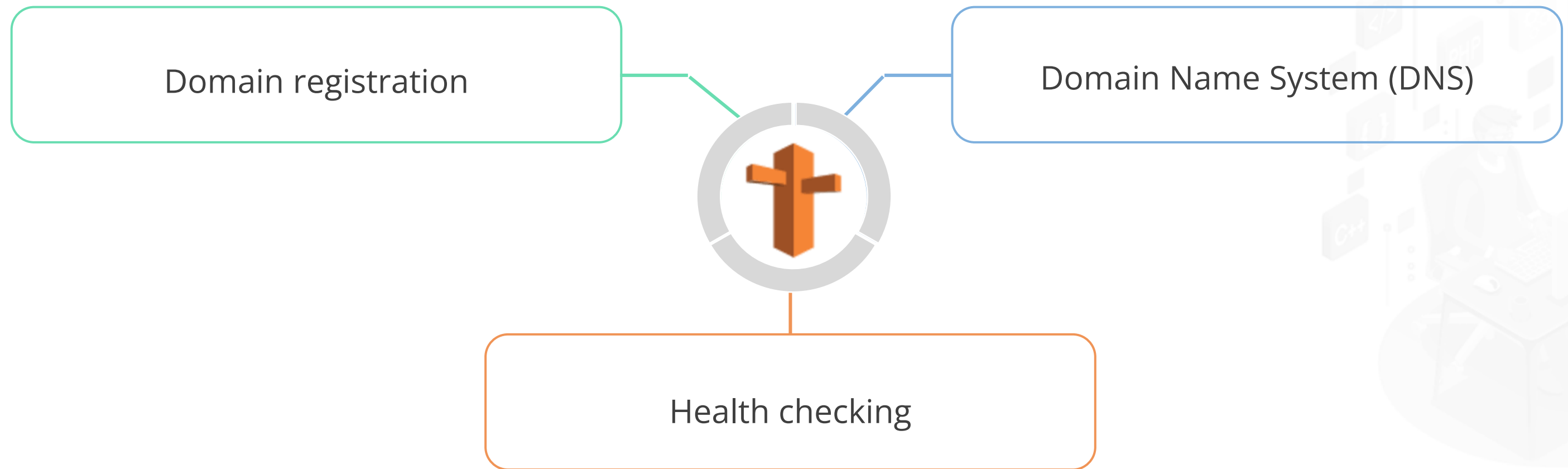


Image source: [https://aws.amazon.com/route53/what-is-dns/#:~:text=DNS%2C%20or%20the%20Domain%20Name,addresses%20\(for%20example%2C%20192.0.](https://aws.amazon.com/route53/what-is-dns/#:~:text=DNS%2C%20or%20the%20Domain%20Name,addresses%20(for%20example%2C%20192.0.)

Amazon Route 53 Concepts

These concepts are fundamental to understand Amazon Route 53:



Amazon Route 53 Concepts: Domain Registration

The following are the fundamental concepts associated with domain registration:

01 Domain name

02 Domain registrar

03 Domain registry

04 Domain reseller

05 Top-level domain (TLD)

Amazon Route 53 Concepts: Domain Name System (DNS)

The following are the fundamental concepts associated with Domain Name System (DNS):

01 Authoritative name server

02 DNS query

03 DNS resolver

04 IP address

05 Private DNS

Amazon Route 53 Concepts: Domain Name System (DNS)

The following are the fundamental concepts associated with Domain Name System (DNS):

06 Recursive name servers

07 Reusable delegation set

08 Routing policy

09 Subdomain

10 Time to live (TTL)

Amazon Route 53 Concepts: Health Checking

The following concepts are related to health checking:

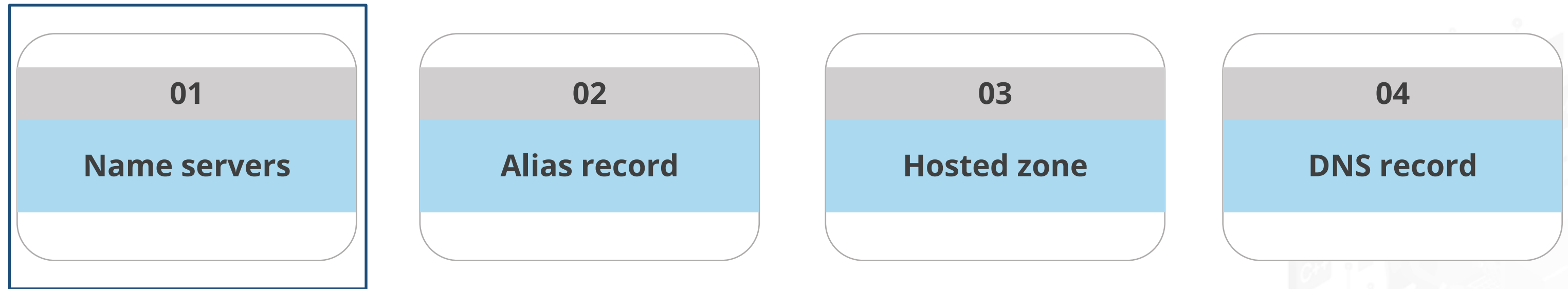
01 DNS failover

02 Endpoint

03 Health check

Amazon Route 53: Terminologies

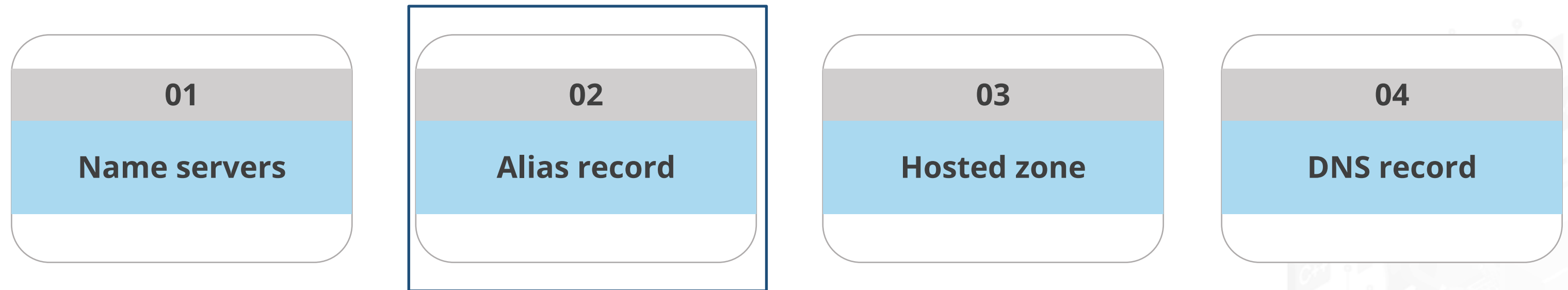
The following terminologies are used in Amazon Route 53:



It helps to translate domain names into the IP addresses that computers use to communicate with one another.

Amazon Route 53: Terminologies

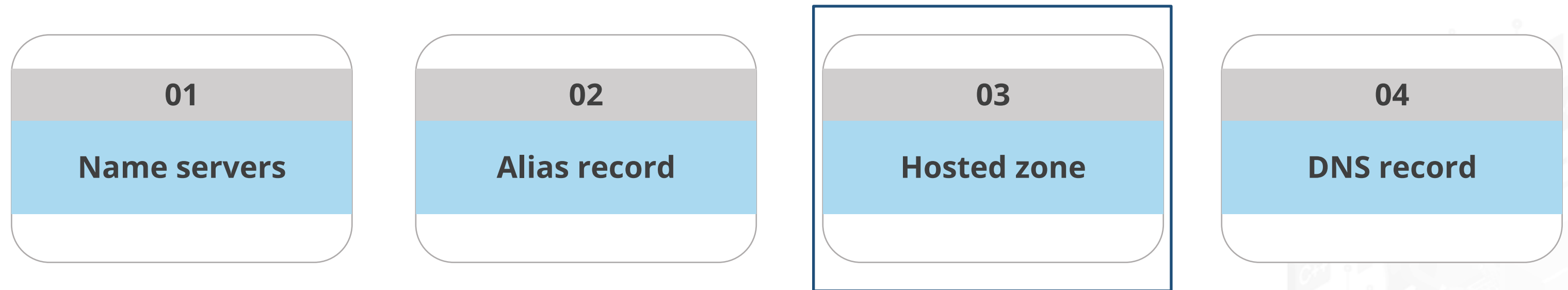
The following terminologies are used in Amazon Route 53:



It is a type of record that users can create with Amazon Route 53 to route traffic to AWS resources such as Amazon CloudFront distributions and Amazon S3 buckets.

Amazon Route 53: Terminologies

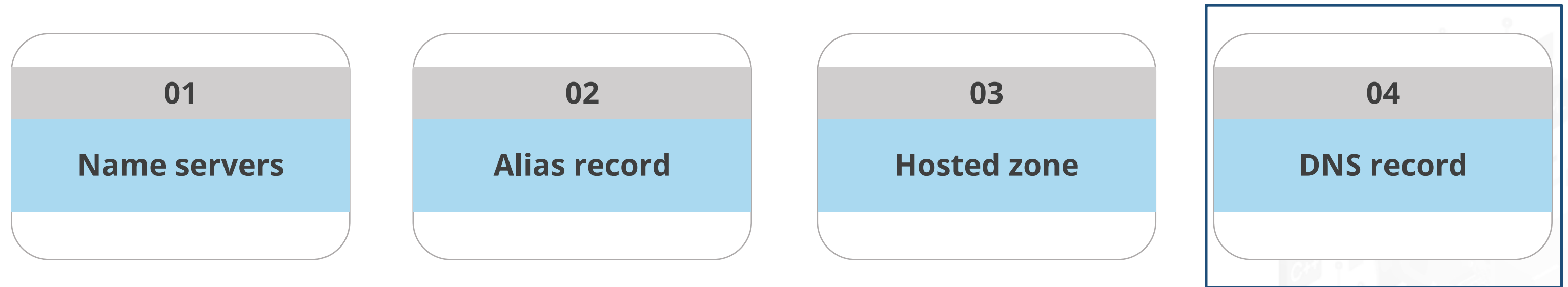
The following terminologies are used in Amazon Route 53:



It is a container for records, which includes information about how the users want to route traffic for a domain and all its subdomains.

Amazon Route 53: Terminologies

The following terminologies are used in Amazon Route 53:



It is an object in a hosted zone that the users use to define how they want to route traffic for the domain or a subdomain.

Amazon Route 53: Policies

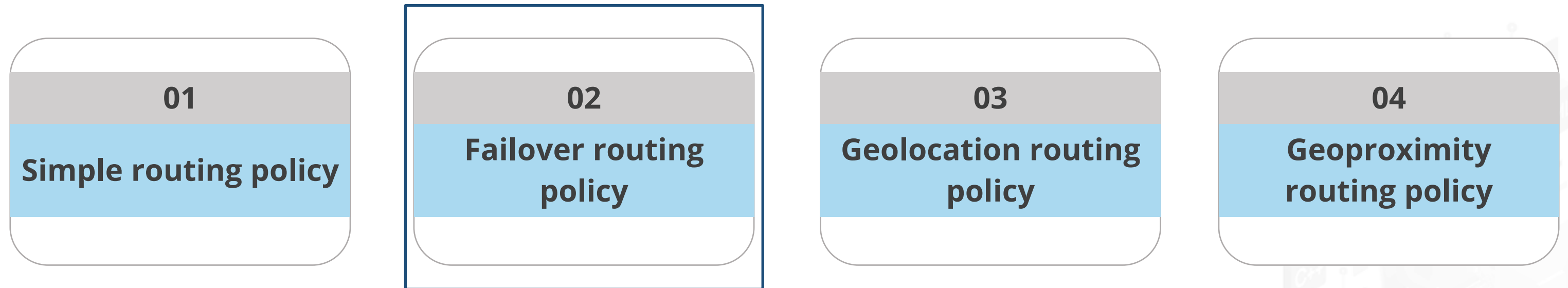
Amazon Route 53 supports the following routing policies:



It is used to route internet traffic to a single resource that performs a given function for the domain, for example, a web server that serves content for the example.com website.

Amazon Route 53: Policies

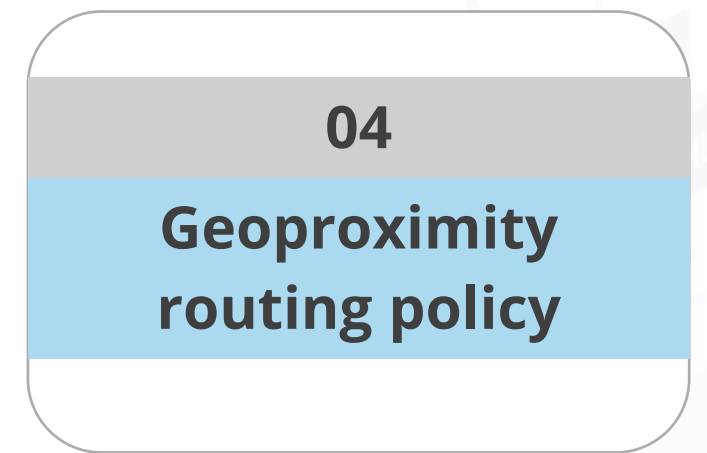
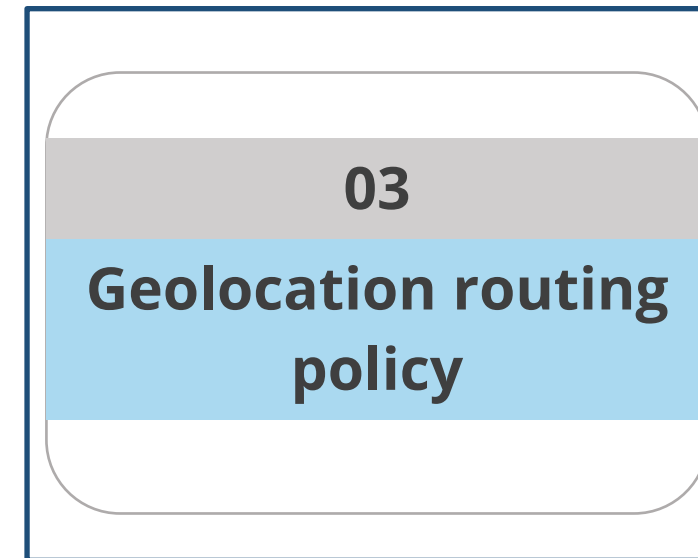
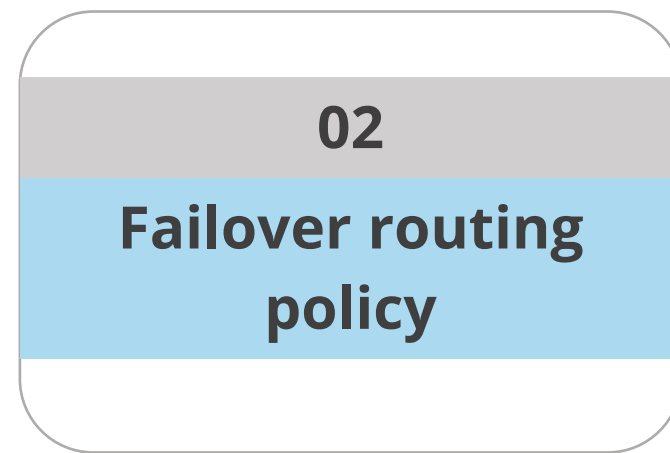
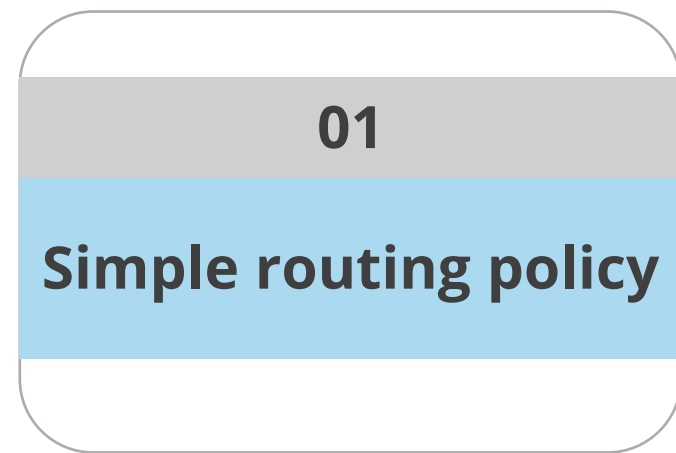
Amazon Route 53 supports the following routing policies:



It is used when users want to configure active-passive failover.

Amazon Route 53: Policies

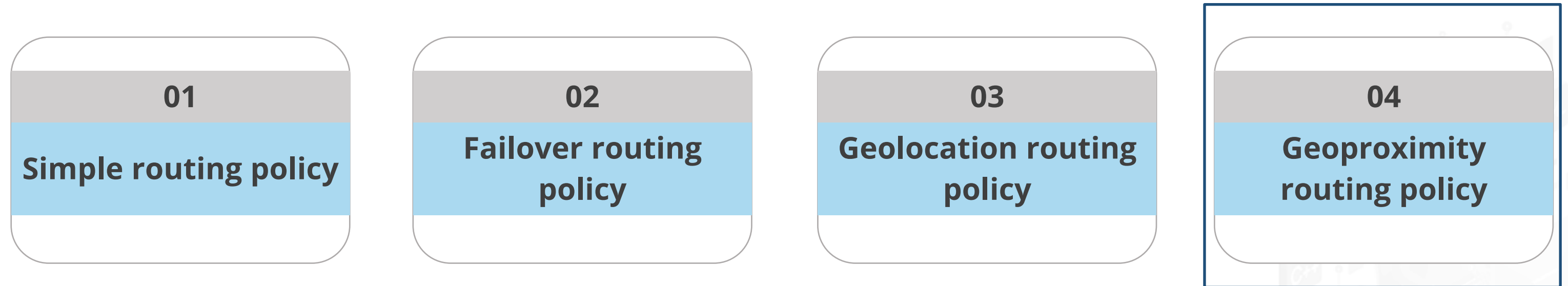
Amazon Route 53 supports the following routing policies:



It is used when users want to route internet traffic to resources based on their location.

Amazon Route 53: Policies

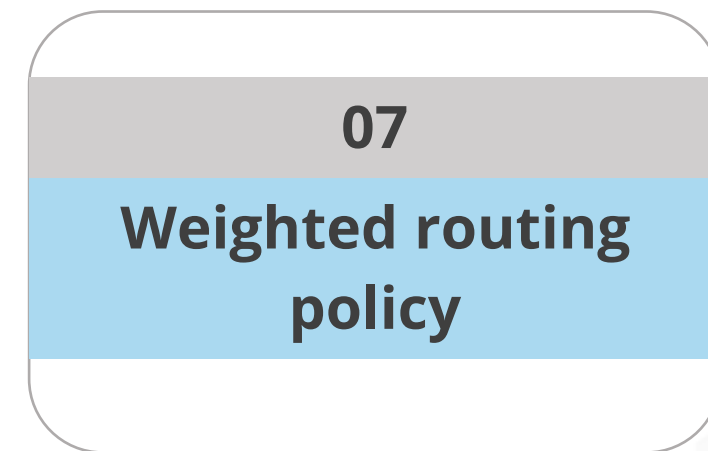
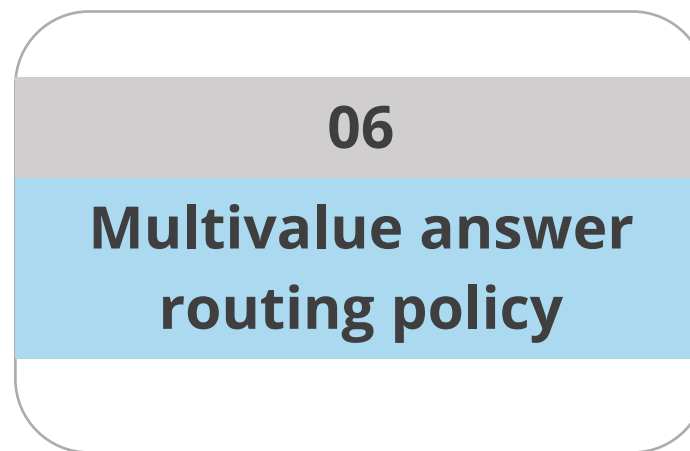
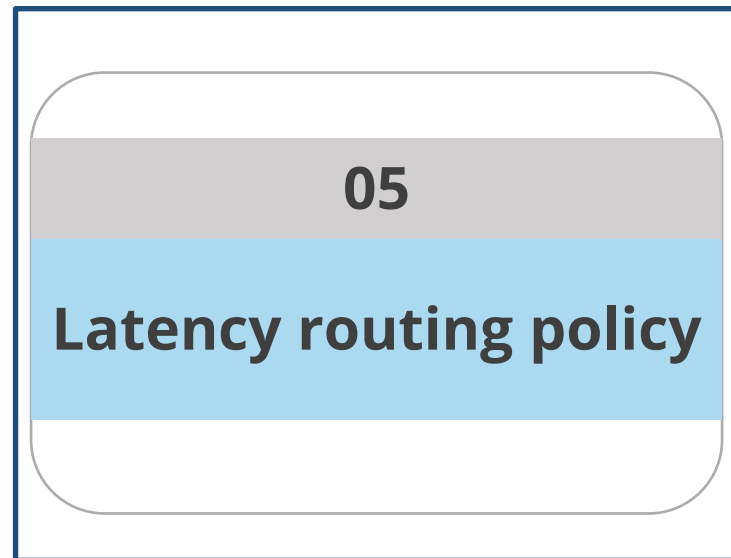
Amazon Route 53 supports the following routing policies:



It is used when the users want to route traffic based on the location of the resources and, optionally, shift traffic from resources in one location to resources in another.

Amazon Route 53: Policies

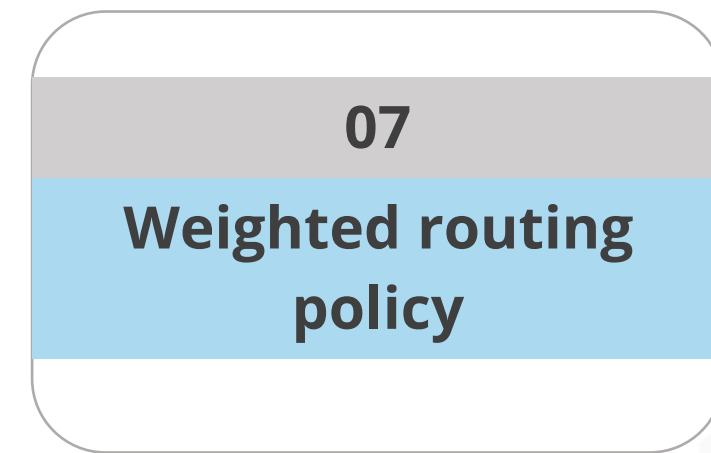
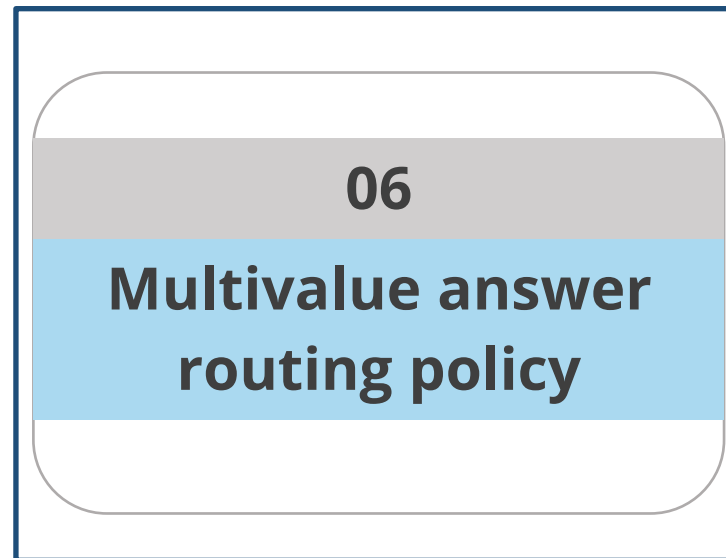
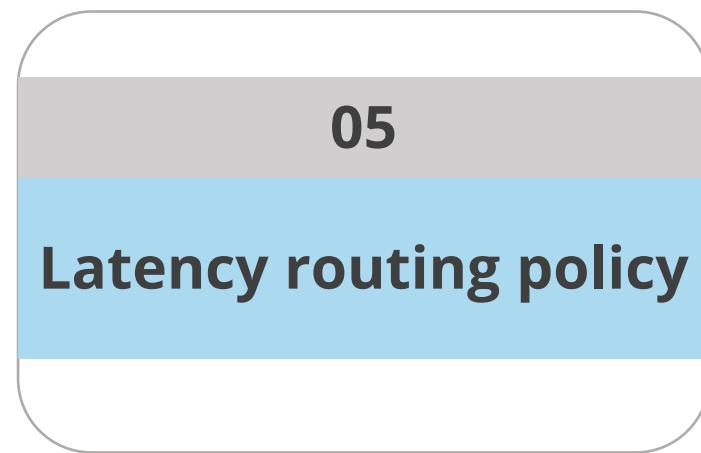
Amazon Route 53 supports the following routing policies:



It is used when the users want to route the traffic based on the location of the resources and, optionally, shift the traffic from the resources in one location to the resources in another.

Amazon Route 53: Policies

Amazon Route 53 supports the following routing policies:



It helps the users check the health of each resource, so Amazon Route 53 returns only values for healthy resources.

Amazon Route 53: Policies

Amazon Route 53 supports the following routing policies:

05

Latency routing policy

06

Multivalue answer routing policy

07

Weighted routing policy

It is used to route traffic to multiple resources in proportions that the users specify.

Amazon Route 53: Health Checks

It keeps track of the status and performance of the web applications, web servers, and other services.

Every health check can monitor one or more of the following:

- The health of a specified resource, such as a web server
- The status of other health checks
- The status of an Amazon CloudWatch alarm
- Amazon Route 53 Application Recovery Controller helps the user to set up routing control health checks with DNS failover records



Amazon Route 53: Best Practices

The following are the best practices for Amazon Route 53:

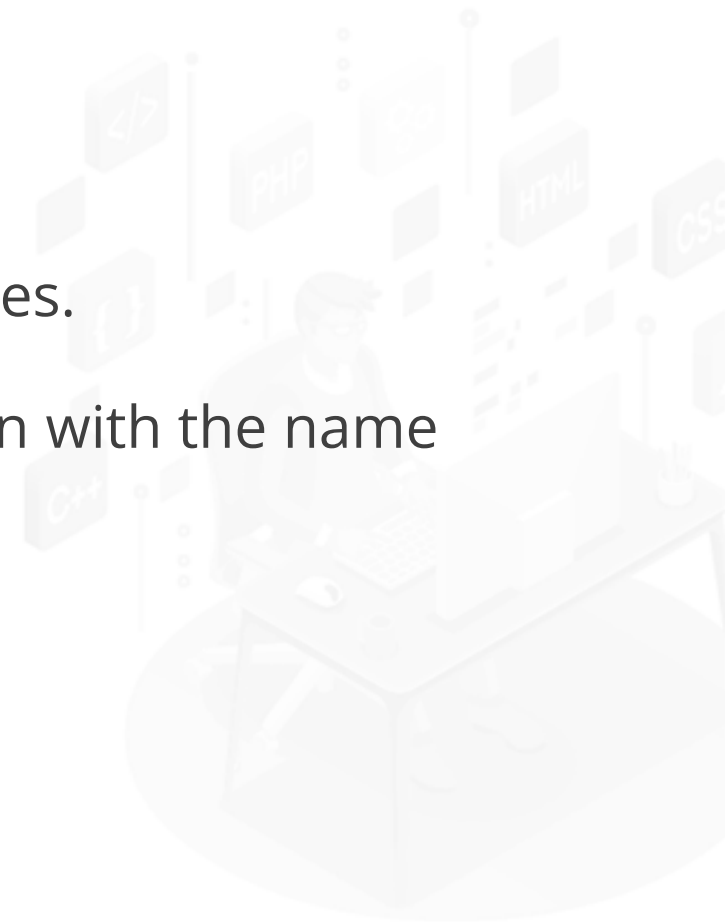
Name server delegation

TTL resource record sets

Alias record

Minimize latency

- It helps to translate domain names into IP addresses.
- It updates the registrar's name server configuration with the name servers to provide maximum availability.



Amazon Route 53: Best Practices

The following are the best practices for Amazon Route 53:

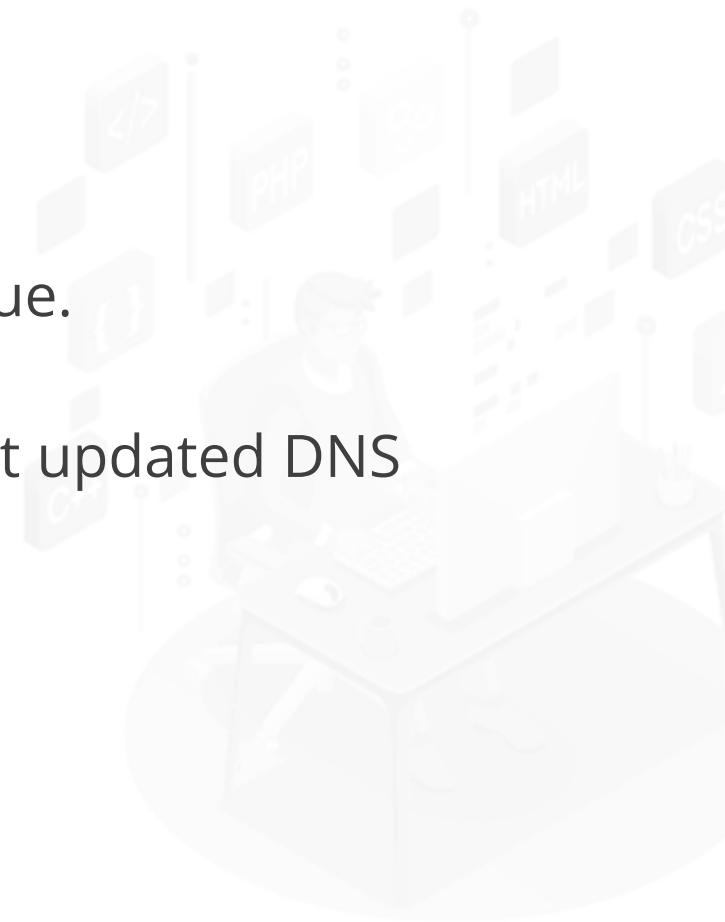
Name server delegation

TTL resource record sets

Alias record

Minimize latency

- Resource records can benefit from a lower TTL value.
- For long TTLs, DNS resolvers take longer to request updated DNS records.



Amazon Route 53: Best Practices

The following are the best practices for Amazon Route 53:

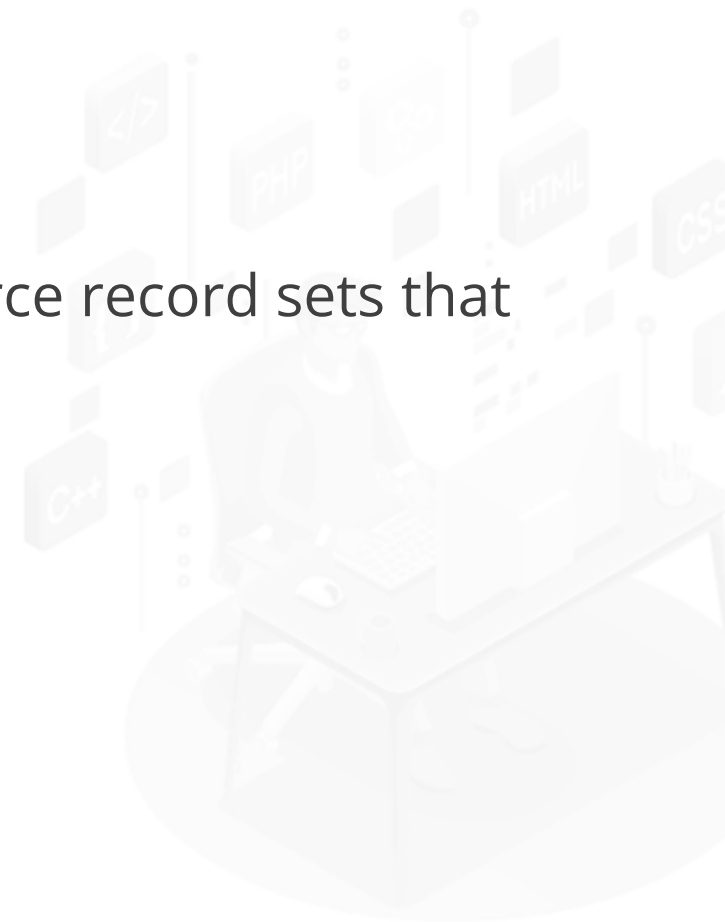
Name server delegation

TTL resource record sets

Alias record

Minimize latency

- Alias record sets are used while configuring resource record sets that route DNS queries to AWS resources.
- These are available at free of cost.



Amazon Route 53: Best Practices

The following are the best practices for Amazon Route 53:

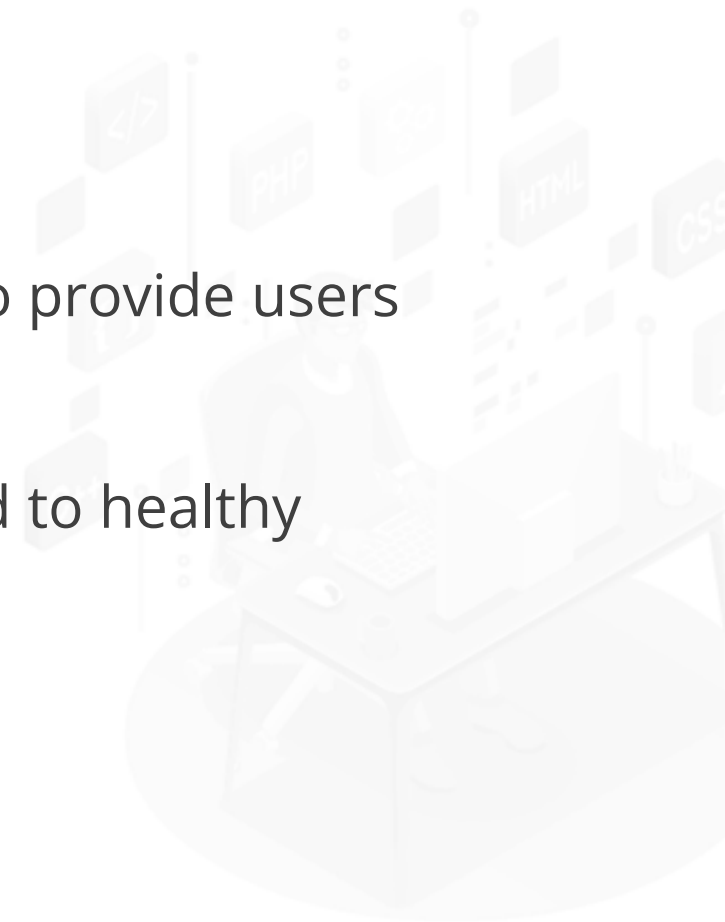
Name server delegation

TTL resource record sets

Alias record

Minimize latency

- Latency or Geolocation routing policies are used to provide users with the best response times.
- A health check ensures that the users are rerouted to healthy instances during an outage.



Limitations of Amazon Route 53

While Amazon Route 53 offers powerful features, it's essential to be aware of its limitations for effective usage.

1

Amazon Route 53 private endpoints are not available via VPN or Direct Connect.

2

It provides no forwarding or conditional forwarding options for domains used on an on-premise network.

3

It does not support private zone transfers.

Pricing

Hosted zones deleted within 12 hours of creation are not charged, allowing for testing and experimentation.



- The first 50 AWS endpoint health linked to an AWS account or paid to the same AWS account as the health check are free.
- Free DNS queries can be made to other AWS services' alias records.

Pricing

Each domain name registered through or transferred to Route 53 is subject to an annual fee. Per AWS account, a maximum of 20 domain names can be registered.



Pricing

The price for the hosted zone depends on the following:



- Charges for a hosted zone are made at the time of creation and on the first of each succeeding month.
- A hosted zone deleted within 12 hours after creation is not charged to facilitate testing.

Pricing

The routing strategy chosen when creating a new record for the hosted zone determines the charges for these query types:



- Alias queries
- Standard DNS
- Geolocation-based queries
- Latency-based queries

Quick Check



You want to distribute traffic evenly across multiple servers but also need the flexibility to assign different weights to each server based on their capacity. Which Route 53 routing policy should you use?

- A. Geolocation Routing
- B. Simple Routing
- C. Latency Routing
- D. Weighted Routing

Key Takeaways

- AWS Identity and Access Management (IAM) is a web service that allows the user to control the authorization of AWS resources.
- Amazon VPC is a service that helps users launch AWS resources into a defined virtual network.
- Amazon Route 53 automatically creates a name server (NS) record that has the same name as the hosted zone.
- VPC flow logs capture information about the IP traffic going to and from the network interfaces in the VPC.



Key Takeaways

- Amazon Global Accelerator is a powerful solution designed to enhance application performance and availability for local and global users.
- VPC flow logs can be published to two destinations: CloudWatch Logs and Amazon S3.
- CloudFront provides seamless integration with Amazon S3, bolstering the security of S3 bucket access by utilizing CloudFront Origin Access Identity (OAI).



Customizing EC2 Instance for Web Instance

Duration: 30 mins



Project agenda: To create a VPC and launch a web instance

Description: You are required to create a custom VPC and a subnet and attach the subnet to the VPC. Additionally, create an internet gateway route table, and launch an EC2 instance.

Perform the following:

1. Create a custom VPC and enable DNS hostname
2. Create an internet gateway and attach it to VPC
3. Create a subnet and a route table
4. Configure a route table
5. Launch the EC2 instance

TECHNOLOGY

Thank You